# Two readings of $X$

Lagrangians, the four-momentum, and the field octants
in the split-signature arena $\mathrm{Cl}(3,3)$

Investigation note to *The electromagnetic sphere*

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**Abstract**

The letter $X$ of the split-signature arena $\mathrm{Cl}(3,3)$ has two readings: a length-valued displacement $X_{\mathrm{cl}}$, with converted times on the temporal triple and lengths on the spatial one, and the field strength $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$, the pair labelling the six components of the electromagnetic two-form. We type the two readings and show that no equivariant linear map relates them; we write the Lagrangians of the worldline, of the interior shifted dynamics, of the real Dirac field and of the energies, and find that the rest energy $E_0 = \frac{1}{2}|q|$ is a Lorentz-invariant and conformal but not a duality-invariant Lagrangian and not an admissible Hamiltonian; we construct the four-momentum for a clock $n$ on the temporal triple, with the momenta conjugate to the three spatial lengths, and show that the mass read by any clock is bounded by the six-dimensional mass; and we place the pair $(\mathbf{B}, \mathbf{E})$ on the eight octants of the spinor module, $\mathbf{E}$ on the Fock weight one and $\mathbf{B}$ on weight two, where Maxwell's equations become one first-order equation with the Gauss laws on the poles while boosts and duality act only through odd operators. The temporal volume $T = e_1e_2e_3 = nJ(n)$ is the element both readings share: the Dirac mass form for every clock, the lift of the frame parity, and the duality operator on the octants; at lowest order, linear in the field, the Dirac field couples to the field octants covariantly only through the Pauli term and its dual. Every finite-dimensional and symbolic claim is verified by script.

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# 1 Introduction

The letter $X$ is used in the arena of [1] for two different things. In the momentum–force construction it is a displacement, a vector of $V = \mathbb{R}^{3,3}$ with the dimension of a length; in the field-strength reading of Section 12 of [1] the same vector space carries the electric and magnetic fields, $X = \sum_i (B_i e_i + E_i e_{i+3})$. A working note of 24 October 2026 [3] proposed to keep the two readings strictly apart, $X_{\mathrm{cl}}$ for the displacement and $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$ for the six components of the field two-form, and to build the four-momentum of each separately. This note answers four questions put on that basis:

(1) construct the Lagrangians for $X$, for the Dirac field and for the energies;

(2) differentiate between the two $X$'s, the first a length and the second the pair $(\mathbf{B}, \mathbf{E})$;

(3) construct the four-momentum, the length-valued $X$ being taken along the three directions of the spatial triple $V^-$, of signature $(-,-,-)$;

(4) place $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$ on the octants, each of the six components $E_x, E_y, E_z, B_x, B_y, B_z$ on an octant.

Question (4) is read as the pair: $X_{\mathrm{EM}}$ is not a sum of two vectors but the six independent components of one field strength. Questions (1) and (2) are answered in Section 2, question (3) in Section 3 and question (4) in Section 4. Section 5 takes up the element that the answers share: the temporal volume $T = e_1 e_2 e_3$, which is the Dirac mass form of the classical reading for every clock $n$, through $T = nJ(n)$ with $J(n)$ the Dirac complex structure of that clock, and the duality operator of the field reading on the octants. Section 6 collects the answers.

**What is established, in brief.** The two readings share a vector space and a Clifford action, but their units, their natural Lorentz algebras, their canonical conjugates and their observables differ, and no equivariant linear map relates them (Theorem 2.3). The Lagrangians are the six-dimensional worldline and the interior shifted Lagrangian for $X_{\mathrm{cl}}$, the real Grassmann Lagrangian for the Dirac field, and the Maxwell Lagrangian $-\frac{1}{2}\operatorname{Re} q$ for the field, whose Hamiltonian is the undivided energy $E_6$; the rest energy $E_0 = \frac{1}{2}|q|$ is a Lorentz-invariant and conformal Lagrangian, but not a duality-invariant one and not an admissible Hamiltonian (Theorem 2.10

and Proposition 2.11). With a clock $n$ on the temporal triple, the four-momentum $(E/c, \mathbf{p})$ has $\mathbf{p}$ conjugate to the three lengths of $V^-$ and $E/c$ conjugate to the time read by $n$, and the four-dimensional mass read by $n$ never exceeds the six-dimensional one (Propositions 3.7 and 3.13); the field's four-momentum comes from its stress tensor and is a four-vector only for an isolated whole (Section 3.4). On the octants, the joint eigenlines of the pole operators, $\mathbf{E}$ sits on the three octants of Fock weight one and $\mathbf{B}$ on the three of weight two; Maxwell's four equations become one first-order equation with the two Gauss laws on the two poles; and the boosts and the duality rotation of the field act on the octants only through odd operators, so that the octant representation of the field is not the spin representation in which a Dirac field transforms (Theorem 4.11 and Proposition 4.12).

**Conventions and classes.** The conventions are those of [1]: $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the temporal triple $V^+ = \langle e_1, e_2, e_3\rangle$ carries the converted times of the classical reading and $\mathbf{B}$ of the field reading, the spatial triple $V^- = \langle e_4, e_5, e_6\rangle$ the lengths and $\mathbf{E}$; the paper's $\mathbf{B}$ is minus the textbook magnetic field. Each section states its units. Every claim carries a class stated in words: *fixed* (an identity or theorem that could have failed), *constructed* (admitted and exhibited, not selected), *obstructed* (impossible under stated hypotheses), *open*, *inherited* (true of ordinary physics in any notation, cited), *interpretive* (a reading), or *restated* (established in [1], [2] or [3] and re-verified here). Where [3] states something, the text says whether it is confirmed, corrected or refuted. Every finite-dimensional and symbolic claim is checked by a script; the checks are described in Section 7.

## 2 Two readings of $X$ and their Lagrangians

The question asked in the introduction contains two requests that this section answers together: that the two uses of the letter $X$ be told apart, the first being a length and the second the pair $(\mathbf{B}, \mathbf{E})$ of three-vectors, and that Lagrangians be written for $X$, for the Dirac field and for the energies. The section types the two readings (Section 2.1), writes the Lagrangians of the classical reading (Section 2.2), of the Dirac field (Section 2.3) and of the energies (Section 2.4), assembles them (Section 2.5) and states what is fixed and what is not (Section 2.6). Identifiers r1-$n$ refer to the lines `PASS r1-`$n$ printed by `verify/r1_lagrangians.py`, which checks every finite-dimensional and symbolic claim in exact arithmetic; Euler–Lagrange equations, Noether currents and stress tensors are derived there on general symbolic fields, and the Grassmann-valued Dirac field is handled by an exact Grassmann jet algebra.

Conventions are those of [1]: $V = \mathbb{R}^{3,3}$ with $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the temporal triple $V^+ = \langle e_1, e_2, e_3\rangle$ and the spatial triple $V^- = \langle e_4, e_5, e_6\rangle$, the realisation of Appendix A of [1], the binding root $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$, which satisfies $G^2 = -\mathbf{1}$, $\eta G = (\eta G)^\top$ and $G^\top \eta G = -\eta$ (r1-1), the interior Lorentz algebra $\mathfrak{l}_{\mathrm{int}}$ on $E = \langle e_1, e_4, e_5, e_6\rangle$ with $\gamma^\mu = (e_1, e_4, e_5, e_6)$, $J = e_2 e_3$ and $T = e_1 e_2 e_3 = \gamma^0 J$. Spacetime signature is $(+,-,-,-)$. The paper's $\mathbf{B}$ is minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section 1.4 of [1]); every Maxwell formula below names which of the two it uses. Each claim carries a class in words: *fixed*, *constructed*, *obstructed*, *open*, *inherited*, *interpretive* or *restated*.

### 2.1 The two readings, typed

**Definition 2.1** (The classical reading)**.** A classical displacement in the arena is

$$X_{\mathrm{cl}} = Y = \sum_{i=1}^{3} c\,t^i\, e_i + \sum_{a=1}^{3} x^a\, e_{a+3}, \qquad \eta(Y,Y) = \sum_i (ct^i)^2 - \sum_a (x^a)^2, \tag{1}$$

with $[t^i] = \mathrm{s}$, $[x^a] = \mathrm{m}$, $[c\,t^i] = \mathrm{m}$ and $[\eta(Y,Y)] = \mathrm{m}^2$. Its length-valued part $\sum_a x^a e_{a+3}$ lies on the spatial triple: this is the precise content of the phrase that $X$ is taken in the three directions

of sign $(-,-,-)$. The temporal components are times, converted to lengths by the constant $c$.

The conversion is not a choice. A map $C : V^+ \to V^-$ that commutes with the common rotations of the two triples is $C = c\,\mathbf{1}$ with $[c] = \mathrm{m\,s^{-1}}$ (Theorem 3.2 of [1]; Proposition 2.7 of [2]), and the binding root acts on the unconverted coordinates $(t; x)$ as

$$G_C = \begin{pmatrix} 0 & -C^{-1} \\ C & 0 \end{pmatrix} = \Phi^{-1} G\, \Phi, \qquad \Phi = \mathrm{diag}(c, c, c, 1, 1, 1), \qquad \Phi(t; x) = Y. \tag{2}$$

Hence $G$ is dimensionless on $Y$, while $G_C$ carries entries of dimension $\mathrm{m\,s^{-1}}$ and $\mathrm{s\,m^{-1}}$: $G$ maps lengths to lengths only after the conversion, and on $Y$ it sends the length $x^a$ to the converted time slot $e_a$ (r1-2; restated from [1, 2], the commutation criterion being fixed there).

**Definition 2.2** (The field reading)**.** The field-strength reading of Theorem 12.3 of [1] is, with SI units and $c$ explicit,

$$X_{\mathrm{EM}} = \sum_{i=1}^{3} (cB_i\, e_i + E_i\, e_{i+3}), \qquad z_c = c\mathbf{B} + i\mathbf{E}, \qquad \eta(X_{\mathrm{EM}}, X_{\mathrm{EM}}) = c^2|\mathbf{B}|^2 - |\mathbf{E}|^2 = \mathrm{Re}\,(z_c \cdot z_c), \tag{3}$$

with $[E_i] = [cB_i] = \mathrm{V\,m^{-1}}$, so $[\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})] = \mathrm{V^2\,m^{-2}}$. $X_{\mathrm{EM}}$ is the pair $(c\mathbf{B}; \mathbf{E})$ of three-vectors, the coordinates of one vector of $\mathbb{R}^{3,3}$ in the basis $e_1, \ldots, e_6$, and not a sum of $\mathbf{E}$ and $\mathbf{B}$: its six entries label the six components of the field two-form, $F_{0i} = E_i$ and $F_{jk} = \epsilon_{ijk} B_i$ in Heaviside–Lorentz units with $c = 1$. In these units $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E}$, $q = z \cdot z$, $E_6 = \frac{1}{2}|z|^2$, $E_0 = \frac{1}{2}|q|$, and $E_6^2 = E_0^2 + |\mathbf{S}|^2$ with $\mathbf{S} = \mathbf{E} \times \mathbf{B}$ (Proposition 2.4 of [2]). The same matrix $\Phi$ of (2) converts $(\mathbf{B}; \mathbf{E})$ to $(c\mathbf{B}; \mathbf{E})$.

**Theorem 2.3** (The type rule)**.** *The two readings share the vector space $\mathbb{R}^{3,3}$, its form $\eta$ and the Clifford action on $S = \mathbb{R}^8$, and differ in the following four respects.*

*(i)* Units. *A linear map $X_{\mathrm{EM}} \mapsto X_{\mathrm{cl}}$ has entries of dimension $\mathrm{m^2\,V^{-1}}$. No monomial $c^a \ell^b \hbar^d$, with real exponents, has this dimension, where $\ell$ is the length of the shift operator (equation (7) of [1]); with the vacuum permittivity admitted, the only such monomial is $\ell^3(\varepsilon_0/\hbar c)^{1/2}$.*

*(ii)* Lorentz action. *Once the clock $e_1$ is chosen, $\mathfrak{l}_{\mathrm{int}}$ acts on $Y$ as $E \oplus 2 \cdot \mathbf{1}$: the four-vector $(ct^1; x^1, x^2, x^3)$, which complexifies to $(\frac{1}{2}, \frac{1}{2})$, and the two scalars $ct^2, ct^3$. The algebra that acts naturally on $X_{\mathrm{EM}}$ is $\mathfrak{l}_G$, the centraliser of $G$ in $\mathfrak{so}(3,3)$, under which $V$ is irreducible and complexifies to $(1,0) \oplus (0,1)$ (Section 12 of [1]). The two subalgebras of $\mathfrak{so}(3,3)$ meet in $0$.*

*(iii)* No equivariant map. *For every Lie algebra isomorphism $\phi : \mathfrak{l}_{\mathrm{int}} \to \mathfrak{l}_G$ let $V_\phi$ be $V$ with $\mathfrak{l}_{\mathrm{int}}$ acting through $\phi$. Then $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(E, V_\phi) = \mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(V_\phi, E) = 0$ and $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(E \oplus 2 \cdot \mathbf{1}, V_\phi) = \mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(V_\phi, E \oplus 2 \cdot \mathbf{1}) = 0$. By contrast $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(\Lambda^2 E, V_\phi) = \{(a + bG) M_0\}$ is two-dimensional: the field strength of the interior four-plane, at grade two (Proposition 12.7 of [1]), is carried to $X_{\mathrm{EM}}$ uniquely up to a scale and a duality angle.*

*(iv)* Conjugates and observables. *The six components of $Y$ are configuration coordinates that Poisson-commute, with conjugate covector $K_A$ of dimension $\mathrm{kg\,m\,s^{-1}}$; its observables are a worldline, a four-momentum and a mass. In Maxwell's theory the configuration variable is the potential $\mathbf{a}$ with $\mathbf{B} = \nabla \times \mathbf{a}$, and $\mathbf{E}$ is (up to $\varepsilon_0$) the momentum conjugate to $\mathbf{a}$, so $X_{\mathrm{EM}}$ is a point of phase space, not of configuration space: $\{\int \mathbf{f} \cdot \mathbf{E}, \int \mathbf{g} \cdot \mathbf{B}\} = -\int \mathbf{f} \cdot \nabla \times \mathbf{g}$ (Heaviside–Lorentz) for test fields $\mathbf{f}, \mathbf{g}$, while the $\mathbf{E}$–$\mathbf{E}$ and $\mathbf{B}$–$\mathbf{B}$ brackets vanish. Its observables are $E_6$, $\mathbf{S}$, the invariants $q$ and the stress tensor.*

*Proof.* (i) Over the base $(\mathrm{m}, \mathrm{kg}, \mathrm{s}, \mathrm{A})$ the volt is $\mathrm{kg\,m^2\,s^{-3}\,A^{-1}}$; none of $c$, $\ell$, $\hbar$ contains the ampere, so the ampere exponent of $\mathrm{m^2V^{-1}}$ cannot be matched, and with $\varepsilon_0$ the linear system for the exponents has the unique solution stated (r1-3). (ii) $\mathfrak{l}_{\mathrm{int}}$ is spanned by $R_i = \frac{1}{2}e_{j+3}e_{k+3}$ and $B_i = \frac{1}{2}e_1e_{i+3}$ ($(i,j,k)$ cyclic), acting on vectors by the commutator; they annihilate $e_2, e_3$ and preserve $E$. The centraliser of $G$ in $\mathfrak{so}(3,3)$ has dimension 6 and is spanned by $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$ and $K_i = \frac{1}{2}(e_ke_{j+3} - e_je_{k+3})$; its commutant on $V$ is $\mathrm{span}\{\mathbf{1}, G\} \cong \mathbb{C}$, a division algebra, and $V$ is irreducible (Corollary 12.5 of [1]). The spans of $\{R_i, B_i\}$ and $\{L_i, K_i\}$ inside $\mathfrak{so}(3,3)$ have trivial intersection (r1-4). (iii) Both triples satisfy $[R_i, R_j] = \epsilon_{ijk}R_k$, $[R_i, B_j] = \epsilon_{ijk}B_k$, $[B_i, B_j] = -\epsilon_{ijk}R_k$ and the same with $(L, K)$, so $R_i \mapsto L_i$, $B_i \mapsto K_i$ is an isomorphism (r1-4). For any isomorphism $\phi$, $V_\phi$ is irreducible of dimension 6 and $E$ irreducible of dimension 4. The image of an equivariant $M : E \to V_\phi$ is a submodule of dimension at most 4, hence 0; the kernel of $M : V_\phi \to E$ is a submodule of dimension at least 2, hence all of $V_\phi$. On the trivial summand an equivariant map lands in the fixed vectors of $V_\phi$, of which there are none, and a map out of $V_\phi$ into it is zero by irreducibility. An intertwiner $\Lambda^2E \to V_\phi$ exists for every $\phi$: the six bivectors of $E$ are $2R_i$ and $2B_i$, so $\Lambda^2E$ is the adjoint module of $\mathfrak{l}_{\mathrm{int}}$ (Proposition 12.7 of [1]), $\phi$ intertwines it with the adjoint module of $\mathfrak{l}_G$ pulled back by $\phi$, and that module is isomorphic to $V$ (r1-5). Schur's lemma [11] on the irreducible $\Lambda^2E \cong V_\phi$ then gives an intertwiner space equal to the commutant $\mathrm{span}\{\mathbf{1}, G\}$. The exact solution of the intertwining equations confirms all these dimensions for $\phi$ and for its parity twist $B_i \mapsto -K_i$ (r1-5), and the Casimirs separate the modules: $\sum R_i^2 - \sum B_i^2 = -3$, $\sum R_iB_i = 0$ on $E$, against $-4$ and $-2G$ (eigenvalues $\pm 2i$) on $V$ (r1-6). (iv) The bracket follows from $\{a_i(\mathbf{x}), E_j(\mathbf{y})\} = \delta_{ij}\delta^3(\mathbf{x}-\mathbf{y})$, read off the first-order action of Proposition 2.12(ii), and from $\mathbf{g}\cdot\nabla\times\mathbf{a} - \mathbf{a}\cdot\nabla\times\mathbf{g} = \nabla\cdot(\mathbf{a}\times\mathbf{g})$ (r1-31). □

Parts (i) and (iii) are fixed: (iii) could have failed for a different pair of subalgebras, and (i) obstructs only conversions built from $c$, $\ell$ and $\hbar$ in SI units. Since $(\varepsilon_0/\hbar c)^{1/2}\mathbf{E}$ has dimension $\mathrm{m^{-2}}$, which is the field of Heaviside–Lorentz units with $\hbar = c = 1$, in those units $\ell^3$ alone supplies the dimension (r1-3); what obstructs a dictionary between the two readings is (iii), not (i). Part (ii) is restated from [1] (Propositions 12.2 and 12.6) except its last sentence, which is fixed (r1-4), and (iv) is inherited electrodynamics [12]. The identity map of $\mathbb{R}^{3,3}$ is linear but not equivariant between the two actions, so sharing the vector space identifies nothing.

**Remark 2.4** (The working note)**.** Definition 2.1 of [3] is confirmed, with one precision: its surplus coordinates $u^a$, there said to be "also length-valued", are the converted times $ct^2, ct^3$ of the temporal triple, length-valued only through $C = c\,\mathbf{1}$. Its distinction between the cotangent momentum $K$ and the momentum $\pi$ of the interior Lagrangian is a distinction between two Lagrangians for the same coordinates, as Section 2.2 shows: applied to $Y$, the interior Lagrangian has canonical momentum $p = \eta\pi$. Definition 2.2 is confirmed as a statement about $\mathfrak{l}_{\mathrm{int}}$-modules: $V_{\mathrm{EM}} \cong \Lambda^2E_n^*$ holds through an isomorphism $\phi$, unique up to the factor $a + bG$, but $\Lambda^2E$ is not a subspace of $V$, and $\mathfrak{l}_G$, under which $V$ is irreducible (Corollary 12.5 of [1]), preserves no proper subspace of $V$, while $G$ itself preserves no Lorentzian four-plane (Proposition 6.2 of [3]). The rule of interpretation of Section 2 of [3] is confirmed and sharpened by parts (i) and (iii): no linear equivariant map exists in either direction, for any isomorphism of the two Lorentz algebras, and in SI units no monomial in $c$, $\ell$, $\hbar$ supplies its dimension; the decomposition (18) of [3] and its consequence "no linear equivariant map $\Lambda^2E_n^* \to E_n$" are confirmed in both directions.

## 2.2 Lagrangians for the classical reading

**Proposition 2.5** (The free worldline in the arena)**.** *Let $Y(\lambda)$ be a curve with $\eta(Y', Y') > 0$, $\lambda$ an arbitrary parameter, $[M] = \mathrm{kg}$, and*

$$L_{\mathrm{free}} = -Mc\sqrt{\eta(Y', Y')}, \qquad S = \int L_{\mathrm{free}}\, d\lambda, \quad [S] = \mathrm{J\,s}. \tag{4}$$

*(i)* $K_A = \partial L/\partial Y'^A = -Mc\,\eta_{AB}Y'^B/\sqrt{\eta(Y',Y')}$, *with* $\eta^{AB}K_AK_B = M^2c^2$ *and* $K_AY'^A - L = 0$*: the canonical Hamiltonian vanishes,* $L$ *is homogeneous of degree one in* $Y'$*, and the dynamics is the constraint* $H = \frac{N}{2}(\eta^{AB}K_AK_B - M^2c^2)$*. The kinetic six-momentum is* $P^A = -\eta^{AB}K_B = McY'^A/\sqrt{\eta(Y',Y')}$*.*

*(ii) The Euler–Lagrange equations are* $dK_A/d\lambda = 0$*; their solutions are the straight lines, in every parametrisation.*

*(iii) The timelike set* $\{\eta > 0\}$ *and the shell* $\{\eta(K,K) = M^2c^2\}$ *are connected: the curve* $(\cos s, \sin s, 0; 0, 0, 0)$ *joins* $e_1$ *to* $-e_1$ *with* $\eta = 1$*, so there is no past and no future. With the clock* $e_1$ *and* $P = (\mathsf{E}/c, q_1, q_2; \mathbf{p})$*, the shell reads* $\mathsf{E}^2 = c^2|\mathbf{p}|^2 + M^2c^4 - c^2|\mathbf{q}|^2$*; for* $M = c = 1$ *and* $\mathbf{q} = (2, 0)$ *the four-dimensional mass squared is* $-3$*.*

*Proof.* (i)–(ii) by differentiation (r1-7). A straight line $Y_0 + Vf(\lambda)$ with $f' > 0$ solves the equations, while the timelike curve $(2\cos\lambda, 2\sin\lambda, 0; \lambda, 0, 0)$, with $\eta(Y',Y') = 3$ and $Y''$ not parallel to $Y'$, does not (r1-8). (iii) Write $Y = (u; v)$ with $u \in V^+$, $v \in V^-$. Along $(u; (1-s)v)$, $0 \le s \le 1$, $\eta$ stays positive, so $\{\eta > 0\}$ retracts onto $\{(u; 0) : u \neq 0\}$ and hence onto the unit sphere of $V^+$, which is connected; the shell is the image of $S^2 \times \mathbb{R}^3$ under $(\omega, v) \mapsto (\sqrt{M^2c^2 + |v|^2}\,\omega; v)$, with inverse $(u; v) \mapsto (u/|u|, v)$. Finally $\eta(P,P) = \mathsf{E}^2/c^2 + |\mathbf{q}|^2 - |\mathbf{p}|^2$ (r1-9). □

The form (4) is inherited [18]; (iii) is restated from Propositions 27.1–27.2 and Example 27.3 of [1], where the surplus momenta $\mathbf{q}$ produce the unstable band and the first-class constraints $q_a \approx 0$ (Proposition 23.4, Theorem 30.1 of [1]) remove it. Equation (8) of [3] is confirmed, and its Option II is this constraint.

The interior Lagrangian of [1] is the split-form Lagrangian of equation (26) there. Applied to the length-valued $Y$ it reads

$$L_\kappa = \frac{m}{2}\,\eta(Y' + \kappa GY,\, Y' + \kappa GY), \qquad \kappa = \frac{c}{\ell}, \quad [\tau] = \mathrm{s}, \quad [m] = \mathrm{kg}, \quad [L_\kappa] = \mathrm{J}, \tag{5}$$

with $Y' = dY/d\tau$; the sum $Y' + \kappa GY$ is homogeneous in dimension only because $G$ acts on the converted coordinates ((2), r1-2, r1-36).

**Proposition 2.6** (The interior Lagrangian on lengths)**.** *(i)* $p = \partial L_\kappa/\partial Y' = m\eta(Y' + \kappa GY)$*,* $\pi := \eta^{-1}p = m(Y' + \kappa GY)$*, and the Euler–Lagrange equation is* $Y'' = -\kappa^2 Y$ *on all six components; on shell* $p' = \kappa G^\top p$ *and* $\pi' = \kappa G\pi$*.*

*(ii)* $L_\kappa = \frac{m}{2}\big(\eta(Y',Y') - \kappa^2\eta(Y,Y)\big) + \frac{d}{d\tau}\Lambda$ *with* $\Lambda = \frac{m\kappa}{2}\eta(Y,GY) = -m\kappa\sum_a (ct^a)\,x^a$*, and* $p - m\eta Y' = \partial\Lambda/\partial Y$*.*

*(iii) The energy* $\mathcal{E} = p\cdot Y' - L_\kappa = \frac{m}{2}\big(\eta(Y',Y') + \kappa^2\eta(Y,Y)\big)$ *is conserved and equals* $H(Y,p) = \frac{1}{2m}\eta^{-1}(p,p) - \kappa\, p^\top GY$*, which generates the flow. Explicitly* $\mathcal{E} = \frac{m}{2}\sum_i((c\dot t^i)^2 + \kappa^2(ct^i)^2) - \frac{m}{2}\sum_a((\dot x^a)^2 + \kappa^2(x^a)^2)$*: the three converted-time oscillators carry positive and the three length oscillators negative energy, and both* $\mathcal{E}$ *on* $(Y,Y')$ *and* $H$ *on* $(Y,p)$ *have signature* $(6,6)$*.*

*Proof.* (i) Differentiation and $G^\dagger = G$ (r1-10). (ii) $\eta(GY,GY) = -\eta(Y,Y)$, $\frac{d}{d\tau}\eta(Y,GY) = 2\eta(Y',GY)$ because $\eta G$ is symmetric, and $\eta(Y,GY) = -2\sum_i y_iy_{i+3}$ (r1-11). (iii) Substitution and Hamilton's equations; the inertia of the two $12\times12$ Hessians is computed exactly (r1-12). □

Item (i) is restated (equations (26)–(27) and Theorem 16.2 of [1]); the claim of [3] (below its equation (15)) that the raised momentum obeys $\pi' = \kappa G\pi$ for constant $G$ is confirmed. Item (ii) is Theorem 16.6 of [1] read on lengths: the boundary term that is $-m\kappa\,\mathbf{E}\cdot\mathbf{B}$ in the field reading is $-m\kappa\sum_a(ct^a)x^a$, the correlation of converted time and length along each paired axis. Item (iii) is fixed; the indefinite sign is that of Theorem 5.14 of [2].

**Proposition 2.7** (The spatial triple is not invariant)**.** *Every solution is* $Y(\tau) = e^{-\kappa G\tau}Y_0 + \frac{\sin\kappa\tau}{\kappa m}\pi_0$, $\pi(\tau) = e^{\kappa G\tau}\pi_0$, *with* $e^{sG} = \cos s\,\mathbf{1} + \sin s\,G$.

*(i) Phase-space data of lengths and spatial momenta,* $Y_0 = \sum_a x_0^a e_{a+3}$ *and* $\pi_0 \in V^-$*, acquire the temporal part* $\sin(\kappa\tau)\sum_a x_0^a e_a$*: lengths are rotated into converted times at once.*

*(ii) Configuration data* $(Y_0, Y_0') \in V^- \times V^-$ *stay in* $V^-$*, since* $Y'' = -\kappa^2 Y$ *is diagonal, but their canonical momentum has the temporal part* $\pi(0)^+ = -m\kappa\sum_a x_0^a e_a$*, nonzero for* $Y_0 \neq 0$*, and* $\pi' = \kappa G\pi$ *maps* $V^-$ *into* $V^+$*; for example* $\pi(0) = e_5$ *gives* $\pi(\tau) = \cos(\kappa\tau)e_5 - \sin(\kappa\tau)e_2$.

*(iii) The zero-momentum orbits* $e^{-\kappa G\tau}Y_0$ *are duality rotations;* $e^{-\kappa G\tau}e_4 = \cos(\kappa\tau)e_4 + \sin(\kappa\tau)e_1$.

*Proof.* The formula solves $Y'' = -\kappa^2 Y$ and $m(Y' + \kappa GY) = \pi$ with $Y(0) = Y_0$; the components are read off (r1-13). □

This is fixed. Restricting $X_{\mathrm{cl}}$ to the spatial triple is therefore not preserved by the interior dynamics as a statement about phase space, while it is preserved as a statement about configurations. The example in (ii) is the one of Section 6 of [3] (equations (16)–(17)), confirmed, and Proposition 3.23 extends the rank of the leakage to every clock. Its reading needs one correction. By (ii) of Proposition 2.6 the rotation of $\pi$ out of a subspace is carried by the gradient $m\kappa\eta GY$ of a boundary term, which a canonical transformation removes (Theorem 16.6 of [1]); the kinetic momentum $mY'$ obeys $mY'' = -m\kappa^2 Y$ and preserves $V^-$ and every four-plane $E_n$. The obstruction of Proposition 6.2 and equation (16) of [3] is therefore an obstruction to taking the canonical $\pi$ as the observable momentum, not a property of the motion. Neither $\pi$ nor $mY'$ is conserved, so neither is a four-momentum.

## 2.3 The Dirac Lagrangian

Section 4 of [1] fixes the kernels and the conditions they must meet, not a normalised Lagrangian: the mass form $T = \gamma^0 J$ is antisymmetric, the kinetic kernels $K^\mu = (\gamma^0 J)(J\gamma^\mu) = -\gamma^0\gamma^\mu$ are symmetric with $K^0 = -\mathbf{1}$, and $(\gamma^0 J)\gamma^\mu$ is antisymmetric (Proposition 4.5 of [1]; r1-14, which also checks $T^2 = -\mathbf{1}$, $[K^\mu, J] = [T, J] = 0$ and that $K^\mu J$ is antisymmetric). The normalised one-constituent Lagrangian is that of Proposition 3.6(iii) of [2]. For a real Grassmann field $\psi : \mathbb{R}^{1,3} \to S \otimes \Lambda$, in Heaviside–Lorentz units with $\hbar = c = 1$,

$$L_D = \tfrac{1}{2}\psi^\top(K^\mu D_\mu - mT)\psi = \tfrac{1}{2}\psi^\top T(J\gamma^\mu D_\mu - m)\psi, \qquad D_\mu = \partial_\mu + qA_\mu J, \tag{6}$$

the real form of $\bar\psi(i\gamma^\mu D_\mu - m)\psi$ with $J$ in the place of $i$ [13], the charge $q$ being dimensionless in these units ($D_\mu = \partial_\mu + \frac{q}{\hbar}A_\mu J$ with $[q] = \mathrm{C}$ in SI); in SI units the density is $\frac{1}{2}\hbar c\,\psi^\top(K^\mu\partial_\mu - \frac{mc}{\hbar}T)\psi$ with $x^0 = ct$ and $[\psi] = \mathrm{m}^{-3/2}$ (r1-36).

**Proposition 2.8** (The real Dirac field)**.** *(i) For arbitrary real* $8 \times 8$ *kernels, the Euler–Lagrange expression of* $\frac{1}{2}\psi^\top A^\mu\partial_\mu\psi - \frac{1}{2}\psi^\top M\psi$ *is* $A^\mu_{\mathrm{sym}}\partial_\mu\psi - M_{\mathrm{anti}}\psi$*. A symmetric mass kernel gives a vanishing bilinear and an antisymmetric kinetic kernel a total derivative.*

*(ii) The field equation of* (6) *at* $q = 0$ *is* $K^\mu\partial_\mu\psi = mT\psi$*, that is* $T(J\gamma^\mu\partial_\mu - m)\psi = 0$*, equivalently* $J\gamma^\mu\partial_\mu\psi = m\psi$ *or* $\gamma^\mu\partial_\mu\psi = -mJ\psi$*; since* $(\gamma^\mu\partial_\mu - mJ)(\gamma^\nu\partial_\nu + mJ) = \Box + m^2$*, each component obeys* $(\Box + m^2)\psi = 0$.

*(iii)* $L_D$ *is invariant under* $\psi \mapsto e^{\theta J}\psi$*;* $D_\mu$ *is covariant under* $\psi \mapsto e^{-q\lambda J}\psi$*,* $A \mapsto A + d\lambda$*, since* $e^{-\theta J}$ *preserves every* $K^\mu$ *and* $T$*. The Noether current* $j^\mu = \frac{1}{2}\psi^\top K^\mu J\psi$ *satisfies* $\partial_\mu j^\mu = \psi^\top J\,\mathcal{E}_D$ *identically, with* $\mathcal{E}_D = K^\mu\partial_\mu\psi - mT\psi$*;* $j^0 = -\frac{1}{2}\psi^\top J\psi$.

*(iv) The canonical tensor* $\Theta^\mu{}_\nu = \frac{1}{2}\psi^\top K^\mu\partial_\nu\psi - \delta^\mu_\nu L_D$ *satisfies* $\partial_\mu\Theta^\mu{}_\nu = -(\partial_\nu\psi)^\top\mathcal{E}_D$ *identically, and its energy density is* $\Theta^0{}_0 = \frac{1}{2}\psi^\top(\gamma^0\gamma^a\partial_a + mT)\psi$*; on a mode with* $\partial_a\psi = Jp_a\psi$ *it is* $\frac{1}{2}\psi^\top JH(p)\psi$*, with* $H(p) = \gamma^0\gamma^a p_a + m\gamma^0$ *the free Hamiltonian of equation (8) of [1].*

(v) *On shell* $\Theta^{\mu\nu} - \Theta^{\nu\mu} = \partial_\rho S^{\rho\mu\nu}$ *with the totally antisymmetric spin tensor* $S^{\rho\mu\nu} = \frac{1}{4}\psi^\top \gamma^0 \gamma^{[\rho}\gamma^\mu\gamma^{\nu]}\psi$*, so the Belinfante tensor* $T_B^{\mu\nu} = \Theta^{\mu\nu} - \frac{1}{2}\partial_\rho S^{\rho\mu\nu} = \frac{1}{4}\psi^\top(K^\mu\partial^\nu + K^\nu\partial^\mu)\psi - \eta^{\mu\nu}L_D$ *is symmetric, conserved on shell, and has* $T_B^{00} = \Theta^{00}$*.*

*Proof.* For a Grassmann $\psi$, $\psi^\top M\delta\psi = \delta\psi^\top M\psi$ when $M$ is antisymmetric and $\psi^\top A\partial\delta\psi = \partial(\psi^\top A\delta\psi) + \delta\psi^\top A^\top\partial\psi$; collecting the left variations gives (i), which the jet algebra confirms for random integer kernels (r1-15). (ii) $K^\mu = TJ\gamma^\mu$ and $T$ is invertible (r1-16). (iii) $(K^\mu J)^\top = -K^\mu J$, and $\partial_\mu j^\mu = \psi^\top K^\mu J\partial_\mu\psi = \psi^\top J(\mathcal{E}_D + mT\psi)$ with $\psi^\top JT\psi = 0$ because $JT$ is symmetric (r1-17). (iv) Exchanging Grassmann factors and using the symmetry of $K^\mu$; $\gamma^0\gamma^a J = J\gamma^0\gamma^a$ and $mT = Jm\gamma^0$ give the mode form (r1-18). (v) The coefficient $\frac{1}{4}$ is solved for, and the identity holds for all six pairs $(\mu,\nu)$ after the time derivatives are eliminated with the field equation (r1-19). □

Item (i) is fixed and turns the admissibility conditions of Proposition 4.5 of [1] into a theorem of the variation: a Grassmann field sees only the symmetric part of a first-order kinetic kernel and only the antisymmetric part of a mass kernel. Items (ii)–(iii) are restated (Section 5 of [1], where the Dirac operator is written $J\gamma^\mu\partial_\mu - m$); the sign of the mass term in $\gamma^\mu\partial_\mu\psi = -mJ\psi$ follows from $K^0 = -\mathbf{1}$ and $T = \gamma^0 J$, and $m \mapsto -m$ passes to the opposite sign. Items (iv)–(v) are inherited [6, 17]. Equation (35) of [1], on the rank-64 block of $P_*$ in the three-constituent module, uses the kinetic kernels $P_*\gamma^0\gamma^\mu P_*$, which are minus the compressed $K^\mu$, with $K^0 = +\mathbf{1}$ on its block, and the Yukawa term $+\frac{y}{2}\psi^\top\Phi\psi$ in place of a mass term: with $y\Phi$ in the place of $mT$ its fermion Lagrangian is minus (6), and its field equation $P_*\gamma^0\gamma^\mu P_* D_\mu\psi + y\Phi\psi = 0$ has the same form (r1-16). For Grassmann fields $\Theta^0{}_0$ is not a number, and the sign of the energy is decided only after quantisation. On the state space that Theorems 5.6–5.7 of [1] construct from outside the real module, the one-momentum Hamiltonian has the spectrum $\{0, \pm E, \pm 2E\}$ with multiplicities $6, 4, 4, 1, 1$, symmetric about zero and bounded below (r1-37); a positive energy requires measuring it from the lowest state, the usual choice of vacuum.

## 2.4 Lagrangians for the energies

In this subsection units are Heaviside–Lorentz with $c = 1$, $A_\mu$ is covariant, $\mathbf{a} = (A_1, A_2, A_3)$, and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ with $F_{0i} = E_i$ and $F_{jk} = \epsilon_{ijk}B_i$ (Theorem 12.3 of [1]). In SI every density below is multiplied by $\varepsilon_0$ with $\mathbf{B} \to c\mathbf{B}$, which makes it a density in $\mathrm{J\,m^{-3}}$ (r1-36).

**Proposition 2.9** (Maxwell's Lagrangian in the field reading)**.** *(i)* $\mathbf{E} = \partial_t\mathbf{a} - \nabla A_0$*,* $\mathbf{B} = \nabla\times\mathbf{a}$*, and* $L_M = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) = -\frac{1}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}}) = -\frac{1}{2}\operatorname{Re} q$*; in SI,* $L_M = \frac{\varepsilon_0}{2}(|\mathbf{E}|^2 - c^2|\mathbf{B}|^2)$*.*

*(ii) The Euler–Lagrange equations of* $L_M - A_\mu j^\mu$ *are* $\partial_\mu F^{\mu\nu} = j^\nu$*:* $\nabla\cdot\mathbf{E} = \rho$ *and* $\partial_t\mathbf{E} = -\nabla\times\mathbf{B} - \mathbf{j}$ *(in textbook variables* $\partial_t\mathbf{E} = \nabla\times\mathbf{B}_{\mathrm{t}} - \mathbf{j}$*), together with the identities* $\partial_t\mathbf{B} = \nabla\times\mathbf{E}$*,* $\nabla\cdot\mathbf{B} = 0$*.*

*(iii) The symmetric tensor* $T^{\mu\nu} = F^{\mu\lambda}F_\lambda{}^\nu + \frac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}$ *is traceless, with* $T^{00} = E_6$ *and* $T^{0i} = -(\mathbf{E}\times\mathbf{B})_i = (\mathbf{E}\times\mathbf{B}_{\mathrm{t}})_i$*: the momentum density is* $-\mathbf{S}$ *in the paper's variables. It obeys* $\partial_\mu T^{\mu\nu} = -F^\nu{}_\lambda\,\partial_\mu F^{\mu\lambda}$ *identically, and* $T^\mu{}_\lambda T^\lambda{}_\nu = E_0^2\delta^\mu_\nu$*, so that* $E_0 = \frac{1}{2}\sqrt{T_{\mu\nu}T^{\mu\nu}}$*.*

*(iv) The second invariant* $\mathbf{E}\cdot\mathbf{B} = \frac{1}{2}\operatorname{Im} q = -\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$ *is the divergence* $\mathbf{E}\cdot\mathbf{B} = -\partial_\mu K^\mu$ *of the Chern–Simons current* $K^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}A_\nu\partial_\rho A_\sigma$ *(*$\varepsilon_{0123} = +1$*), with* $K^0 = -\frac{1}{2}\mathbf{a}\cdot\nabla\times\mathbf{a}$*; its Euler–Lagrange expression vanishes identically, so* $\theta\,\mathbf{E}\cdot\mathbf{B}$ *changes no field equation.*

*Proof.* Symbolic differentiation on general $A_\mu(t,\mathbf{x})$ and $j^\mu(t,\mathbf{x})$ (r1-20, r1-21, r1-23); (iii) on general $(\mathbf{E},\mathbf{B})$ (r1-22). □

All of this is inherited [14, 18, 15]; the Rainich identity and $E_6^2 = E_0^2 + |\mathbf{S}|^2$ are restated from Propositions 2.4–2.5 of [2]. Two signs are fixed by the paper's orientation and are easy to lose:

the second invariant is $+\frac{1}{2}\operatorname{Im} q$ in the paper's variables and $-\frac{1}{2}\operatorname{Im} q$ in textbook ones, and the momentum density $T^{0i}$ is $-\mathbf{S}$, not $\mathbf{S}$. Maxwell's Lagrangian is the $\eta$-form of $X_{\mathrm{EM}}$, a member of the Lorentz class of Theorem 17.11 of [1].

**The rest energy as a Lagrangian.** Write $S = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}_{\mathrm{t}}|^2)$ and $P = \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$, so that $E_0 = \frac{1}{2}|q| = \sqrt{S^2+P^2}$, and set $L_0 := E_0$. For a Lagrangian $L(S,P)$ put $\mathbf{D} = \partial L/\partial\mathbf{E}$ and $\mathbf{H} = -\partial L/\partial\mathbf{B}_{\mathrm{t}}$.

**Theorem 2.10** (The rest-energy Lagrangian)**.** *(i) $L_0$ is Lorentz invariant (annihilated by $\mathfrak{l}_G$), homogeneous of degree two, and invariant pointwise under the duality rotation $z \mapsto e^{i\theta}z$, which does not leave $L_M = -\frac{1}{2}\operatorname{Re} q$ invariant. It is not differentiable at null fields: along $\mathbf{B}_{\mathrm{t}} = (1,0,0)$, $\mathbf{E} = (0,e,0)$ it equals $\frac{1}{2}|1-e^2|$, with one-sided derivatives $-1$ and $+1$ at $e=1$.*

*(ii) $\mathbf{D} = (S\mathbf{E} + P\mathbf{B}_{\mathrm{t}})/E_0$, $\mathbf{H} = (S\mathbf{B}_{\mathrm{t}} - P\mathbf{E})/E_0$, and the Hamiltonian density is $\mathbf{D}\cdot\mathbf{E} - L_0 = \frac{S}{E_0}E_6 = -\cos(\arg q)\,E_6$, which ranges over $[-E_6, E_6]$ and equals $-\frac{1}{2}$ on the magnetic field $\mathbf{B}_{\mathrm{t}} = (1,0,0)$, $\mathbf{E} = 0$, where $L_0$ is not convex in $\mathbf{E}$.*

*(iii) For every $L(S,P)$ the gauge-invariant tensor $T^{\mu}{}_{\nu} = \frac{\partial L}{\partial(\partial_\mu A_\lambda)}F_{\nu\lambda} - \delta^\mu_\nu L$ equals $L_S T_M{}^{\mu}{}_{\nu} + \delta^\mu_\nu(SL_S + PL_P - L)$ and obeys $\partial_\mu T^{\mu}{}_{\nu} = -\mathcal{E}^\lambda F_{\nu\lambda}$, with $\mathcal{E}^\lambda$ the Euler–Lagrange expression of $A_\lambda$. Since $L_0$ is homogeneous of degree one in $(S,P)$, its tensor $T_0 = (S/E_0)T_M$ is traceless: $L_0$ is conformal.*

*(iv) For any $L(S,P)$, $\mathbf{D}\cdot\mathbf{H} - \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = P(L_S^2 - L_P^2 - 1) - 2SL_SL_P$. The defect is $-2P$ for $L_0$ and $-(1+\lambda^2)P$ for $\lambda L_0$: no multiple of $E_0$ is a duality-invariant Lagrangian in the sense of Gaillard and Zumino.*

*(v) The ModMax Lagrangians $L_\gamma = \cosh\gamma\, S + \sinh\gamma\, E_0 = \cosh\gamma\, L_M + \sinh\gamma\, L_0$ have vanishing defect for every $\gamma$, are conformal, and for $\gamma \geq 0$ have $L_S \geq e^{-\gamma} > 0$ and $\partial_\gamma L_\gamma = \frac{1}{2}\sqrt{T^{(\gamma)}_{\mu\nu}T^{(\gamma)\mu\nu}}$; in particular $L_0 = \partial_\gamma L_\gamma|_{\gamma=0} = \frac{1}{2}\sqrt{T^M_{\mu\nu}T^{M\mu\nu}}$. As $\gamma\to\infty$, $L_\gamma/\sinh\gamma \to S + E_0 = L_M + L_0$, whose defect is $-P$.*

*Proof.* (i) Under $L_i, K_i$ the variations of $S$ and $P$ vanish; $|q|$ is invariant under $q \mapsto e^{2i\theta}q$ (r1-24). (ii) Differentiation and $\mathbf{D}\cdot\mathbf{E} - L_0 = (SE^2 + P^2 - S^2 - P^2)/E_0$ with $E^2 - S = E_6$; the Hessian in $\mathbf{E}$ at the stated point has a negative eigenvalue (r1-25). (iii) With $G^{\mu\lambda} = \partial L/\partial(\partial_\mu A_\lambda) = L_S\partial S/\partial(\partial_\mu A_\lambda) + L_P\partial P/\partial(\partial_\mu A_\lambda)$ the identity is polynomial in $(\partial A, L_S, L_P, L)$; conservation uses $G^{\mu\lambda}\partial_\mu F_{\nu\lambda} = \frac{1}{2}G^{\mu\lambda}\partial_\nu F_{\mu\lambda}$, which holds for antisymmetric $G$ by the Bianchi identity (r1-27). (iv) $\mathbf{D} = L_S\mathbf{E} + L_P\mathbf{B}_{\mathrm{t}}$, $\mathbf{H} = L_S\mathbf{B}_{\mathrm{t}} - L_P\mathbf{E}$ (r1-25, r1-26). (v) $L_S = \cosh\gamma + \sinh\gamma\,(S/E_0)$ with $S/E_0 \in [-1,1]$; by (iii) and Proposition 2.9(iii), $\frac{1}{2}\sqrt{T^{(\gamma)}\cdot T^{(\gamma)}} = L_S E_0 = \partial_\gamma L_\gamma$ (r1-26, r1-28). □

The general defect formula in (iv) is restated from Section 14 of [1]; every other computation in the theorem is fixed. The duality invariance and conformality of ModMax are inherited [5], as is the Gaillard–Zumino criterion [9, 10] (Section 14 of [1]) and the generation of ModMax by the root-$T\bar{T}$ flow [4]; the scripts re-derive them. Calling $L_0$ "the rest-energy Lagrangian" is interpretive, and the theorem says what the name costs. $L_0$ is an admissible Lagrangian, Lorentz invariant and conformal, but it is not duality invariant although $E_0$ is pointwise duality invariant; its energy density is indefinite; and it is singular exactly on the null fields that carry radiation. A positive-energy field theory with Lagrangian $\lambda E_0$ is obstructed. What the rest energy is, in the Lagrangian formalism, is the operator that deforms Maxwell's theory into ModMax: $L_0 = \partial_\gamma L_\gamma|_0$, equal to $\frac{1}{2}\sqrt{T_{\mu\nu}T^{\mu\nu}}$ by Rainich's identity. The limit $\gamma\to\infty$ does not isolate it but gives $L_M + L_0$.

**Hamiltonians.** In Hamiltonian variables $(\mathbf{D}, \mathbf{B}_\mathrm{t})$ put $w = \mathbf{B}_\mathrm{t} + i\mathbf{D}$, $s = E_6(w) = \frac{1}{2}(|\mathbf{D}|^2 + |\mathbf{B}_\mathrm{t}|^2)$, $p = |\mathbf{D} \times \mathbf{B}_\mathrm{t}| = |\mathbf{S}(w)|$ and $E_0(w) = \frac{1}{2}|w \cdot w| = \sqrt{s^2 - p^2}$, the last by the identity of Proposition 2.4 of [2]. For $\mathcal{H}(\mathbf{D}, \mathbf{B}_\mathrm{t})$ the field equations are $\dot{\mathbf{D}} = \nabla \times \mathbf{H}$, $\dot{\mathbf{B}}_\mathrm{t} = -\nabla \times \mathbf{E}$ with $\mathbf{E} = \partial\mathcal{H}/\partial\mathbf{D}$, $\mathbf{H} = \partial\mathcal{H}/\partial\mathbf{B}_\mathrm{t}$; the energy flux is $\mathbf{E} \times \mathbf{H}$, the momentum density $\mathbf{D} \times \mathbf{B}_\mathrm{t}$, and Lorentz invariance requires $\mathbf{E} \times \mathbf{H} = \mathbf{D} \times \mathbf{B}_\mathrm{t}$; duality invariance is $\mathbf{E} \cdot \mathbf{B}_\mathrm{t} = \mathbf{D} \cdot \mathbf{H}$ and conformal invariance $\mathbf{D} \cdot \mathbf{E} + \mathbf{B}_\mathrm{t} \cdot \mathbf{H} = 2\mathcal{H}$ [7, 5].

**Proposition 2.11** (Which energies are Hamiltonians)**.** *For $\mathcal{H} = a\,s + b\,E_0(w)$ one has $\mathbf{E} \times \mathbf{H} = (a^2 - b^2)\,\mathbf{D} \times \mathbf{B}_\mathrm{t}$, and $\mathcal{H}$ is duality invariant and conformal for all $(a, b)$. Hence $\mathcal{H}$ is Lorentz invariant exactly when $a^2 - b^2 = 1$; with $a > 0$ these are $\cosh\gamma\,E_6 - \sinh\gamma\,E_0$, the Maxwell Hamiltonian $s = E_6$ at $\gamma = 0$ and the ModMax Hamiltonians for $\gamma > 0$. The rest energy alone, $\mathcal{H} = E_0$, gives $\mathbf{E} \times \mathbf{H} = -\mathbf{D} \times \mathbf{B}_\mathrm{t}$ and is not Lorentz invariant. The Bialynicki-Birula Hamiltonian $\mathcal{H} = p = |\mathbf{S}(w)| = \sqrt{E_6^2 - E_0^2}$ satisfies all three conditions.*

*Proof.* With $\xi = \frac{1}{2}(|\mathbf{D}|^2 - |\mathbf{B}_\mathrm{t}|^2)$, $\eta_w = \mathbf{D} \cdot \mathbf{B}_\mathrm{t}$ and $E_0 = \sqrt{\xi^2 + \eta_w^2}$, $\mathbf{E} = a\mathbf{D} + b(\xi\mathbf{D} + \eta_w\mathbf{B}_\mathrm{t})/E_0$ and $\mathbf{H} = a\mathbf{B}_\mathrm{t} + b(\eta_w\mathbf{D} - \xi\mathbf{B}_\mathrm{t})/E_0$; the cross terms cancel and the $b^2$ term is $-(\xi^2 + \eta_w^2)\mathbf{D} \times \mathbf{B}_\mathrm{t}/E_0^2$ (r1-29). □

The classification of conformal duality-invariant Hamiltonians, ModMax and Bialynicki-Birula's, is inherited [5, 7]. What is fixed here is its form in the variables of [2]: the differentiation $E_6^2 = E_0^2 + |\mathbf{S}|^2$ of the undivided energy supplies all three duality invariants, the admissible Hamiltonians are $\cosh\gamma\,E_6 - \sinh\gamma\,E_0$ and $|\mathbf{S}|$, and the rest energy is admissible only together with $E_6$, never alone.

**Proposition 2.12** (The undivided energy is a Hamiltonian, not a Lagrangian)**.** *(i) Used as a Lagrangian of the potential, $E_6 = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2)$ with $\mathbf{E} = \partial_t\mathbf{a} - \nabla A_0$, $\mathbf{B} = \nabla \times \mathbf{a}$ gives $\nabla \cdot \mathbf{E} = 0$ and $\partial_t\mathbf{E} = +\nabla \times \mathbf{B}$; a transverse plane wave then needs $\omega^2 + k^2 = 0$. The equations are elliptic.*

*(ii) The first-order action $L_H = \mathbf{E} \cdot \partial_t\mathbf{a} - E_6 + A_0\nabla \cdot \mathbf{E}$, with $\mathbf{E}$ independent, has Euler–Lagrange equations $\mathbf{E} = \partial_t\mathbf{a} - \nabla A_0$, $\partial_t\mathbf{E} = -\nabla \times \mathbf{B}$, $\nabla \cdot \mathbf{E} = 0$, and on the first of them $L_H = L_M + \nabla \cdot (A_0\mathbf{E})$: $E_6$ is the Legendre transform of $L_M$, with $\mathbf{E}$ conjugate to $\mathbf{a}$.*

*(iii) With two potentials, $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = \nabla \times \mathbf{C}$, $X_A = (\mathbf{A}; \mathbf{C})$, the Schwarz–Sen Lagrangian $L_{SS} = \frac{1}{2}(\nabla \times X_A) \cdot G\dot{X}_A - \frac{1}{2}|\nabla \times X_A|^2 = \frac{1}{2}(\mathbf{E} \cdot \dot{\mathbf{A}} - \mathbf{B} \cdot \dot{\mathbf{C}}) - E_6$ is first order in time with Hamiltonian $E_6$, is exactly invariant under $X_A \mapsto e^{\theta G}X_A$, and the duality charge density is $-\frac{1}{2}X_A \cdot \nabla \times X_A$, minus the optical helicity density. $L_{SS} - (\mathbf{E} \cdot \dot{\mathbf{A}} - E_6)$ has identically vanishing Euler–Lagrange expression.*

*Proof.* Symbolic Euler–Lagrange equations on general fields, with the plane wave $\mathbf{a} = (0, e^{i(kx-\omega t)}, 0)$ and $A_0 = 0$ in (i) (r1-30, r1-31, r1-32). □

Item (i) is fixed, (ii) inherited [12], and (iii) restated from equation (17) and Theorem 13.10 of [1], the duality-symmetric form of Deser and Teitelboim and of Schwarz and Sen [8, 16]. In (iii) duality is a symmetry off shell because both terms are $G$-invariant: $E_6 = \frac{1}{2}|z|^2$ has duality weight zero, and $G$ is orthogonal for the Euclidean product of the symplectic term. This is the field form of the dichotomy of Theorem 17.11 of [1] and Theorem 2.1 of [2]: the Lagrangian $-\frac{1}{2}\eta(X, X)$ lies in the Lorentz class, the Hamiltonian $E_6$ in the duality class, and the two classes meet in 0, so no quadratic form is both. $E_0$ enters neither $L_M$ nor $L_H$ nor $L_{SS}$; it enters as the deformation of Theorem 2.10(v) and of Proposition 2.11.

## 2.5 The total Lagrangian

With $\hbar = c = 1$ the total Lagrangian of the field theory and of a classical charged test particle is

$$\begin{aligned} \mathcal{L}_{\mathrm{tot}} &= -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} + \tfrac{1}{2} \psi^{\top} (K^{\mu} D_{\mu} - mT) \psi, \\ L_{\mathrm{pp}} &= -M \sqrt{\eta(Y', Y')} - e\, A_{\mu}(x)\, x'^{\mu}, \qquad x = (Y^1; Y^4, Y^5, Y^6). \end{aligned} \tag{7}$$

**Proposition 2.13** (Coupled equations and conservation)**.** *(i) $\partial \mathcal{L}_{\mathrm{tot}} / \partial A_{\nu} = q\, j^{\nu}$, so that Maxwell's equations read $\partial_{\mu} F^{\mu\nu} = J_e^{\nu} := -q\, j^{\nu}$; the Dirac equation is $K^{\mu} D_{\mu} \psi = mT\psi$.*

*(ii) The gauge-covariant tensor $\Theta_D{}^{\mu}{}_{\nu} = \tfrac{1}{2} \psi^{\top} K^{\mu} D_{\nu} \psi - \delta^{\mu}_{\nu} L_D$ satisfies $\partial_{\mu} \Theta_D{}^{\mu}{}_{\nu} = q F_{\lambda\nu} j^{\lambda}$ on the Dirac shell, and $\partial_{\mu} T_M{}^{\mu}{}_{\nu} = -F_{\nu\lambda} \partial_{\mu} F^{\mu\lambda}$, which equals $q F_{\nu\lambda} j^{\lambda}$ on the Maxwell shell. Hence $\partial_{\mu} (T_M + \Theta_D)^{\mu}{}_{\nu} = 0$ on the coupled shell.*

*(iii) For $L_{\mathrm{pp}}$ the surplus momenta $K_2, K_3$ are conserved, and the interior equations are the Lorentz force $\frac{d}{d\lambda} \big( M x'_{\mu} / \sqrt{\eta(Y', Y')} \big) = e F_{\mu\nu} x'^{\nu}$.*

*Proof.* (i)–(ii) in the Grassmann jet algebra with coefficients depending on general $A_{\mu}(t, \mathbf{x})$, the time derivatives of $\psi$ being eliminated with the Dirac equation (r1-33, r1-34). (iii) Symbolic Euler–Lagrange equations for general $A_{\mu}(x)$ (r1-35). □

The coupled statement is inherited [17]; in the arena it is the one-constituent version of Theorem 13.19 of [1], with the Grassmann kernels of (6) in place of the commuting ones there, and the checks make it fixed for this realisation. Equation (19) of [3] is confirmed. The four-momentum of the field is $P^{\mu} = \int T^{0\mu}_{\mathrm{tot}}\, d^3x$, developed in Section 3.4.

Each reading enters $\mathcal{L}_{\mathrm{tot}}$ in its own place. The classical reading is the argument: the coordinates $x^{\mu} = (ct^1; x^a)$ on which $A_{\mu}$ and $\psi$ depend, and the worldline $Y(\lambda)$, whose surplus components $ct^2, ct^3$ couple to nothing but the six-dimensional mass shell. The field reading is a value: $X_{\mathrm{EM}}$ is the field strength $F = dA$ of the interior potential, entering through $L_M = -\tfrac{1}{2} \eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})$, through $T^{\mu\nu}$ and through the Lorentz force, and it becomes a vector of $\mathbb{R}^{3,3}$ with the action of $\mathfrak{l}_G$ only through the two-parameter dictionary of Theorem 2.3(iii). The two meet only in the derivative $\partial_{\mu}$, of dimension $\mathrm{m}^{-1}$ in the classical reading, which carries the potential to the field strength.

## 2.6 Answer

The Lagrangians asked for in the introduction, and the status of each statement, are the following.

(1) *The two $X$'s.* $X_{\mathrm{cl}} = Y$ is a displacement in metres whose length-valued part lies on the spatial triple, the temporal part being times converted by the unique $c$; $X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$ is a field strength in volts per metre. They share $\mathbb{R}^{3,3}$, $\eta$ and the Clifford action, and nothing else: no nonzero linear equivariant map joins them for any identification of their Lorentz algebras, and in SI units no monomial in $c$, $\ell$, $\hbar$ supplies the unit $\mathrm{m}^2\mathrm{V}^{-1}$ such a map would need (Theorem 2.3; fixed, the first statement obstructing a dictionary; the second holds only in SI units, since in Heaviside–Lorentz units with $\hbar = c = 1$ the monomial $\ell^3$ supplies the conversion). They are kept apart by treating $X_{\mathrm{cl}}$ as the argument and $X_{\mathrm{EM}}$ as a value of the fields (Section 2.5; interpretive).

(2) *Lagrangians for $X_{\mathrm{cl}}$.* The free worldline (4) (inherited form; its split-signature consequences restated) and the interior Lagrangian (5) on lengths, with $\kappa = c/\ell$, Euler–Lagrange equation $Y'' = -\kappa^2 Y$, $\pi' = \kappa G \pi$ (restated, confirming [3]) and an energy of signature $(6, 6)$ (fixed). The interior flow does not preserve the spatial triple in phase space (fixed); the obstruction concerns the canonical momentum, which differs from the kinetic one by a boundary-term gradient (fixed; corrects the reading of Section 6 of [3]).

| symbol | reading | SI unit | natural Lorentz action | Lagrangian | canonical conjugate | Noether charges |
|---|---|---|---|---|---|---|
| $Y = X_{\mathrm{cl}}$ | displacement; lengths on $V^-$ | m | $\mathfrak{l}_{\mathrm{int}}$ on $(ct^1; x)$ after the clock; $ct^2, ct^3$ scalars | $-Mc\sqrt{\eta(Y', Y')}$ | $K_A$, $\mathrm{kg\,m\,s^{-1}}$ | $K_A$ (translations), $\mathfrak{so}(3,3)$ charges $Y_A K_B - Y_B K_A$ |
| $Y$, interior | configuration | m | none preserving a Lorentzian four-plane | $\frac{m}{2}\eta(Y' + \kappa GY, Y' + \kappa GY)$ | $p = m\eta(Y' + \kappa GY)$ | energy, signature $(6,6)$ |
| $A_\mu = (A_0; \mathbf{a})$ | potential | $\mathrm{V\,s\,m^{-1}}$ ($A_0 = \phi/c$) | $\mathfrak{l}_{\mathrm{int}}$, $(\frac{1}{2}, \frac{1}{2})$ | $L_M$ through $F = dA$ | $\varepsilon_0 \mathbf{E}$ conjugate to $\mathbf{a}$ | $P^\mu = \int T^{0\mu} d^3x$ |
| $X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$ | field strength | $\mathrm{V\,m^{-1}}$ | $\mathfrak{l}_G$, $(1,0) \oplus (0,1)$ | $L_M = -\frac{\varepsilon_0}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})$ | none: a phase-space point | observables $E_6$, $\mathbf{S}$, $q$, $T^{\mu\nu}$ |
| $X_A = (\mathbf{A}; \mathbf{C})$ | dual potentials | $\mathbf{A}$: $\mathrm{V\,s\,m^{-1}}$, $\mathbf{C}$: V | boosts only on shell; duality $e^{\theta G}$ | $L_{SS}$ | $\frac{1}{2}G^\top \nabla \times X_A$ (second class) | energy $\int E_6$, helicity |
| $\psi$ | Dirac field | $\mathrm{m^{-3/2}}$ | $\mathfrak{l}_{\mathrm{int}}$, $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ | $\frac{1}{2}\psi^\top (K^\mu D_\mu - mT)\psi$ | $\frac{1}{2}\psi^\top$ (second class) | $\int j^0$, $P^\mu$ |
| $E_6$ | undivided energy | $\mathrm{J\,m^{-3}}$ | $T^{00}$, a tensor component | Hamiltonian of $L_M$, $L_{SS}$; not a Lagrangian | – | generates time translation |
| $E_0$ | rest energy | $\mathrm{J\,m^{-3}}$ | scalar, duality invariant | $L_0$ (interpretive); ModMax deformation | – | – |

Table 1: The objects of Section 2: reading, unit, Lorentz action, Lagrangian, conjugate and charges.

(3) *The Dirac Lagrangian.* (6), with the Grassmann variation rule (fixed), the field equation $J\gamma^\mu \partial_\mu \psi = m\psi$ (restated), the current $\frac{1}{2}\psi^\top K^\mu J\psi$ and the canonical and Belinfante tensors (inherited, checked).

(4) *Lagrangians for the energies.* Maxwell's $L_M = -\frac{1}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})$ (inherited); $E_6$ is its Hamiltonian, in the first-order and the duality-symmetric forms (inherited, restated), and as a Lagrangian it gives elliptic equations (fixed); $E_0$ is a Lorentz-invariant conformal Lagrangian with indefinite energy, pointwise duality invariant but not duality invariant in the sense of Gaillard and Zumino (fixed), and it is the generator of ModMax and the partner of $E_6$ in the ModMax Hamiltonian (fixed; the ModMax facts inherited). Reading $L_0$ as "the rest-energy Lagrangian" is interpretive, and as the Lagrangian of a positive-energy theory it is obstructed.

(5) *The total Lagrangian.* (7), with total conservation on the coupled shell (inherited, checked) and Table 1.

(6) *Open.* A dynamical relation between the two readings, which would have to be a new, dimensionful, covariant map, and the quantisation of either reading in the arena's own terms, remain open.

## 3 The four-momentum with $X$ on the spatial triple

The third question asked in the introduction is how the four-momentum is constructed when $X$ is taken in the three directions $(-,-,-)$. We read the sign triple as the values $\eta(e_{a+3}, e_{a+3}) = -1$

of the metric on the spatial triple $V^- = \langle e_4, e_5, e_6\rangle$: the classical configuration $X_{\mathrm{cl}}$ has its three lengths there, and its three remaining components are times converted by $c$ on the temporal triple $V^+$. The starting point is the working note [3]. Every statement of it used below is re-derived and marked as confirmed, corrected or refuted. In brief: (1) a clock $n \in V^+$ selects the four-plane $E_n = \mathbb{R}n \oplus V^-$, and these are exactly the Lorentzian four-planes that contain the three lengths; Theorem 4.1 of [1] holds at every clock, with complex structure $J(n) = e_1e_2e_3\, n$ (Section 3.1); (2) the four-momentum of a particle is the projection $P_n = (E/c, \mathbf{p})$ of the six-momentum, conjugate to $(ct_n, x^a)$ and conserved by every dynamics that is translation invariant along $E_n$; of the three mass shells the arena's metric fixes the sign of the ambient one, and the gauge principle of Section 30 of [1] selects the reduced one (Section 3.2); (3) rotating the clock reads a fixed six-momentum as a four-dimensional mass with $m_4(n)^2 \le \mu^2$, with equality exactly at the clocks parallel to the temporal momentum (Section 3.3); (4) the field $X_{\mathrm{EM}}$ has no linear, and no quadratic, covariant map to a four-vector: its four-momentum is the integrated first row of the stress tensor, a four-vector only for an isolated whole (Section 3.4); (5) the no-go for the binding root concerns $G$ on the vector module $E_n$, where it leaks two of the three spatial directions into the surplus plane at every clock; it does not concern $G = \star$ on the field module (Section 3.5). Section 3.6 collects the verdicts.

**Conventions and units.** As in Sections 2, 3 and 12 of [1]: $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of equation (1) of [1], $s_a := e_{a+3}$, the binding root $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$ ($G^2 = -\mathbf{1}$, $\eta$-self-adjoint, an anti-isometry) and the para-complex $\mathsf{K}e_i = e_{i+3}$, $\mathsf{K}e_{i+3} = e_i$ ($\mathsf{K}^2 = \mathbf{1}$, $\eta$-skew; this is $-\mathcal{K}$ for the grading root $\mathcal{K}$ of Proposition 3.3 of [1]). A bivector $a \wedge b$ acts on $V$ by $v \mapsto a\,\eta(b,v) - b\,\eta(a,v) = \frac{1}{2}[ab, v]$. In the particle reading units are SI with $c$ explicit: times $t^i$ in s, lengths $x^a$ and the coordinates $Y^A$ in m, $[c] = \mathrm{m\,s^{-1}}$, momenta $p_a, q_a$ and $K$ in $\mathrm{kg\,m\,s^{-1}}$, energies in J, masses $M_6, m_4, \mu$ in kg; clocks $n$ and frames are dimensionless. In the field reading the computations use Heaviside–Lorentz units with $c = 1$, with $\mathbf{B}$ measured as $c\mathbf{B}$; the SI form is restored through $z_c = c\mathbf{B} + i\mathbf{E}$, $[\mathbf{E}] = [c\mathbf{B}] = \mathrm{V\,m^{-1}}$ (Proposition 2.8 of [2]). The paper's $\mathbf{B}$ is minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section 1.4 of [1]); the frame Poynting vector is $\mathbf{S} = \mathbf{E} \times \mathbf{B} = -\mathbf{S}_{\mathrm{t}}$, with $\mathbf{S}_{\mathrm{t}} = \mathbf{E} \times \mathbf{B}_{\mathrm{t}}$ the textbook one (equation (1) of [2]). Identifiers r2-$n$ refer to `verify/r2_fourmomentum.py`; r2-0a–c check these conventions, the Fock modes $a_i^{\pm} = \frac{1}{2}(e_i \pm e_{i+3})$ (CAR, $N$ counting the minus signs of the pole operators $\omega_i = e_ie_{i+3}$, which are diagonal) and the generators $L_i, K_i$ of $\mathfrak{l}_G$.

### 3.1 Configuration, clock and the selected four-plane

**Definition 3.1** (Configuration and clock)**.** The classical configuration is $Y = \sum_i c\,t^i e_i + \sum_a x^a s_a$, three converted times on $V^+$ and three lengths on $V^-$, with $\eta(Y,Y) = c^2|t|^2 - |x|^2$ in $\mathrm{m^2}$. A *clock* is a unit vector $n \in V^+$. It defines

$$E_n = \mathbb{R}n \oplus V^-, \qquad U_n = n^\perp \cap V^+, \qquad V = E_n \oplus U_n,$$

an oriented orthonormal frame $(n, u_1, u_2)$ of $V^+$, the time $t_n := \eta(n,Y)/c = n \cdot t$ read by the clock and the surplus coordinates $u^a := \eta(u_a, Y)$, in m. With $\pi_\pm$ the projectors onto $V^\pm$ put $\mathsf{P}_n = n\,\eta(n,\cdot) + \pi_-$ and $\mathsf{Q}_n = \pi_+ - n\,\eta(n,\cdot)$. The clocks of non-negative weight form the Fisher–Rao octant $n_i = \sqrt{w_i}$, $w \in \Delta^2$, which is the octant of radius two of Section 3.3 of [1] scaled by one half; the symmetric clock is $\widehat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$ (Proposition 3.5 of [1]).

This is constructed: a reading rule, the content of Definition 3.1 of [3], which is confirmed in that form. The splitting $V^+ \oplus V^-$ is background structure, not invariant under $SO(3,3)$.

**Proposition 3.2** (The four-planes that contain the three lengths)**.** *(i) $\mathsf{P}_n$ and $\mathsf{Q}_n$ are complementary, $\eta$-self-adjoint, $\eta$-orthogonal projectors onto $E_n$ and $U_n$; $\eta|_{E_n}$ has signature $(1,3)$ and $\eta|_{U_n}$ is positive definite.*

*(ii) Every four-plane $E \subset V$ that contains $V^-$ is $E_n$ for exactly one pair $\pm n$. Hence the four-planes containing the three lengths are parametrised by the clock sphere modulo $\pm$, a two-dimensional family inside the eight-dimensional space of Lorentzian four-planes of Theorem 28.1(iii) of [1].*

*Proof.* (i) $\mathsf{P}_n + \mathsf{Q}_n = \pi_+ + \pi_- = \mathbf{1}$, $\mathsf{P}_n n = n$, $\mathsf{P}_n s_a = s_a$, $\mathsf{Q}_n u_a = u_a$, and the Gram matrices are $\mathrm{diag}(1,-1,-1,-1)$ on $(n, s_1, s_2, s_3)$ and $\mathbf{1}$ on $(u_1, u_2)$. (ii) Since $V^-$ is nondegenerate, $E = V^- \oplus (E \cap V^+)$, and $E \cap V^+$ is a line on which $\eta$ is positive, spanned by a unit $n$ determined up to sign. □

Fixed (r2-1a, symbolic for a general frame in the quaternion parametrisation of $SO(3)$; r2-1b). Taking $X$ "in the three directions $(-,-,-)$" therefore leaves exactly one choice, that of a clock, and every observer of the three lengths is an $E_n$.

**The six-momentum.** The covector $K = K_A\, dY^A$ conjugate to $Y$ has the raised form (equation (6) of [3])

$$K^\sharp = \eta^{-1} K = \frac{E}{c} n + \sum_a p_a s_a + \sum_a q_a u_a, \tag{8}$$

so that $E/c = \eta(n, K^\sharp)$, $p_a = -\eta(s_a, K^\sharp)$ and $q_a = \eta(u_a, K^\sharp)$. With the phase-space form $K_A\, dY^A = E\, dt_n - \sum_a p_a\, dx^a + \sum_a q_a\, du^a$ and $\{Y^A, K_B\} = \delta^A_B$ one finds

$$\{ct_n, E/c\} = 1, \qquad \{x^a, p_b\} = -\delta_{ab}, \qquad \{u^a, q_b\} = \delta_{ab},$$

and all other brackets among these twelve functions vanish (fixed; r2-5a, for the clocks $e_1$, $(2,3,6)/7$ and $(-1,4,8)/9$). So $\mathbf{p}$ is conjugate to the three lengths, $E/c$ to the converted time $ct_n$ of the clock and $q_a$ to the surplus coordinates; the relative sign between the first two is the Lorentzian lowering $K_{a+3} = -p_a$. This is the convention of Theorem 30.1 of [1] and of equation (9) of [3]; the action $-Mc\int ds$ has the opposite overall sign, which exchanges which pair carries the minus sign and changes no equation of motion.

**Proposition 3.3** (Covariance of the selected projection)**.** *(i) An element of $O(V,\eta)$ commutes with $\mathsf{P}_n$ iff it preserves $E_n$ and $U_n$; the commutant of $\mathsf{P}_n$ in $\mathfrak{so}(3,3)$ is $\mathfrak{l}_{\mathrm{int}}(n) \oplus \mathfrak{so}(U_n)$, of dimension 7, where $\mathfrak{l}_{\mathrm{int}}(n)$ is spanned by the six bivectors of $E_n$.*

*(ii) For $\Lambda \in SO^+(E_n)$ extended by the identity on $U_n$, $\mathsf{P}_n(\Lambda K^\sharp) = \Lambda\, \mathsf{P}_n K^\sharp$ and $q_a(\Lambda K^\sharp) = q_a(K^\sharp)$: the projection is a four-vector and the $q_a$ are Lorentz scalars.*

*(iii) The rotations $R_\theta = \exp(\theta\, u_1 \wedge u_2)$ of $U_n$ commute with $\mathsf{P}_n$ and turn $(u^1, u^2)$ and $(q_1, q_2)$ through the same angle; their cotangent lift preserves $K_A\, dY^A$.*

*(iv) The eight generators of $E_n \wedge U_n$ do not commute with $\mathsf{P}_n$, and their eight commutators are linearly independent. For instance at $n = e_1$ the rotation $\exp(\varphi\, e_1 \wedge e_2)$ carries the rest momentum $K^\sharp = (E/c)\, e_1$ to one with $|q_1| = (E/c)|\sin\varphi|$.*

*Proof.* (i) A map commuting with a projector preserves its image and kernel; an isometry preserving $E_n$ preserves $E_n^\perp = U_n$. On Lie algebras this is $\mathfrak{so}(E_n) \oplus \mathfrak{so}(U_n)$, of dimension $6 + 1$. (ii) $\Lambda$ commutes with $\mathsf{P}_n$ by (i), and $\eta(u_a, \Lambda K^\sharp) = \eta(\Lambda^{-1} u_a, K^\sharp) = \eta(u_a, K^\sharp)$. (iii) $\eta$ is positive on $U_n$, so contravariant and covariant components of $U_n$ rotate alike; for an isometry $R$ the contragredient on $K_A$ is $R^{-\top} = \eta R \eta^{-1}$, so $K^\sharp$ moves as $Y$ and $K_A\, dY^A$ is invariant. (iv) A generator of $E_n \wedge U_n$ maps $U_n$ into $E_n$; the example is a direct computation. □

Restated: this is Proposition 3.2 of [3], confirmed, including its stated limitation that nothing here gives covariance under the whole of $SO(3,3)$ (r2-2a for seven clocks including $\widehat{W}$, the dimension count for six rational clocks; r2-2b for a finite boost and the $SO(2)$ at $n = (2,3,6)/7$; r2-2c). Part (i) is the case $C = J(n)$ of Theorem 28.1(iii) of [1].

**Proposition 3.4** (The clock, the surplus plane and $J(n)$)**.** *Let* $I_T = e_1e_2e_3$.

*(i)* $I_T^2 = -\mathbf{1}$*;* $I_T$ *commutes with* $e_1, e_2, e_3$ *and anticommutes with* $e_4, e_5, e_6$.

*(ii)* $J(n) := I_T n = n_1e_2e_3 + n_2e_3e_1 + n_3e_1e_2$ *satisfies* $J(n)^2 = -|n|^2\mathbf{1}$ *and commutes with* $n$ *and with* $s_1, s_2, s_3$*, hence with all of* $\mathfrak{l}_{\text{int}}(n)$.

*(iii) For every oriented orthonormal frame* $(n, u_1, u_2)$ *of* $V^+$*,* $nu_1u_2 = I_T$*; hence* $J(n) = u_1u_2$ *is the oriented unit bivector of* $U_n$*, and* $J(e_1) = e_2e_3 = J$.

*(iv) Under time reversal* $n \mapsto -n$*:* $\mathsf{P}_{-n} = \mathsf{P}_n$*,* $\mathsf{Q}_{-n} = \mathsf{Q}_n$*,* $E(-n) = -E(n)$ *for the energy* $E(n) := c\,\eta(n, K^\sharp)$ *read by the clock,* $J(-n) = -J(n)$ *and* $\Omega_{-n} = -\Omega_n$ *for* $\Omega_n := ns_1s_2s_3$*, so that* $\Gamma_n = J(n)\Omega_n$ *is unchanged.*

*Proof.* (i) is the blade rule (Lemma 2.1 of [1]): $|I||J| - |I \cap J|$ equals 2 for $e_i$ and 3 for $s_a$. (ii) $I_T$ and $n$ commute, so $J(n)^2 = I_T^2 n^2 = -|n|^2$; $I_T$ and $n$ both commute with $n$ and both anticommute with each $s_a$. (iii) $nu_1u_2$ is the volume element of an oriented orthonormal frame of $V^+$, which is $\det(n, u_1, u_2)\, e_1e_2e_3 = I_T$; then $u_1u_2 = n^{-1}I_T = nI_T = I_Tn$. (iv) $\mathsf{P}_n$ is even in $n$, $J$ and $\Omega$ are odd. □

Restated: Proposition 5.1 of [3], confirmed, with a direct proof in place of the rotation argument (r2-3a and r2-3c for symbolic $n$, r2-3b for a symbolic frame, r2-3d). The formula $J(n) = I_Tn$ is the identity $J = \gamma^0 T$ of Remark 4.2 of [1] with $\gamma^0 = n$. The sentence of [3] that time orientation and the sign of the complex structure are tracked together is (iv).

**Theorem 3.5** (The selected complex structure at every clock)**.** *Let* $\gamma^0 = n$*,* $\gamma^a = s_a$ *and* $\Omega_n = ns_1s_2s_3$.

*(i) The joint commutant of* $\mathfrak{l}_{\text{int}}(n)$ *and of the surplus bivector* $J(n)$ *in* $\mathrm{End}_{\mathbb{R}}(S) = M_8(\mathbb{R})$ *is* $\mathrm{span}\{\mathbf{1}, \Omega_n, J(n), \Omega_nJ(n)\} \cong \mathbb{C} \oplus \mathbb{C}$*. Its solutions of* $x^2 = -\mathbf{1}$ *are* $\pm\Omega_n, \pm J(n)$*, and only* $\pm J(n)$ *commute with the four* $\gamma^\mu$.

*(ii) The commutant of the four* $\gamma^\mu$ *alone is* $\mathrm{span}\{\mathbf{1}, J(n), u_1\Omega_n, u_2\Omega_n\} \cong \mathbb{H}$*, and the elements of it that commute with* $I_T$ *are* $\mathrm{span}\{\mathbf{1}, J(n)\}$.

*(iii) The mass form* $\gamma^0 J(n) = nI_Tn = I_T$ *is the same for every clock; the kinetic kernels are* $I_TJ(n)\gamma^\mu = -n\gamma^\mu$*; and* $H_n(\mathbf{p}) = c\,ns_ap_a + mc^2n$ *is symmetric, commutes with* $J(n)$ *and satisfies* $H_n(\mathbf{p})^2 = (c^2|\mathbf{p}|^2 + m^2c^4)\mathbf{1}$.

*Proof.* A rotor $R$ of $\mathrm{Spin}(V^+)$ with $RnR^{-1}$ prescribed exists for every unit $n$, and since temporal bivectors commute with each $s_a$ by the blade rule, $\mathrm{Ad}_R$ fixes $s_1, s_2, s_3$ and $I_T$. It therefore carries $(e_1, e_2, e_3)$ to an oriented frame $(n, u_1, u_2)$, $\mathfrak{l}_{\text{int}}(e_1)$ to $\mathfrak{l}_{\text{int}}(n)$, $e_2e_3$ to $u_1u_2 = J(n)$ and $\Omega$ to $\Omega_n$. Conjugation is an automorphism of $M_8(\mathbb{R})$, so (i) and (ii) are Theorem 4.1 and Remark 4.2 of [1] transported. For (ii), $I_T$ commutes with $u_a$ and anticommutes with $\Omega_n$. (iii) $n$ commutes with $I_T$ and $n^2 = \mathbf{1}$; the square of $H_n$ is that of equation (8) of [1] with $\gamma^0 = n$. □

Fixed. Each part is checked directly, not only through the transport: r2-4a,b compute the commutants and solve $x^2 = -\mathbf{1}$ exactly at $n = e_1, e_2, (1,2,2)/3, (2,3,6)/7, (-1,4,8)/9$; r2-4c is symbolic in a general frame; r2-4d checks the transport by the rotor $\cos\frac{\theta}{2} - \sin\frac{\theta}{2}\, e_1e_2$. So the Dirac spinor read by the clock $n$ has imaginary unit $J(n)$, its sign fixed by the orientation of $V^+$ and of $n$, and its energy for momentum $\mathbf{p}$ is the $E$ of the four-momentum below. The test of the commutant conditions for changing $n$ asked for in Section 5 of [3] is passed. That the mass form does not depend on the clock is consistent with Corollary 18.4 of [1], where $T = e_1e_2e_3$ is the mass kernel invariant under all of $\mathfrak{so}(3,3)$.

**Remark 3.6** (Three readings of $(-,-,-)$)**.** The sign triple has three meanings in the arena, and only the first bears on the four-momentum. (a) The values of $\eta$ on $V^-$, the reading used here (interpretive: a reading of the question). (b) The sign sector $(-,-,-)$ of the temporal $\mathbb{R}^3$, which was proposed, in an earlier and superseded version of the programme, as the carrier of the Lorentzian interior: by Proposition 3.4(iv) the clocks of that sector give the same four-planes and surplus planes as those of the Fisher–Rao octant $(+,+,+)$, with the opposite time orientation, the opposite energy and the opposite complex structure, so the sector carries nothing the Fisher–Rao octant does not (fixed). (c) In the spinor module the joint eigenline of the pole operators with signs $(-,-,-)$ is the filled Fock line $N=3$ (r2-0b), a spinor and not a direction of $V$. The octant readings are the subject of Section 4.

### 3.2 The four-momentum of a particle

The four-momentum read by the clock $n$ is the projection

$$P_n := \mathsf{P}_n K^\sharp = \frac{E}{c} n + \sum_a p_a s_a, \qquad P_n^\mu = (E/c, \mathbf{p}), \qquad P_{n,\mu} = (E/c, -\mathbf{p}), \tag{9}$$

whose three spatial components are the conjugates of the three lengths and whose time component is the conjugate of the clock's time.

**Proposition 3.7** (Four-vector and conservation)**.** *(i) Under the boost* $\exp(\chi\, s_1 \wedge n)$*, extended by the identity on* $U_n$*, the components* $(E/c, \mathbf{p})$ *transform by the* $4\times 4$ *boost of rapidity* $\chi$ *along the first spatial axis, an element of* $SO^+(1,3)$*.*

*(ii) The Hamiltonian flows of* $E/c$ *and of* $-p_a$ *translate* $ct_n$ *and* $x^a$ *by one unit and leave every other coordinate fixed.*

*(iii) For every Hamiltonian* $H(K_A, u^1, u^2)$ *that does not depend on* $(ct_n, x^a)$*,* $\{E/c, H\} = \{p_a, H\} = 0$*. Adding a term* $V(ct_n)$ *destroys only the conservation of* $E$*;* $q_a$ *is conserved iff* $H$ *does not depend on* $u^a$*.*

*Proof.* (i) is Proposition 3.3(ii) in the basis $(n, s_a)$. (ii) is the bracket table above. (iii) $\{P_\mu, H\} = -\partial H/\partial x^\mu$ in the adapted canonical coordinates. □

Fixed (r2-5c at $n = (1,2,2)/3$; r2-5b for a symbolic frame and an arbitrary function $H$). Part (iii) completes Proposition 3.2 of [3]. Its limitation, that the axiom alone does not make $E/c$ a Noether energy, is confirmed, since the axiom fixes no dynamics; (iii) supplies what is missing: $E/c$ and $\mathbf{p}$ are the generators of the translations of $E_n$, hence the Noether charges of every dynamics invariant under them. Whether the energy component of the May paper cited in [3] equals $E/c$ cannot be decided here, since that paper is not among the sources.

**Three mass shells.** The six-momentum is free until a dynamics is chosen. Propositions 3.8, 3.10 and 3.11 treat the three options of Section 4 of [3].

**Proposition 3.8** (Option I: the ambient shell)**.** *For every clock,* $\eta(K^\sharp, K^\sharp) = E^2/c^2 - |\mathbf{p}|^2 + |\mathbf{q}|^2$*. On the shell* $\eta(K^\sharp, K^\sharp) = M_6^2 c^2$*,*

$$E^2 = c^2|\mathbf{p}|^2 + M_6^2 c^4 - c^2|\mathbf{q}|^2, \qquad m_{\text{eff}}^2 = M_6^2 - |\mathbf{q}|^2/c^2,$$

*so the four-dimensional mass vanishes at* $|\mathbf{q}| = M_6 c$ *and is imaginary (tachyonic) beyond.*

*Proof.* $(n, s_a, u_a)$ is $\eta$-orthonormal with norms $(+1, -1, +1)$. □

Fixed (r2-6, symbolic frame); equations (7) and (8) of [3] are confirmed. The sign of $|\mathbf{q}|^2$ is not a choice: the normal plane of a four-plane of signature $(1,3)$ in $\mathbb{R}^{3,3}$ has the sign of time (Theorem 28.1(ii) of [1]). The option is a statement of [2] in another notation.

**Proposition 3.9** (Option I is the momentum reading of [2])**.** *Let* $K^\sharp = \sum_i (\mathsf{E}_i/c) e_i + \sum_a p_a s_a$ *with* $\eta(K^\sharp, K^\sharp) = \mu^2 c^2$*, as in Proposition 2.10 of [2], and write* $\mathbf{E} = (\mathsf{E}_1, \mathsf{E}_2, \mathsf{E}_3)$*, in* J*. For every clock* $n$*,*

$$E = n \cdot \mathbf{E}, \qquad c\, q_a = u_a \cdot \mathbf{E}, \qquad c^2 |\mathbf{q}|^2 = |\mathbf{E}|^2 - (n \cdot \mathbf{E})^2 = |n \times \mathbf{E}|^2,$$

*so Option I holds with* $M_6 = \mu$ *and* $m_{\mathrm{eff}}$ *equal to the mass* $m_4(n)$ *read by the clock. At* $n = \widehat{W}$*,* $c\,|\mathbf{q}| = |\mathbf{E}_\perp|$ *and* $m_4^2 c^4 = \mu^2 c^4 - |\mathbf{E}_\perp|^2$*, which is the statement of [2] that splitting the temporal energy lowers the mass. For* $\mathbf{E} = (3, 4, 0)$*,* $\mathbf{p} = (5, 0, 0)$*,* $c = 1$*:* $\mu = 0$*,* $m_4^2 = -16$ *at* $e_1$ *and* $m_4^2 = -\frac{26}{3}$ *at* $\widehat{W}$*.*

*Proof.* $E = c\,\eta(n, K^\sharp)$*,* $q_a = \eta(u_a, K^\sharp)$*,* and $(n, u_1, u_2)$ is orthonormal; the last identity is Lagrange's. □

Restated from [2] (Proposition 2.10 there), re-verified for a symbolic clock (r2-7). The surplus momenta of [3] are the relative-time energies of [2] divided by $c$, and the tachyonic shell of [3] is realised by the example of [2]. The obstruction stated there, that in the momentum reading the differentiation of the temporal energy subtracts from the mass, is the same obstruction as the sign in Proposition 3.8.

**Proposition 3.10** (Option II: the first-class reduction)**.** *In adapted canonical coordinates take the action of equation (9) of [3],*

$$S_{\mathrm{red}} = \int d\tau \Big( P_\mu \dot{x}^\mu + q_a \dot{u}^a - \tfrac{N}{2} (P_\mu P^\mu - M_6^2 c^2 + |\mathbf{q}|^2) - \lambda^a q_a \Big),$$

*with* $[N] = \mathrm{kg}^{-1}$ *and* $[\lambda^a] = \mathrm{m\,s}^{-1}$ *for* $\tau$ *in seconds. Then* $\{q_a, q_b\} = 0$ *and* $\{q_a, H\} = 0$*, so the constraints* $q_a \approx 0$ *are first class;* $\varepsilon^a q_a$ *generates* $\delta u^a = \varepsilon^a$ *and nothing else;* $S_{\mathrm{red}}$ *is exactly invariant under* $\delta u^a = \varepsilon^a(\tau)$*,* $\delta\lambda^a = \dot{\varepsilon}^a$*; the reduced phase space is* $T^* E_n$*, of dimension* $12 - 2 \times 2 = 8$*, with the shell* $P_\mu P^\mu = M_6^2 c^2$*. With a normal connection* $A_\mu(x)$*,* $k_\mu = P_\mu - A_\mu \ell$ *and* $\ell = v p_u - u p_v$ *at* $n = e_1$*, where* $\mathbf{q} = (p_u, p_v)$*, the brackets become* $\{p_u, H\} = -C p_v$*,* $\{p_v, H\} = C p_u$ *with* $C = A_\mu h^{\mu\nu} k_\nu$*, and the constraints stay first class.*

*Proof.* $H$ does not depend on $u$; the variation of the surplus part of the Lagrangian is $q_a \dot{\varepsilon}^a - \dot{\varepsilon}^a q_a = 0$. The last statement is the computation of Proposition 23.4 of [1]. □

Restated: at $A = 0$ this is Theorem 30.1 and Proposition 23.4 of [1] with the clock $e_1$ replaced by $n$ (r2-8a; r2-8b re-verifies the brackets with $A \neq 0$). Proposition 3.10 confirms Option II of [3] and sharpens its "closely related to the first-class surplus-momentum idea in October Section 30": it is that construction. Two consequences are restated from [1]: gauging only the rotation of $U_n$ leaves the unstable band (Proposition 30.2), and in Dirac quantisation the physical states are functions on $E_n$ obeying the Klein–Gordon equation (Theorem 30.1 and Corollary 30.3). By Proposition 3.9, $q \approx 0$ means $\mathbf{E} \parallel n$: the reduction does not select a clock in $V^+$, it ties the temporal momentum to whichever clock is chosen.

**Proposition 3.11** (Option III: a separate internal mass)**.** *For the action of equation (10) of [3],* $S_n = \int d\tau\, [P_\mu \dot{x}^\mu + q_a \dot{u}^a - \frac{N}{2}(P_\mu P^\mu - c^2 \mathcal{M}^2(u, q))]$ *with an arbitrary function* $\mathcal{M}^2$ *(dimension* $\mathrm{kg}^2$*):*

*(i)* $\dot{P}_\mu = 0$*,* $\frac{d}{d\tau}\mathcal{M}^2 = 0$*, and the constraint is* $E^2 = c^2 |\mathbf{p}|^2 + c^4 \mathcal{M}^2$*;*

*(ii) for the toy* $\mathcal{M}^2 = m_0^2 + |\mathbf{q}|^2/c^2$ *the surplus velocity is* $\dot{u}^a = -N q_a$*, against* $+N q_a$ *on the ambient shell, and the constraint form is* $P_\mu P^\mu - |\mathbf{q}|^2 = \eta(R_n K^\sharp, K^\sharp)$ *with* $R_n = \mathsf{P}_n - \mathsf{Q}_n = \exp(\pi\, u_1 \wedge u_2) \in SO(3, 3)$*, a quadratic form of signature* $(1, 5)$*.*

*Proof.* (i) $H$ has no $x$-dependence, and $\{\mathcal{M}^2, H\} = -\frac{N c^2}{2}\{\mathcal{M}^2, \mathcal{M}^2\} = 0$. (ii) $\partial H/\partial q_a$; $R_n$ is the identity on $E_n$ and $-\mathbf{1}$ on $U_n$, the half-turn of $U_n$. □

Proposition 4.1 of [3] is confirmed, and Option III is constructed: $\mathcal{M}^2$ is supplied from outside the arena (r2-9a for an arbitrary function; r2-9b at $n = e_1, (2,3,6)/7$). The remark of [3] that the positive sign of the toy is opposite to the ambient one is confirmed and made precise by (ii): the toy is the free particle of the twisted form $\eta(R_n\cdot,\cdot)$, which has one time and five space directions, so its sign is that of surplus *lengths*, the Kaluza–Klein sign, and not that of the arena's surplus times.

**Remark 3.12** (Which option the arena prefers)**.** The arena's metric fixes the sign of Option I. For fields the same sign is the ill-posed, unbounded-below baseline of Proposition 27.1 of [1], whose unstable band Option II removes together with every other nonzero surplus momentum (Figure 19 and Theorem 30.1 there; r2-8a). Option III needs a functional whose surplus sign is opposite to that of the metric; the clock supplies one, the twisted form $\eta(R_n\cdot,\cdot)$ of Proposition 3.11(ii), but nothing in the arena selects it in place of $\eta$. So the arena prefers Option II, as the only one of the three that has the arena's sign and no instability; the preference is conditional, because the gauge principle of the normal translations is chosen and not derived (Section 30 of [1]). This is interpretive; whether a principle makes the normal translations gauge is open.

### 3.3 Rotating the clock

Fix a six-momentum $K^\sharp = \sum_i (\mathsf{E}_i/c)e_i + \sum_a p_a s_a$ with temporal energy vector $\mathbf{E}$ and six-dimensional mass $\mu^2c^4 = |\mathbf{E}|^2 - |\mathbf{p}|^2c^2$, and let the clock vary. The energy and the four-dimensional mass read by $n$ are

$$E(n) := c\,\eta(K^\sharp, n), \qquad m_4(n)^2c^4 := E(n)^2 - |\mathbf{p}|^2c^2.$$

**Proposition 3.13** (The mass read by a rotating clock)**.** *(i) On the clock sphere $E(n) = n\cdot\mathbf{E}$ and*

$$m_4(n)^2c^4 = \mu^2c^4 - |n\times\mathbf{E}|^2 = \mu^2c^4 - |\mathbf{E}|^2\sin^2\angle(n,\mathbf{E}).$$

*Hence $m_4(n)^2 \le \mu^2$, with equality iff $n \parallel \mathbf{E}$ (in either orientation), and $E(n) \le |\mathbf{E}|$, with equality iff $n = \mathbf{E}/|\mathbf{E}|$: the clock that reads the largest energy is the direction of the temporal part of $K$, and there $m_4 = \mu$.*

*(ii) On the Fisher–Rao octant, $E(n) = \sum_i \sqrt{w_i}\,\mathsf{E}_i$ and $m_4^2c^4 = \left(\sum_i \sqrt{w_i}\,\mathsf{E}_i\right)^2 - |\mathbf{p}|^2c^2$. Let $\mathbf{E}^+$ be the componentwise positive part. If $\mathbf{E}^+ \neq 0$, $\max E(n) = |\mathbf{E}^+|$, attained at $w_i = (\mathsf{E}_i^+)^2/|\mathbf{E}^+|^2$; if $\mathbf{E}^+ = 0$, the maximum is $\max_i \mathsf{E}_i \le 0$, at a vertex, and no clock of the octant reads a positive energy. If $\mathbf{E}$ has no negative component the maximising clock of (i) lies in the octant; otherwise, if $\mathbf{E}^+ \neq 0$, the largest mass read by a future-reading clock of the octant is $m_4^2c^4 = \mu^2c^4 - |\mathbf{E} - \mathbf{E}^+|^2 < \mu^2c^4$.*

*(iii) At $\widehat{W}$, $E(\widehat{W}) = (\mathsf{E}_1+\mathsf{E}_2+\mathsf{E}_3)/\sqrt{3}$ and $m_4^2c^4 = \mu^2c^4 - |\mathbf{E}_\perp|^2$, the $\mathsf{E}_W$ of [2].*

*Proof.* (i) $|n| = 1$ and Lagrange's identity $|\mathbf{E}|^2 = (n\cdot\mathbf{E})^2 + |n\times\mathbf{E}|^2$. (ii) For $n_i \ge 0$, $n\cdot\mathbf{E} \le n\cdot\mathbf{E}^+ \le |\mathbf{E}^+|$, with equality iff $n = \mathbf{E}^+/|\mathbf{E}^+|$ when $\mathbf{E}^+ \neq 0$; if $\mathbf{E}^+ = 0$, then $\sum_i n_i \ge |n| = 1$ gives $n\cdot\mathbf{E} \le \max_i \mathsf{E}_i \sum_i n_i \le \max_i \mathsf{E}_i$, with equality at the vertex of the largest component; $m_4^2$ increases with $E(n) \ge 0$, and $|\mathbf{E}^+|^2 = |\mathbf{E}|^2 - |\mathbf{E} - \mathbf{E}^+|^2$. (iii) is (i) at $n = \widehat{W}$. □

Fixed (r2-10a symbolic; r2-10c a random test of (ii), 200 vectors $\mathbf{E}$ against 4000 octant clocks each, in floating point with tolerance $10^{-12}$; r2-10d the case $\mathbf{E}^+ = 0$, exactly at $\mathbf{E} = (-1,-2,-3)$ and by a random test). An exact example (r2-10b): $\mathbf{E} = (2,3,6)$, $\mathbf{p} = (0,0,3)$, $c = 1$ give $\mu^2 = 40$; the aligned clock $(2,3,6)/7$ lies in the Fisher–Rao octant with weights $(4,9,36)/49$ and reads $m_4^2 = 40$, as does the opposite clock, which reads the energy $-7$; the clock $e_1$ reads $m_4^2 = -5$, and $\widehat{W}$ reads $\frac{94}{3} = 40 - |\mathbf{E}_\perp|^2$.

**Remark 3.14** (The mass as a projection)**.** Interpretive. In the six-momentum the temporal energy is one vector $\mathbf{E}$, undivided; a clock reads its component $E(n)$, and what the reading leaves out, $c\,\mathsf{Q}_n K^\sharp$, is the part orthogonal to the clock. The energy read by the clock is the projection of the undivided temporal energy onto it; the four-dimensional mass computed from it is maximal, and equal to the six-dimensional $\mu$, only for the clocks parallel to $\mathbf{E}$, and is lowered by every misalignment, which is Proposition 3.9 read on the clock sphere. Nothing in this reading prefers a direction of $V^+$: the aligned clock is fixed relative to the momentum, and for several particles read by one clock the reduction of Proposition 3.10 requires all their temporal momenta to be parallel to it.

### 3.4 The four-momentum of the field

In the field reading the field is a function on spacetime $E_n$, with coordinates $x^\mu = (ct_n, x^a)$, valued in a *second* copy $V_{\mathrm{EM}}$ of $\mathbb{R}^{3,3}$ on which $X_{\mathrm{EM}} = \sum_i (B_i e_i + E_i e_{i+3})$ is read as a two-form through $\Phi : X \mapsto F_X$, $(F_X)_{0i} = E_i$, $(F_X)_{jk} = \epsilon_{ijk} B_i$, with $\mathfrak{l}_G$ as Lorentz algebra (Theorem 12.3 of [1]). Both copies are needed: by Proposition 12.6 of [1] one six-dimensional space cannot carry $\mathfrak{l}_{\mathrm{int}}(n)$ and $\mathfrak{l}_G$ as its Lorentz action at once.

**Proposition 3.15** (The dictionary is equivariant)**.** *$\Phi$ intertwines $\mathfrak{l}_G$ on $V_{\mathrm{EM}}$ with $\mathfrak{l}_{\mathrm{int}}(n)$ on $\Lambda^2 E_n^*$ through a unique isomorphism $\phi : \mathfrak{l}_G \to \mathfrak{l}_{\mathrm{int}}(n)$, namely $L_i \mapsto s_j \wedge s_k$ and $K_i \mapsto s_i \wedge n$ for cyclic $(i,j,k)$. Moreover $F_{GX} = \star F_X$ with $\varepsilon_{0123} = +1$; on $\Lambda^2 E_n^*$ the Hodge star satisfies $\star^2 = -\mathbf{1}$, commutes with $\mathfrak{l}_{\mathrm{int}}(n)$, and is self-adjoint and anti-isometric for $\langle F, F\rangle = \frac{1}{2} F_{ab} F^{ab} = |\mathbf{B}|^2 - |\mathbf{E}|^2 = \eta(X,X)$, a form of signature $(3,3)$.*

Restated from Theorem 12.3 and Proposition 12.7 of [1] (r2-12a, which also finds $\phi$; r2-12b). With $\phi$, $E_n$ becomes a four-vector module of $\mathfrak{l}_G$, and the questions of Section 7 of [3] become questions about $\mathfrak{l}_{\mathrm{int}}(n)$-modules.

**Proposition 3.16** (No covariant map from the field to a four-vector)**.** *(i) For $n = e_1$ and $n = (2,3,6)/7$ the space $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}(n)}(\Lambda^2 E_n^*, E_n)$ is zero, and so is $\mathrm{Hom}_{\mathfrak{l}_G}(V_{\mathrm{EM}}, E_n)$ for the module structure given by $\phi$.*

*(ii) For quadratic maps, $\mathrm{Hom}_{\mathfrak{l}_G}(\mathrm{Sym}^2 V_{\mathrm{EM}}, E_n) = 0$, $\dim \mathrm{Hom}_{\mathfrak{l}_G}(\mathrm{Sym}^2 V_{\mathrm{EM}}, \mathbb{R}) = 2$ (the invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E}\cdot\mathbf{B}$), $\dim \mathrm{Hom}_{\mathfrak{l}_G}(\mathrm{Sym}^2 V_{\mathrm{EM}}, \mathrm{Sym}^2 E_n) = 3$, and the traceless part of the last space is one-dimensional.*

*Proof.* Complexified, $V_{\mathrm{EM}} \cong (1,0) \oplus (0,1)$ and $E_n \cong (\frac{1}{2}, \frac{1}{2})$, and $\mathrm{Sym}^2 V_{\mathrm{EM}} \cong (2,0) \oplus (0,2) \oplus (1,1) \oplus 2(0,0)$ contains no $(\frac{1}{2}, \frac{1}{2})$; Schur's lemma [11] gives the dimensions, and the exact computation of the intertwiner spaces confirms them. □

Part (i) is inherited representation theory, computed (r2-11a, with two controls: the endomorphisms of $\Lambda^2 E^*$ commuting with $\mathfrak{l}_{\mathrm{int}}$ are spanned by the identity and the Hodge star, and $\dim \mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(V, E) = 1$); it confirms the corresponding statement of Section 7 of [3]. Part (ii) is inherited as well, computed (r2-11b), and sharpens that section: [3] says that field momentum "can be quadratic in the field and integrated over space", and both halves are needed. Quadratic alone is not enough: the only traceless symmetric quadratic covariant is the stress tensor, up to scale, and a four-vector arises only after it is contracted with the unit normal of a spacelike hyperplane, which is a clock, and integrated.

**Proposition 3.17** (Pointwise energy–momentum is a tensor row)**.** *For the stress tensor $\Theta^{\mu\nu} = \eta^{\mu\alpha} F_{\alpha\lambda} F^{\lambda\nu} + \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$ of $F_X$ [14]:*

*(i) $\Theta^{00} = E_6$ and $\Theta^{0a} = (\mathbf{S}_{\mathrm{t}})_a = -S_a$; $\Theta$ is symmetric and traceless, $\Theta^\mu{}_\lambda \Theta^\lambda{}_\nu = E_0^2 \delta^\mu{}_\nu$ and $E_6^2 = E_0^2 + |\mathbf{S}|^2$.*

*(ii) The rest field $X = e_1$ ($E_6 = E_0 = \frac{1}{2}$, $\mathbf{S} = 0$) boosted by $\exp(\chi \operatorname{ad}_{K_3})$ has $\mathbf{B} = (\cosh\chi, 0, 0)$, $\mathbf{E} = (0, \sinh\chi, 0)$, $E_6 = \frac{1}{2}\cosh 2\chi$ and $\mathbf{S} = -\frac{1}{2}\sinh 2\chi\, e_3$, while a four-vector with rest components $(\frac{1}{2}, 0)$ would have time component $\frac{1}{2}\cosh\chi$.*

*(iii) For every $X$, $\Theta[\exp(\chi \operatorname{ad}_{K_3})X] = L\,\Theta[X]\,L^{\top}$ with $L$ the boost of rapidity $\chi$ in the $(0,3)$ plane, the passage to the frame moving with velocity $-\tanh\chi\, e_3$.*

Restated from Propositions 2.4 and 2.5 and Theorem 2.6 of [2], re-verified symbolically (r2-13, r2-14a,b); the identities are inherited [15, 18]. In the paper's variables the momentum density is $-\mathbf{S} = -\mathbf{E}\times\mathbf{B}$, the textbook $\mathbf{E}\times\mathbf{B}_{\mathrm{t}}$, that is minus the frame Poynting vector. So $(E_6, -\mathbf{S})$ at a point is the first row of a tensor, quadratic in the boost, and not a four-vector, as [2] states; the density seen by the clock $n$ is $\Theta(\cdot, n)$, different for each clock and of Lorentz length $E_0$ (Proposition 2.5 of [2]).

**Proposition 3.18** (The four-momentum of an isolated field)**.** *(i) If $\partial_\mu \Theta^{\mu\nu} = 0$ and the Cauchy data have compact support, $P^\mu = (U/c, \mathbf{P})$ with $U = \int u\, d^3x$ and $\mathbf{P} = \int \mathbf{g}\, d^3x$, $\mathbf{g} = \varepsilon_0\, \mathbf{E}\times\mathbf{B}_{\mathrm{t}}$ in SI, is independent of the spacelike hyperplane and is a future causal four-vector, with $M^2c^4 = U^2 - c^2|\mathbf{P}|^2 \geq 0$.*

*(ii) For two free solutions the interference part $\Theta[X_1 + X_2] - \Theta[X_1] - \Theta[X_2]$ is divergence free. Hence $P[X_1 + X_2] = P[X_1] + P[X_2]$ whenever the two fields have disjoint supports at some time, and two null wholes with momenta at angle $\vartheta$ have $M^2c^4 = 2U_1U_2(1-\cos\vartheta)$, timelike unless $\vartheta = 0$, that is unless the two four-momenta are proportional.*

*(iii) In Heaviside–Lorentz units with $c = 1$ and $f(s) = (1 - s^2)^2$ on $|s| \leq 1$, zero elsewhere, the field $E_x = af(z - t) + bf(z + t)$, $B_{ty} = af(z - t) - bf(z + t)$ has, per unit transverse area, $U = \frac{256}{315}(a^2 + b^2)$ and $P_z = \frac{256}{315}(a^2 - b^2)$ at every $t$, so $M = \frac{512}{315}|ab|$. Boosting the fields along $z$ with rapidity $\chi$ gives exactly $(U\cosh\chi - P_z\sinh\chi,\ P_z\cosh\chi - U\sinh\chi)$. For $|t| \geq 1$ the pulses are disjoint, $q \equiv 0$ and $\int E_0 = 0$, while $M > 0$.*

*Proof.* (i) Integrate the conservation law over the region between two spacelike hyperplanes; the flux through the timelike boundary vanishes by finite propagation speed [20, 18]. Each $\Theta(\cdot, n)$ is future causal (the dominant energy condition [19]), hence so is the integral. (ii) $\Theta$ is quadratic and divergence free on every solution, so its polarisation is divergence free on pairs of solutions; the interference charge is conserved and vanishes when the supports are disjoint. The mass formula is that of two future null vectors. (iii) The cross terms of $u$ and of $S_{\mathrm{t}z}$ cancel pointwise; a right-moving pulse is multiplied by $e^{-\chi}$ and stretched by $e^{\chi}$ under the boost, and a left-moving one the reverse. □

Item (i) is inherited (von Laue [20]) and is the mass of a whole of Proposition 2.9 of [2], restated; items (ii) and (iii) are fixed by computation (r2-16: the interference part for two Maxwell plane waves with wave vectors $(0, 0, 3)$ and $(4, 0, 0)$, the null sum $(3; 0, 0, 3) + (4; 4, 0, 0)$ of mass squared 24 and the antiparallel sum $(3; 0, 0, 3) + (4; 0, 0, -4)$ of mass squared 48; r2-15, by a genuine Lorentz transformation of the fields and exact integration). With $a = b = 1$, (iii) is item (4) of Proposition 2.9 of [2]. The mass of the whole comes from the non-collinearity of the two parts' four-momenta (their three-momenta are antiparallel), not from their interference, which is absent at $|t| \geq 1$. The four-momentum of the field is the integrated row $\int \Theta^{0\mu} d^3x$, which uses one hyperplane, that is one clock; the result does not depend on it.

**Remark 3.19** (The photon)**.** Inherited, and open in the arena. Quantising the free field gives translation generators with $\widehat{P}^\mu|\mathbf{k}, \lambda\rangle = \hbar k^\mu |\mathbf{k}, \lambda\rangle$, $k$ null and $\lambda = \pm 1$ [22, 17]; this is equation (20) of [3], and its statement that the quantisation is not derived in the arena is confirmed. The arena supplies the classical counterparts: a null field such as $z = (1, i, 0)e^{ik(x_3 - t)}$ of Example 12.11 of [1] has $E_6 = |\mathbf{S}|$ and momentum density along its propagation (r2-17), and on positive-frequency modes the eigenvalue of $G$ is $i$ times the helicity (Example 13.11 of [1]), where the quantum of

$\hbar$ per photon is put in by hand. Nor are the arena's own Fock modes photons: $a_i^{\pm}$ obey the anticommutation relations on $\mathbb{R}^8$, and no finite matrices obey $[a, a^\dagger] = \mathbf{1}$, since the trace of a commutator vanishes (r2-0b, r2-17).

**Remark 3.20** (Base and fibre)**.** The field's four-momentum is a vector of the base $E_n$, the spacetime of the classical reading; the six components of $X_{\mathrm{EM}}$ are values in the fibre $V_{\mathrm{EM}} \cong \Lambda^2 E_n^*$, with units of field strength, and its temporal triple carries $\mathbf{B}$, not time. So $X_{\mathrm{EM}}$ is not a displacement and supplies no extra time components of momentum, as Section 7 of [3] states (confirmed; constructed: the field is a function on the base $E_n$, so its translation charges carry indices of $E_n$; covariance alone would allow more, since the invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E} \cdot \mathbf{B}$ map covariantly to the trivial module $U_n$, r2-11b). A field that depended on the surplus coordinates would carry surplus translation charges, the field analogue of $q_a$; these are what the gauge principle of Proposition 3.10 removes (Corollary 30.3 of [1]). The two readings of $X$ thus meet in one place, which is a reading (interpretive): the classical reading supplies the base $E_n$, and the field reading the fibre over it.

### 3.5 The no-go in its scope

**Lemma 3.21** (Anti-isometry)**.** *If $G^\dagger = G$ and $G^2 = -\mathbf{1}$ then $\eta(Gv, Gw) = -\eta(v, w)$.*

*Proof.* $\eta(Gv, Gw) = \eta(v, G^\dagger G w) = \eta(v, G^2 w)$. □

Restated (Proposition 3.3 and Theorem 16.2 of [1]; Lemma 6.1 of [3], confirmed; r2-0a, r2-18).

**Proposition 3.22** (No Lorentzian plane is $G$-invariant)**.** *Every nondegenerate $G$-invariant subspace of $V$ has neutral signature $(k, k)$. In particular no four-plane of signature $(1, 3)$, and no $E_n$, is invariant under an $\eta$-self-adjoint $G$ with $G^2 = -\mathbf{1}$.*

*Proof.* If $GW \subset W$, then $G|_W$ is invertible and by Lemma 3.21 makes $\eta|_W$ congruent to $-\eta|_W$; by Sylvester's law of inertia [21] a signature $(p, q)$ then equals $(q, p)$, so it is $(k, k)$. For the binding root the Gram matrix of $\mathrm{span}(v, Gv)$ is $\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$, of determinant $-(a^2 + b^2) \leq 0$. □

Restated: Proposition 6.2 of [3] is confirmed (r2-18: the symbolic identities, and 300 random $G$-invariant four-planes of signature $(2, 2)$, in floating point with tolerance $10^{-9}$).

**Proposition 3.23** (Leakage at every clock)**.** *For the interior Lagrangian $L_{\mathrm{int}} = \frac{m}{2}\eta(\dot{X} + \kappa G X, \dot{X} + \kappa G X)$ of equation (26) of [1], with $X$ length-valued and $\kappa = c/\ell$ of equation (7) there:*

*(i) $p = m\,\eta(\dot{X} + \kappa G X)$, the Euler–Lagrange equation is $\dot{p} = \kappa G^\top p$, and $\pi = \eta^{-1} p$ obeys $\dot{\pi} = \kappa G \pi$, so $\pi(\lambda) = (\cos\kappa\lambda + \sin\kappa\lambda\, G)\pi(0)$. At $n = e_1$, $\pi(0) = e_5$ gives $\pi(\lambda) = \cos(\kappa\lambda) e_5 - \sin(\kappa\lambda) e_2$ and $\mathsf{P}_{e_1}\pi = \cos(\kappa\lambda) e_5$; along every solution $\eta(\pi, \pi)$ becomes $\cos 2\kappa\lambda\, \eta(\pi, \pi) + \sin 2\kappa\lambda\, \eta(\pi, G\pi)$.*

*(ii) For every clock, $\mathsf{Q}_n G n = 0$ and $\mathsf{Q}_n G \mathsf{P}_n = -(\mathbf{1} - nn^\top) \circ \sigma$, where $\sigma : s_a \mapsto e_a$; hence $\mathrm{rank}\, \mathsf{Q}_n G \mathsf{P}_n = 2$, with kernel $\mathrm{span}(n, Gn)$ in $E_n$.*

*(iii) A vector $v \in E_n$ stays in $E_n$ under $\exp(\theta G)$ for all $\theta$ iff $v \in \mathrm{span}(n, Gn)$, a plane of signature $(1, 1)$; at $n = e_1$ it is $\langle e_1, e_4 \rangle$.*

*Proof.* (i) Direct variation; $\eta^{-1} G^\top \eta = G$, and $e^{\theta G} = \cos\theta + \sin\theta\, G$ since $G^2 = -\mathbf{1}$; $G$ is an anti-isometry, so $e^{\theta G}$ is not an isometry. (ii) $G$ maps $V^+$ into $V^- \subset E_n$, so $\mathsf{Q}_n G n = 0$; on $V^-$, $G = -\sigma$ lands in $V^+$, of which $\mathsf{Q}_n$ keeps the part orthogonal to $n$: an idempotent of trace two composed with an isomorphism. (iii) If $e^{\theta G} v \in E_n$ for all $\theta$, the derivative at 0 gives $\mathsf{Q}_n G v = 0$, so $v \in \mathrm{span}(n, Gn)$; conversely this plane is $G$-invariant, and $\eta(Gn, Gn) = -1$. □

Fixed (r2-19a with sympy Euler–Lagrange equations; r2-19b symbolic in a general frame, and exact ranks at six rational clocks; r2-19c). Equations (16) and (17) of [3], its rank two at $n = e_1$ and its remark that only $\langle e_1, e_4\rangle$ stays are confirmed; (ii) extends the rank to every clock, and the mechanism is that $G$ maps the three spatial directions into $V^+$, and of $V^+$ only the $n$-component returns to $E_n$. If the interior momentum $\pi$ is identified with $K^\sharp$ and $q \approx 0$ is imposed on its flow, the momentum is confined to the $(1,1)$ plane $\mathrm{span}(n, Gn)$ and two of the three spatial momenta are lost: the identification is obstructed. Allowing $q \neq 0$ does not help, since neither $\mathsf{P}_n\pi$ nor even $\eta(\pi,\pi)$ is conserved, which confirms the last sentence of Section 6 of [3].

**Proposition 3.24** (The two Lorentz algebras and the clock)**.** $\mathfrak{l}_{\mathrm{int}}(n)$ *fixes* $U_n$ *pointwise and* $\mathfrak{l}_G$ *fixes no nonzero vector;* $\mathfrak{l}_G \cap \mathfrak{l}_{\mathrm{int}}(n) = 0$*; and* $\mathfrak{l}_G$ *meets the stabiliser* $\mathfrak{so}(E_n) \oplus \mathfrak{so}(U_n)$ *of the clock exactly in the line of the diagonal rotation* $\sum_i n_i L_i$*. G commutes with* $\mathfrak{l}_G$ *and not with* $\mathfrak{l}_{\mathrm{int}}(n)$*.*

The first two statements restate Proposition 12.2 of [1]; the rest is fixed (r2-20 at $n = e_1, (1,2,2)/3$, and $\widehat{W}$ for the fixed planes). Proposition 3.23 is the vector-module face of the exclusion of Proposition 12.6 of [1]: the operator that is the Hodge star for $\mathfrak{l}_G$ moves every four-plane of $\mathfrak{l}_{\mathrm{int}}(n)$, and the only symmetry the two structures share is the rotation about the clock, whose axis plane $\mathrm{span}(n, Gn)$ is the one that survives in Proposition 3.23(iii).

The no-go does not obstruct $G$ as the Hodge star. On the field module $\Lambda^2 E_n^*$ the star acts on the whole module, which is neutral of signature $(3,3)$ (Proposition 3.15); Proposition 3.22 forbids only invariant subspaces that are not neutral, and Theorem 16.2 of [1] requires exactly $P = Q$, which $\Lambda^2 E_n^*$ satisfies. So $G_{\mathrm{EM}} = \star$ is electric–magnetic duality on field strengths, commuting with the Lorentz action, while $G_{\mathrm{cl}}$ on the vector module $E_n$ is obstructed. This is restated (Theorems 12.3 and 16.2 of [1]; r2-12b with Proposition 3.22) and confirms the revised interpretation of Section 6 of [3].

### 3.6 Answer

For the particle reading the four-momentum is constructed as follows. $X$ is taken with its lengths on the three directions of $V^-$; a clock $n \in V^+$ selects the four-plane $E_n = \mathbb{R}n \oplus V^-$, and every four-plane containing the lengths is of this form. The six-momentum conjugate to $Y$ splits into the four-momentum $P_n = (E/c, \mathbf{p})$, conjugate to $(ct_n, x^a)$, and two surplus momenta $q_a$, Lorentz scalars. $P_n$ is a four-vector and the Noether charge of the translations of $E_n$. What it needs beyond the arena is a dynamics: the arena's metric gives the ambient shell, on which $q$ lowers the mass; the gauge principle of [1] gives the reduced shell $q \approx 0$, on which the temporal momentum is parallel to the clock and $m_4 = \mu$; a positive internal mass needs a functional of the opposite sign, that of extra space dimensions. For the field reading the four-momentum is the integrated first row of the stress tensor on the base $E_n$, a four-vector for an isolated whole, with the mass of a whole from parts whose four-momenta are not collinear; no linear or quadratic covariant map produces it pointwise, and $X_{\mathrm{EM}}$ supplies no time components. The two readings share the clock: it fixes $E_n$, the complex structure $J(n)$ of the Dirac spinor and the hyperplane on which the field's momentum is integrated.

| statement | class | checks | on [3] |
|---|---|---|---|
| four-planes containing $V^-$ are the $E_n$ (Proposition 3.2) | fixed | r2-1a,b | new |
| $\mathsf{P}_n$ covariant under $SO^+(1,3) \times SO(2)$ only (Proposition 3.3) | restated | r2-2a–c | Prop. 3.2 confirmed |
| $J(n) = I_T n$, $J(n) = u_1 u_2$, $J(-n) = -J(n)$ (Proposition 3.4) | restated | r2-3a–d | Prop. 5.1 confirmed |
| Theorem 4.1 of [1] at every clock; mass form $I_T$ clock-independent (Theorem 3.5) | fixed | r2-4a–d | test of Sec. 5 passed |

| statement | class | checks | on [3] |
|---|---|---|---|
| $P_n$ four-vector, Noether charge of translations (Proposition 3.7) | fixed | r2-5a–c | limitation confirmed; completed |
| ambient shell, tachyonic for $\|\mathbf{q}\| > M_6 c$ (Proposition 3.8) | fixed | r2-6 | eqs. (7),(8) confirmed |
| Option I is the momentum reading of [2] (Proposition 3.9) | restated | r2-7 | new link |
| first-class reduction = [1, Thm. 30.1] at $A = 0$ (Proposition 3.10) | restated | r2-8a,b | sharpened |
| internal mass functional; toy has signature $(1, 5)$ (Proposition 3.11) | constructed; sign fixed | r2-9a,b | Prop. 4.1 confirmed |
| the arena prefers Option II (Remark 3.12) | interpretive | – | new |
| $m_4(n)^2 \le \mu^2$, equality iff $n \parallel \mathbf{E}$; octant maximum (Proposition 3.13) | fixed | r2-10a–d | new |
| mass as projection onto the clock (Remark 3.14) | interpretive | – | new |
| dictionary equivariant, $\star$ on a neutral module (Proposition 3.15) | restated | r2-12a,b | confirmed |
| no linear map to a four-vector (Proposition 3.16(i)) | inherited | r2-11a,b | confirmed |
| no quadratic map to a four-vector (Proposition 3.16(ii)) | inherited | r2-11b | confirmed, sharpened |
| pointwise $(E_6, -\mathbf{S})$ is a tensor row (Proposition 3.17) | restated | r2-13, r2-14a,b | confirmed |
| von Laue four-vector, mass of a whole (Proposition 3.18(i)) | inherited | – | confirmed |
| interference conserved; exact packets (Proposition 3.18(ii),(iii)) | fixed | r2-15, r2-16 | new |
| photon $P = \hbar k$, two helicities (Remark 3.19) | inherited; open in the arena | r2-17 | Sec. 7 confirmed |
| $X_{\mathrm{EM}}$ supplies no time components (Remark 3.20) | constructed | r2-11b | confirmed |
| no $G$-invariant $(1, 3)$ plane (Proposition 3.22) | restated | r2-18 | Prop. 6.2 confirmed |
| $\operatorname{rank} \mathsf{Q}_n G \mathsf{P}_n = 2$ at every clock; $\pi \equiv K^\sharp$ with $q \approx 0$ (Proposition 3.23) | fixed; obstructed | r2-19a–c | extended |
| $\mathfrak{l}_G \cap (\mathfrak{so}(E_n) \oplus \mathfrak{so}(U_n)) = \mathbb{R} \sum_i n_i L_i$ (Proposition 3.24) | fixed | r2-20 | new |
| $G = \star$ on $\Lambda^2 E_n^*$ not obstructed | restated | r2-12b | confirmed |
| a principle making the normal translations gauge | open | – | open there too |

What remains open is what remained open in [3], now located more precisely: a principle that makes the surplus translations gauge rather than chosen, a dynamics that selects the clock rather than aligning it with the momentum, and a derivation of the photon's Hilbert space from the arena. Nothing in this section identifies $X_{\mathrm{cl}}$ with $X_{\mathrm{EM}}$; they share the clock, and through it the base on which both four-momenta live.

# 4 The field-strength sector $X_{\mathrm{EM}} = (\mathbf{B};\mathbf{E})$ on the Clifford octants

The question asked in the introduction about the octants concerns the field reading $X_{\mathrm{EM}} = (\mathbf{B};\mathbf{E})$, where each of the six components $E_x, E_y, E_z, B_x, B_y, B_z$ is to sit on an octant. In the sense of the working note [3] (its Definition 2.2), $X_{\mathrm{EM}}$ is not a sum of two vectors and not a displacement: the pair $(\mathbf{B};\mathbf{E})$ labels the six components of the field two-form $F$, and the question is how these six components, as six independent components of one field strength, sit on the eight octants of the spinor module. Everything below is a classical field representation: placing $(\mathbf{B};\mathbf{E})$ on six lines of $S$ quantises nothing. A construction of this kind was proposed in an earlier, superseded version of the programme: the eight octants are the joint eigenspaces of the pole operators $\omega_i$; the six mixed octants are field octants, those with one plus sign (temporal weight one there, Fock weight $N = 2$ here) carrying the electric components (as boosts) and those with two plus signs ($N = 1$) the magnetic components (as rotations); the two poles carry "mass" and "statistics"; and Maxwell's equations are "pooled" across the octants. This section establishes in the conventions of [1] what of this is true. In brief: the octants are the eight lines of a Fock basis of the spinor module $S = \mathbb{R}^8$, which is the exterior algebra of a null three-plane; there is a field-octant map $\Phi$ placing the electric member $\mathbf{E}$ of the pair on the Fock weight $N = 1$ and the magnetic member $\mathbf{B}$ on $N = 2$, unique up to a scale and a quarter turn of duality once rotations and Maxwell's equations are required, the quarter turn giving the earlier labelling; it carries the rotations, a parity and the premetric $\mathfrak{gl}(3)$, but no even element of $\mathrm{Cl}(3,3)$ carries the boosts or the duality rotation; the $\mathrm{Spin}(3,3)$-invariant mass form pairs each octant with its antipode and $\mathbf{E}$ with $\mathbf{B}$; and Maxwell's four equations are one first-order equation on $S$, with the two Gauss laws on the poles. In textbook variables the whole construction is Hestenes' Pauli-algebra electrodynamics, written with the three temporal generators acting on the vacuum pole.

**Conventions.** $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the generators $e_1,\dots,e_6$ are those of eq. (1) of [1], $\omega := e_1\cdots e_6 = \Gamma$, $T := e_1e_2e_3$ (the mass form, Corollary 18.4 of [1]), and $\mathsf{P} = \sum_i \omega_i$, $\omega_i := e_ie_{i+3}$. The field point is $X = \sum_i (B_ie_i + E_ie_{i+3})$ (Theorem 12.3 of [1]), with $\mathbf{B}$ minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$; $z = \mathbf{B} + i\mathbf{E}$, $q = z\cdot z$, $E_6 = \frac{1}{2}|z|^2$, $E_0 = \frac{1}{2}|q|$ (eq. (1) of [2]). The binding root is $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$; the grading root of [1] is $\mathcal{K} = \frac{1}{2}\,\mathrm{ad}_{\mathsf{P}}$, $\mathcal{K}e_i = -e_{i+3}$, $\mathcal{K}e_{i+3} = -e_i$ (Proposition 3.3 and Lemma 3.4 of [1]; the para-complex structure of the earlier version is $-\mathcal{K}$). On $V$ we use coordinates $(\mathbf{B};\mathbf{E}) \in \mathbb{R}^6$. Units are Heaviside–Lorentz with $c = 1$ unless SI is stated. Dimensions: in SI $[\mathbf{E}] = \mathrm{V\,m^{-1}}$, $[\mathbf{B}] = \mathrm{T}$, so that $\mathbf{E}$ and $c\mathbf{B}$ share a unit; $[t] = \mathrm{s}$, $[x^a] = \mathrm{m}$, $[\partial_a] = \mathrm{m^{-1}}$; $[\rho] = \mathrm{C\,m^{-3}}$, $[\mathbf{j}] = \mathrm{A\,m^{-2}}$, $[\varepsilon_0] = \mathrm{F\,m^{-1}}$; the generators, the Fock operators, the octant basis vectors and $T$ are dimensionless, and $\psi = \Phi(X)$ carries the dimension of a field strength. Identifiers r3-$n$ refer to the script `verify/r3_octants.py`, which checks every finite-dimensional and symbolic claim below exactly.

## 4.1 The octants and the octant group

**Proposition 4.1** (Octants and Fock grading)**.** *(i) The pole operators are $\omega_i|b\rangle = (-1)^{b_i}|b\rangle$ on the basis $|b_1b_2b_3\rangle$ of $S$; their joint eigenlines, the* octants*, are the eight basis lines, labelled by the sign triples $\sigma = (\sigma_1,\sigma_2,\sigma_3)$, $\sigma_i = (-1)^{b_i}$.*

*(ii) $a_i^{\pm} := \frac{1}{2}(e_i \pm e_{i+3})$ satisfy $\{a_i^+, a_j^+\} = \{a_i^-, a_j^-\} = 0$, $\{a_i^+, a_j^-\} = \delta_{ij}$, and $\omega_i = \mathbf{1} - 2a_i^+a_i^-$. Hence $N := \sum_i a_i^+a_i^-$ is diagonal and counts the minus signs of $\sigma$.*

*(iii) $|\mathrm{vac}\rangle := |000\rangle$, of signs $(+,+,+)$, spans the common kernel of the $a_i^-$ (the Fock vacuum, $N = 0$), and $a_{i_1}^+\cdots a_{i_r}^+|\mathrm{vac}\rangle = |b_I\rangle$ for $i_1 < \cdots < i_r$, where $b_I$ has ones exactly at $I$. The filled pole is $|\mathrm{full}\rangle := |111\rangle$, of signs $(-,-,-)$ and $N = 3$.*

*(iv) The octant $\sigma$ is the pure spinor line of the maximal isotropic plane $N_\sigma := \mathrm{span}\{e_i - \sigma_i e_{i+3}\}$: the vectors $v \in V$ with $v|b\rangle = 0$ are exactly those of $N_\sigma$.*

*(v)* $\Gamma = -\omega_1\omega_2\omega_3$, *so* $(-1)^N = \omega_1\omega_2\omega_3 = -\Gamma$, *which is the chirality* $\gamma_5$ *of Section 5 of [1]; on the octant with* $w = 3 - N$ *plus signs* $\Gamma = (-1)^w$.

*Proof.* (i) is in Proposition 2.5 and Appendix A of [1]: $e_i|b\rangle = (-1)^{b_1+\cdots+b_{i-1}}|b\oplus\delta_i\rangle$ and $e_{i+3}|b\rangle = (-1)^{b_1+\cdots+b_i}|b\oplus\delta_i\rangle$. (ii) The Clifford relations give $\{e_i \pm e_{i+3}, e_j \pm e_{j+3}\} = 2(\delta_{ij} - \delta_{ij}) = 0$ and $\{e_i + e_{i+3}, e_j - e_{j+3}\} = 4\delta_{ij}$; then $\omega_i = (a_i^+ + a_i^-)(a_i^+ - a_i^-) = a_i^-a_i^+ - a_i^+a_i^- = \mathbf{1} - 2a_i^+a_i^-$. (iii)/(iv) By the two formulas of (i), $(e_i - s\,e_{i+3})|b\rangle = (-1)^{b_1+\cdots+b_{i-1}}(1 - s(-1)^{b_i})|b\oplus\delta_i\rangle$, which vanishes iff $s = \sigma_i$. So $N_\sigma$ annihilates the octant $\sigma$; an annihilator is isotropic, since $2\eta(u,v)\phi = (uv + vu)\phi = 0$, hence of dimension at most three, and it equals $N_\sigma$. For $\sigma = (+,+,+)$ this is the statement $a_i^-|000\rangle = 0$. The sign of $a_{i_1}^+\cdots a_{i_r}^+|\mathrm{vac}\rangle$ is a product of factors $(-1)^{b_1+\cdots+b_{i-1}}$ taken while all bits before $i$ are still zero. (v) Reordering $e_1e_4e_2e_5e_3e_6$ into $e_1\cdots e_6$ costs three transpositions (Example 2.6 of [1]). □

All five items are restated from [1] or fixed here (r3-1, r3-2, r3-3). Item (v) confirms the earlier version's statement $I = -\omega_1\omega_2\omega_3$ with eigenvalue $(-1)^w$; note only that its "temporal weight" $w$ counts plus signs, so $w = 3 - N$.

**Proposition 4.2** (The octant group)**.** *For each axis* $a$*, the automorphisms of* $\mathrm{Cl}(3,3)$ *extending the reflection* $R_a : e_a \mapsto -e_a$, $e_b \mapsto e_b$ $(b \neq a)$ *are the conjugations by the multiples of* $g_a := e_a\omega$*. Each* $g_a$ *is a signed permutation matrix; it conjugates* $\omega_i$ *to* $-\omega_i$ *for* $a \in \{i, i+3\}$ *and fixes the other two, so it carries the octant* $\sigma$ *to the octant with* $\sigma_i$ *reversed. The reflections of a temporal axis and of its spatial partner act alike on the octants, and* $g_ig_{i+3} = \pm\omega_i$ *fixes every octant. The lifts* $g_1, g_2, g_3$ *satisfy* $g_i^2 = -\mathbf{1}$*, anticommute pairwise and generate a group of order* 16 *whose centre* $\{\pm\mathbf{1}, \pm g_1g_2g_3\}$ *contains* $g_1g_2g_3 = e_4e_5e_6$*, the element mapping each octant to its antipode; the kernel of its action on the eight octants is* $\{\pm\mathbf{1}\}$*, and it acts through* $\mathbb{Z}_2^3$*, simply transitively.*

*Proof.* $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ is central simple, so every automorphism is inner and its implementing element is unique up to a real factor (Skolem–Noether). Since $\omega$ anticommutes with every generator, $g_ae_bg_a^{-1} = -e_ae_be_a^{-1}$, which is $-e_b$ for $b = a$ and $e_b$ otherwise. Hence $g_a\omega_ig_a^{-1} = R_a(e_i)R_a(e_{i+3})$. Further $g_i^2 = -e_i^2\omega^2 = -\mathbf{1}$, and $g_ig_j = -e_ie_j$ anticommutes with $g_jg_i = -e_je_i$. Moving $g_i$ past the other two lifts costs two signs, so $g_1g_2g_3 = -T\omega = e_4e_5e_6$ commutes with every $g_i$; it is not $\pm\mathbf{1}$, and it reverses all three signs. The images of $|\mathrm{vac}\rangle$ under the eight sign classes are the eight octants (r3-4, r3-5). □

Fixed. The earlier version defined the octant group as the reflections of the temporal $\mathbb{R}^3$ acting on its eight *sign sectors* $\{x \in V^+ : \sigma_ix_i > 0\}$, and identified them with the spinor octants. Both sets are $\mathbb{Z}_2^3$-torsors, so equivariant bijections between them exist, eight of them, one for each choice of the octant assigned to the positive sector. One of them is singled out by an operator of [1]:

**Proposition 4.3** (Clock octant and vacuum)**.** *For a dimensionless* $x \in V^+$ *the conversion operator of Theorem 3.6 of [1] with* $C = \mathrm{diag}(x)$ *is* $B_x = \sum_i x_ie_ie_{i+3} = \sum_i x_i\omega_i$*. It has eigenvalue* $\sum_i \sigma_ix_i$ *on the octant* $\sigma$*; if no* $x_i$ *vanishes, its unique top eigenline is the octant* $\mathrm{sign}(x)$*; and* $g_iB_xg_i^{-1} = B_{R_ix}$*. The Fisher–Rao octant of clocks of Section 3.3 of [1], the closed positive octant* $x_i = 2\sqrt{w_i} \geq 0$ *of the sphere of radius two, therefore corresponds to the Fock vacuum: for* $x$ *in its interior the top eigenline of* $B_x$ *is* $|\mathrm{vac}\rangle$*, and for the symmetric clock* $B_{\widehat{W}} = \mathsf{P}/\sqrt{3}$ *has top eigenvalue* $\sqrt{3}$ *on* $|\mathrm{vac}\rangle$*. The weights* $w_i = x_i^2/4$ *do not see the signs of the amplitudes: the eight octants of the clock sphere are the eight lifts of the clock simplex, which is their quotient by the octant group.*

*Proof.* $B_x$ is diagonal by Proposition 4.1(i), and $\sum_i \sigma_ix_i < \sum_i |x_i|$ unless $\sigma = \mathrm{sign}(x)$. Equivariance follows from Proposition 4.2 (r3-6, with twenty random rational $x$). □

The spectral statements are fixed. The identification of the clock octant with the vacuum pole is constructed: it uses the order in $\omega_i = e_ie_{i+3}$ and the choice of the top rather than the

bottom eigenline, and the opposite choice assigns the filled pole. The two objects are different in kind, a two-dimensional region of the sphere of clock directions in $V^+$ and a line of the spinor module; read physically, the correspondence is interpretive.

### 4.2 The exterior-algebra structure

**Proposition 4.4** (The spinor module as an exterior algebra)**.** *$L := \operatorname{span}\{e_i + e_{i+3}\}$ and $L' := \operatorname{span}\{e_i - e_{i+3}\}$ are maximal isotropic, transverse, and dual to each other under $\eta$. The map $\Lambda^r(L) \to S$, $\ell_1 \wedge \cdots \wedge \ell_r \mapsto 2^{-r}\ell_1 \cdots \ell_r|\mathrm{vac}\rangle$, is an isomorphism $\Lambda^*(L) \cong S$ carrying the degree to $N$, with $\dim \Lambda^w = 1, 3, 3, 1$ for $w = N = 0, 1, 2, 3$: the weights of the octants. Writing $X = X_L + X_{L'}$, a vector acts as exterior multiplication by $2X_L$ plus contraction with the form $\eta(X_{L'}, \cdot)$ on $L$,*

$$X = \sum_i \big[(B_i + E_i)\, a_i^+ + (B_i - E_i)\, a_i^-\big]. \tag{10}$$

*Moreover $a \mapsto a|\mathrm{vac}\rangle$ is an isomorphism from the temporal Clifford algebra $\mathrm{Cl}(V^+) = \langle e_1, e_2, e_3\rangle \cong \mathrm{Cl}(3,0)$ onto $S$, and $e_I|\mathrm{vac}\rangle = a_I^+|\mathrm{vac}\rangle$: the octant weight $N$ is the Clifford grade of the temporal blade $e_I$.*

*Proof.* $\eta(e_i \pm e_{i+3}, e_j \pm e_{j+3}) = \delta_{ij} - \delta_{ij} = 0$, while $\eta(e_i + e_{i+3}, e_j - e_{j+3}) = 2\delta_{ij}$. Elements of $L$ anticommute, so the map is well defined, and it is bijective by Proposition 4.1(iii). Under it $a_i^+$ is $\ell_i\wedge$ with $\ell_i = e_i + e_{i+3} = 2a_i^+$, and $\ell_i' = e_i - e_{i+3} = 2a_i^-$ is the contraction with $\eta(\ell_i', \cdot)$, since $\eta(\ell_i', \ell_j) = 2\delta_{ij}$. (10) is $e_i = a_i^+ + a_i^-$, $e_{i+3} = a_i^+ - a_i^-$. Finally $e_j = a_j^+ + a_j^-$ and $a_j^-$ annihilates $a_I^+|\mathrm{vac}\rangle$ for $j \notin I$, which gives $e_I|\mathrm{vac}\rangle = a_I^+|\mathrm{vac}\rangle$ by induction (r3-7). □

Inherited: this is Chevalley's realisation of the spinors of a space $L \oplus L'$ with $L' \cong L^*$ on $\Lambda^*(L)$ [24, 30], and the pure spinor lines of Proposition 4.1(iv) are the lines of Cartan and Chevalley (see Section 2 of [26]). The last statement is the classical identification of the spinors of $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ restricted to $\mathrm{Cl}(3,0)$ with the Pauli algebra itself.

**Proposition 4.5** (The commutant of the grading root)**.** *The elements of $\mathfrak{so}(3,3)$ commuting with $\mathcal{K}$ form a subalgebra of dimension nine, the adjoint image of the bivectors $E_{ij} := a_i^+ a_j^- - \frac{1}{2}\delta_{ij}\mathbf{1}$ (so $E_{ij} = a_i^+ a_j^-$ for $i \neq j$ and $E_{ii} = -\frac{1}{2}\omega_i$). They commute with $N$; $\operatorname{ad} E_{ij}$ acts on $L$ by the matrix unit $e_{ij}$ and on $L'$ by $-e_{ji}$, so the subalgebra is $\mathfrak{gl}(L) \cong \mathfrak{gl}(3,\mathbb{R})$; and the spin lift of $A \in \mathfrak{gl}(3)$ acts on $\Lambda^w(L)$ as $\Lambda^w(A) - \frac{1}{2}\operatorname{tr} A$. As a $\mathfrak{gl}(3)$-module, $S \cong \Lambda^*(L) \otimes (\det L)^{-1/2}$.*

*Proof.* $\mathcal{K}$ has the eigenspaces $L$ (eigenvalue $-1$) and $L'$ $(+1)$, so an element commuting with it preserves both; $\eta$-skewness then forces $A|_{L'} = -(A|_L)^*$ under $L' \cong L^*$, and every $A|_L \in \mathfrak{gl}(L)$ occurs: dimension nine. Since $\mathcal{K} = \frac{1}{2}\operatorname{ad}_{\mathsf{P}}$ and $\mathsf{P} = 3\,\mathbf{1} - 2N$, an element $\operatorname{ad}\beta$ commutes with $\mathcal{K}$ iff $[N, \beta] = 0$. A bivector is quadratic in the $a^{\pm}$; the number-conserving ones are the $a_i^+ a_j^-$ up to scalars, and $E_{ij}$ removes the scalar part. By the CAR $[a_i^+ a_j^-, a_k^+] = \delta_{jk} a_i^+$ and $[a_i^+ a_j^-, a_k^-] = -\delta_{ik} a_j^-$, and $a_i^+ a_j^-$ acts on $a_K^+|\mathrm{vac}\rangle$ as the derivation replacing $a_j^+$ by $a_i^+$ (r3-8). □

The statement is standard (Example 2.12 of [26] gives $\Lambda^* V^* \otimes (\det V)^{1/2}$ with $V^*$ in the role of $L$); it is fixed here in the realisation. It shows that $\mathfrak{gl}(3)$ does not act on $\Lambda^w$ as the bare exterior power: the density factor $-\frac{1}{2}\operatorname{tr} A$, the same on all weights, is part of the spin representation. The commutant of $\mathcal{K}$ is not stated in [1] or [2].

### 4.3 The field-octant map

The map should intertwine the rotations of the field. Which rotations act on both triples as the field's rotations is fixed first.

**Lemma 4.6** (The diagonal rotations)**.** *For cyclic $(i, j, k)$ the bivector $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_j e_k)$ of [2] acts on $V$ by $\delta\mathbf{B} = e_i \times \mathbf{B}$, $\delta\mathbf{E} = e_i \times \mathbf{E}$ and commutes with $G$ and $\mathcal{K}$. The paired*

*bivector* $e_j e_k + e_{j+3} e_{k+3}$ *rotates the two triples in opposite senses and commutes with neither. The diagonal rotations* $\mathfrak{so}(3)_\Delta = \operatorname{span}\{L_i\}$ *are the compact part of* $\mathfrak{l}_G$ *and the antisymmetric part of* $\mathfrak{gl}(3)$*, and* $\mathfrak{so}(3)_\Delta = \mathfrak{l}_G \cap \mathfrak{gl}(3)$*. In Fock form*

$$L_i = a_k^+ a_j^- - a_j^+ a_k^-, \qquad K_i = \tfrac{1}{2}(e_k e_{j+3} - e_j e_{k+3}) = a_j^- a_k^- - a_j^+ a_k^+.$$

*The spin lift of* $L_i$ *(left multiplication) satisfies* $L_i^3 = -L_i$*, so* $\exp(2\pi L_i) = \mathbf{1}$ *on* $S$*; it annihilates exactly* $\operatorname{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle\}$ *and acts as* $L_i f_m = \epsilon_{imn} f_n$*,* $L_i g_m = \epsilon_{imn} g_n$ *on*

$$f_k := a_k^+ |\mathrm{vac}\rangle, \qquad g_m := \tfrac{1}{2}\epsilon_{mjk}\, a_j^+ a_k^+ |\mathrm{vac}\rangle.$$

*Proof.* $[e_j e_k, e_k] = 2e_j$ and $[e_j e_k, e_j] = -2e_k$, while $[e_{j+3}e_{k+3}, e_{k+3}] = -2e_{j+3}$ because $e_{k+3}^2 = -\mathbf{1}$: the relative minus sign makes the triples turn alike. $G$ and $\mathcal{K}$ map $e_i$ to $\pm e_{i+3}$, so they commute with $R \oplus R'$ iff $R = R'$. The Fock forms follow from $e = a^+ + a^-$, $e' = a^+ - a^-$ and the CAR, and the action on $f, g$ from $[L_i, a_m^+] = \epsilon_{imn} a_n^+$ and $L_i|\mathrm{vac}\rangle = 0$. The intersection is computed in r3-9, which also checks $K_i = GL_i$ on $V$ (Proposition 13.6 of [1]) and the brackets of Theorem 12.3 of [1]. □

Restated and fixed (r3-9). The paired combination $e_j e_k + e_{j+3} e_{k+3}$, a natural first guess, is the wrong one; with the minus sign it is the diagonal rotation of Theorem 12.1 of [1], whose spin lift has integer spin on $S$, as Proposition 13.12 of [1] states.

**Theorem 4.7** (The field-octant map)**.** *(i) The linear maps* $V \to S$ *intertwining* $\mathfrak{so}(3)_\Delta$ *form a four-dimensional space. Those sending* $\mathbf{E}$ *into weight one and* $\mathbf{B}$ *into weight two form the two-parameter family*

$$\Phi_{\alpha,\beta}(X) = \alpha \sum_k E_k f_k + \beta \sum_m B_m g_m,$$

*and* $\Phi := \Phi_{1,1}$ *is an isometry of* $\mathbb{R}^6$ *onto the six field octants.*

*(ii) Under* $a_I^+|\mathrm{vac}\rangle \leftrightarrow dx^I$*,* $\Phi(X) = E_k\, dx^k + \frac{1}{2}\epsilon_{mjk} B_m\, dx^j \wedge dx^k$*:* $\mathbf{E}$ *is a one-form and* $\mathbf{B}$ *a two-form on* $\mathbb{R}^3$*.*

*(iii) The spin lift of* $\mathfrak{gl}(3)$ *preserves the image of* $\Phi$ *and induces* $\mathbf{E} \mapsto (A - \frac{1}{2}\operatorname{tr} A)\mathbf{E}$ *and* $\mathbf{B} \mapsto (\frac{1}{2}\operatorname{tr} A - A^\top)\mathbf{B}$*:* $\mathbf{E}$ *transforms in* $L \otimes (\det L)^{-1/2}$ *and* $\mathbf{B}$ *in* $L^* \otimes (\det L)^{1/2}$*, contragredient up to the determinant.*

*(iv) The same* $\mathfrak{gl}(3) \subset \mathfrak{so}(3,3)$ *acts on the field point through the light-cone components,* $\mathbf{B}+\mathbf{E} \mapsto A(\mathbf{B}+\mathbf{E})$*,* $\mathbf{B}-\mathbf{E} \mapsto -A^\top(\mathbf{B}-\mathbf{E})$*. For symmetric* $A$ *this is* $\mathbf{B} \mapsto A\mathbf{E}$*,* $\mathbf{E} \mapsto A\mathbf{B}$*, a set of boosts, and* $\Phi$ *intertwines the vector action of* $\mathfrak{gl}(3)$ *with its spin lift only on* $\mathfrak{so}(3)_\Delta$*.*

*Proof.* (i) As $\mathfrak{so}(3)_\Delta$-modules $V = 2 \cdot \mathbb{R}^3$ and, by Lemma 4.6, $S = 2 \cdot \mathbb{R}^3 \oplus 2 \cdot \mathbb{R}$ (the $f$'s, the $g$'s and the poles). The vector representation is absolutely irreducible, so by Schur's lemma [11] $\operatorname{Hom}(\mathbb{R}^3, \mathbb{R}^3) = \mathbb{R}$ and $\operatorname{Hom}(\mathbb{R}^3, \mathbb{R}) = 0$; the weight condition leaves one scale on each copy. (ii) $a_k^+$ acts as $dx^k\wedge$, and $g_1 = a_2^+ a_3^+|\mathrm{vac}\rangle \leftrightarrow dx^2 \wedge dx^3$ cyclically. (iii) follows from Proposition 4.5: on $\Lambda^1$ the action is $A - \frac{1}{2}\operatorname{tr} A$, and on $\Lambda^2(L) \cong L^* \otimes \det L$, in the basis $g_m$, it is $\operatorname{tr} A - A^\top$, from which $\frac{1}{2}\operatorname{tr} A$ is subtracted. (iv) is Proposition 4.5 on $V$ with (10) (r3-10, r3-11). □

Items (i), (iii), (iv) are fixed; $\Phi$ itself is constructed. Item (ii) is the premetric statement that the electric field strength is a one-form and the magnetic field strength a two-form on space, $F = \mathbf{E} \wedge d\sigma + \mathbf{B}$ in the notation of Hehl and Obukhov [27] (inherited). Item (iv) makes precise the statement that under $\mathfrak{gl}(3, \mathbb{R})$ $\mathbf{E}$ is a vector and $\mathbf{B}$ a two-form: it is true of the spin lift on the octants and false of the vector action on the field point, where the symmetric part of $\mathfrak{gl}(3)$ mixes $\mathbf{E}$ with $\mathbf{B}$. The two scales of (i) are tied together by Maxwell's equations in Proposition 4.17.

**Proposition 4.8** (Parity on the octants)**.** *(i)* $\mathrm{Ad}_T|_V = \eta$ *is the frame parity, fixing* $\mathbf{B}$ *and reversing* $\mathbf{E}$ *(Proposition 2.11 of [2]), and* $T$ *is, up to a factor, the only element of* $\mathrm{Cl}(3,3)$ *implementing it.* $T$ *carries each octant to its antipode,*

$$T|\mathrm{vac}\rangle = |\mathrm{full}\rangle, \quad Tf_k = g_k, \quad Tg_k = -f_k, \quad T|\mathrm{full}\rangle = -|\mathrm{vac}\rangle,$$

*and* $T\Phi = -\Phi \circ G$*: on the field octants the spinor lift of parity acts as minus the Hodge star, exchanging electric and magnetic octants.*

*(ii)* $(-1)^N\Phi = \Phi\circ\eta$*: the weight parity, which is the chirality* $\gamma_5 = -\Gamma$*, acts as* $-1$ *on the image of* $\mathbf{E}$ *and* $+1$ *on the image of* $\mathbf{B}$*; but* $\mathrm{Ad}_{(-1)^N} = -\mathbf{1}$ *on* $V$*.*

*(iii) The Dirac parity* $P = \gamma^0 = e_1$ *of Theorem 5.3 of [1] acts on the frame field by* $(\mathbf{B};\mathbf{E}) \mapsto (B_1, -B_2, -B_3; -\mathbf{E})$*, and* $e_1 f_1 = |\mathrm{vac}\rangle$*: it leaves the field octants.*

*Proof.* (i) $T$ commutes with $e_1, e_2, e_3$ and anticommutes with $e_4, e_5, e_6$; uniqueness is Skolem–Noether. $T = \prod_i (a_i^+ + a_i^-)$ flips all three signs, and the displayed values are r3-12. With them $T\Phi(X) = \sum_k (E_k g_k - B_k f_k)$, while $GX$ has $\mathbf{B}' = -\mathbf{E}$, $\mathbf{E}' = \mathbf{B}$. (ii) $(-1)^N$ is $-1$ on $f$ and $+1$ on $g$; $\Gamma$ anticommutes with every vector. (iii) $e_1$ commutes with itself and anticommutes with the other generators, and it flips the first sign (r3-12). □

Fixed. The polar and axial characters of $\mathbf{E}$ and $\mathbf{B}$ are present on the octants, but they are carried by the weight parity $(-1)^N = -\Gamma$, whose Clifford action on $V$ is the total reflection $X \mapsto -X$, and not by the spinor lift $T$ of the parity, which acts on the field octants as the duality rotation by $-\pi/2$. This is possible because $\Phi$ intertwines the rotations only, not the Clifford action of $\mathrm{Pin}(3,3)$. The statement that parity acts as $-1$ on the electric and $+1$ on the magnetic octants is therefore false if parity means the spinor lift of the spatial inversion, and true if it means the weight parity.

**Remark 4.9** (The earlier field-octant bijection)**.** The earlier version called the octants with one plus sign electric and those with two plus signs magnetic, the octant with its single plus sign in pair $k$ carrying $E_k$ and the octant with its single minus sign in pair $m$ carrying $B_m$. Its electric octants are therefore the lines of the $g_k$, of weight $N = 2$, and its magnetic ones those of the $f_m$, of weight $N = 1$: up to signs, which it does not fix, its labelling is that of $\Phi \circ G$ (Proposition 4.17), not of $\Phi$. In both labellings $E_k$ and $B_k$ sit on antipodal octants, which the mass form pairs (Proposition 4.14). The earlier bijection mapped octants to a basis of interior bivectors, a bijection of labels and not a linear map; it is admissible as a labelling and equivalent to $\Phi$ by a duality rotation of a quarter turn, and nothing in the octant structure prefers it. Its derivation of $E = cB$ from a count of "temporal kernels" is not reproduced: in the present construction the ratio of the two scales of $\Phi_{\alpha,\beta}$ is $c$ in SI units, the conversion constant of Proposition 2.7 of [2], fixed by Faraday's law (Proposition 4.17 and Theorem 4.16(iii)).

## 4.4 Clifford multiplication does not give the pair reading

The field point is also a Clifford vector, and one may ask what it does to the octants by left multiplication. The answer shows why the pair reading $X_{\mathrm{EM}} = (\mathbf{B};\mathbf{E})$ needs the map $\Phi$ and is not obtained by multiplying.

**Proposition 4.10** (Light-cone amplitudes)**.** *For every* $X = \sum_i (B_i e_i + E_i e_{i+3})$

$$X|\mathrm{vac}\rangle = \sum_i (B_i + E_i)\, f_i, \qquad X|\mathrm{full}\rangle = \sum_i (B_i - E_i)\, g_i = \sum_i (-1)^{i-1}(B_i - E_i)\, |111 \oplus \delta_i\rangle, \quad (11)$$

*and* $X^2 = \eta(X,X)\mathbf{1} = \sum_i (B_i + E_i)(B_i - E_i)\,\mathbf{1}$*. The amplitudes* $B_i \pm E_i$ *are the coordinates of* $X$ *on* $L$ *and* $L'$*, the eigen-coordinates of* $\mathcal{K}$*, with* $\mathbf{B} + \mathbf{E} = \sqrt{2}\,\mathrm{Re}(e^{-i\pi/4}z)$ *and* $\mathbf{B} - \mathbf{E} = -\sqrt{2}\,\mathrm{Im}(e^{-i\pi/4}z)$*. Further*

$$X\big(|\mathrm{vac}\rangle + |\mathrm{full}\rangle\big) = \sqrt{2}\,\Phi\big(e^{\pi G/4}X\big). \quad (12)$$

*No spinor $\chi$ satisfies $X\chi = \Phi(X)$ for all $X$; the Clifford multiplications $X \mapsto X\chi$ that intertwine the rotations are those with $\chi \in \mathrm{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle\}$, and none of them places $\mathbf{E}$ and $\mathbf{B}$ on separate weights.*

*Proof.* (11) follows from (10) with $a_i^-|\mathrm{vac}\rangle = 0$, $a_i^+|\mathrm{full}\rangle = 0$ and $a_i^-|\mathrm{full}\rangle = g_i$; $X^2$ follows from the CAR. The duality rotation $e^{\pi G/4}$ gives $\mathbf{E}' = (\mathbf{E}+\mathbf{B})/\sqrt{2}$, $\mathbf{B}' = (\mathbf{B}-\mathbf{E})/\sqrt{2}$, which is (12). If $X\chi = \Phi(X)$, then $e_1\chi = g_1$ and $e_4\chi = f_1$, i.e. $\chi = e_1|011\rangle = |111\rangle$ and $\chi = -e_4|100\rangle = |000\rangle$, a contradiction. $L_i(X\chi) = [L_i, X]\chi + XL_i\chi$, so equivariance requires $L_i\chi = 0$, and then $X(a|\mathrm{vac}\rangle + b|\mathrm{full}\rangle)$ puts $a(\mathbf{B}+\mathbf{E})$ on weight one (r3-13). □

Fixed; the reading is interpretive. Read as a Clifford vector acting on the vacuum pole, the field point does not keep the two members of the pair apart: it places on the $i$-th electric octant the sum $E_i + B_i$, which in textbook variables is $E_i - B_{\mathrm{t}i}$, and on the filled pole it places $B_i - E_i = -(E_i + B_{\mathrm{t}i})$ on the $i$-th magnetic octant. These sums are the coordinates of $X$ on the null planes $L$ and $L'$, the light-cone components of the para-complex structure; they mix the electric and the magnetic member and are not the field strength. A literal reading "$X = \mathbf{E}+\mathbf{B}$" would therefore be wrong: the pair $(\mathbf{B};\mathbf{E})$ is placed on the octants by $\Phi$, which is not Clifford multiplication by any spinor, and the Clifford action differs from $\Phi$ by a duality rotation of an eighth of a turn, (12). (An eighth of a turn also separates the Klein correspondence from the frame reading in Theorem 13.13 of [1]; the maps are different and we draw no conclusion from the coincidence.)

### 4.5 Boosts and duality cannot act by even elements

**Theorem 4.11** (The even obstruction)**.** *Every even element of* $\mathrm{Cl}(3,3)$ *commutes with* $(-1)^N$ *and maps an octant of weight $w$ into the weights $w$ and $w \pm 2$; this includes* $\mathfrak{so}(3,3)$, $\mathrm{Spin}(3,3)$ *and the spin lifts of* $\mathfrak{l}_G$ *and* $\mathfrak{l}_{\mathrm{int}}$*. No even $Q$ satisfies $Q\Phi = \Phi K_k$ for a boost $K_k$ of* $\mathfrak{l}_G$*, or $Q\Phi = \Phi G$. At the level of groups, no element of* $\mathrm{Cl}(3,3)$ *of definite parity intertwines $e^{sK_k}$ $(s \neq 0)$ with $\Phi$, nor $e^{\vartheta G}$ unless $\vartheta \in \frac{\pi}{2}\mathbb{Z}$.*

*Proof.* $(-1)^N = -\omega$ commutes with even and anticommutes with odd elements, and weights of equal parity differ by 0 or 2. By Example 12.4 of [1], $K_3 e_1 = e_5$; so $\Phi(e_1) = g_1$ has weight two and $\Phi(K_3 e_1) = f_2$ weight one, and an even $Q$ cannot map the first to the second. Likewise $\Phi(e_1) = g_1$ and $\Phi(Ge_1) = \Phi(e_4) = f_1$. For the groups, $e^{sK_3}e_1 = \cosh s\, e_1 + \sinh s\, e_5$, so $\Phi(e^{sK_3}e_1)$ has components of both parities when $s \neq 0$, while an element of definite parity maps $g_1$ to a vector of definite parity; the same holds for $e^{\vartheta G}e_1 = \cos\vartheta\, e_1 + \sin\vartheta\, e_4$ unless $\cos\vartheta \sin\vartheta = 0$ (r3-14, r3-15). □

Obstructed, under the hypothesis that the operator has definite parity; the hypothesis covers every element of the spin group and of its Lie algebra. The intertwiners that do exist are odd:

**Proposition 4.12** (Odd intertwiners and the octant representation)**.** *(i) $-T\Phi = \Phi G$ and $Q_k\Phi = \Phi K_k$ for*

$$Q_k := -TL_k = -\tfrac{1}{2}e_k(\mathbf{1} - \omega_j\omega_l) = -\tfrac{1}{2}(e_k + e_{k+3}\omega), \qquad \{k,j,l\} = \{1,2,3\},$$

*an odd element with parts of grades one and five. The solutions of $Q\Phi = \Phi K_k$ form an affine space of dimension* 16*, of dimension* 8 *among odd elements, and contain no Clifford vector; no Clifford vector implements $G$ on the octants.*

*(ii) On $S = \mathrm{Cl}(V^+)|\mathrm{vac}\rangle$ (Proposition 4.4), $Q_k(a|\mathrm{vac}\rangle) = -\frac{1}{2}[e_k, a]|\mathrm{vac}\rangle$ and $-T$ is left multiplication by the central element $-e_1e_2e_3$ of $\mathrm{Cl}(V^+)$.*

*(iii) $\rho_{\mathrm{oct}}(L_k) = L_k$, $\rho_{\mathrm{oct}}(K_k) = Q_k$ is a representation of $\mathfrak{l}_G$ on $S$. Its Casimir $\sum_k \rho_{\mathrm{oct}}(K_k)^2 - \rho_{\mathrm{oct}}(L_k)^2$ is* 4 *on the field octants, the field-strength value, and* 0 *on the poles, and its intertwiners $V \to S$ are* $\mathrm{span}\{\Phi, \Phi G\}$.

*(iv) The spin lift of $\mathfrak{l}_G$ has Casimir 3 on all of $S$ and admits no nonzero intertwiner $V \to S$. Its boosts carry $f_k$ to $-|\text{full}\rangle$ and $|\text{vac}\rangle$ to $-g_k$: under the genuine Lorentz action the electric octants are the spatial components of the four-vector $S^+$ whose time component is the filled pole, and the magnetic octants those of the four-vector $S^-$ whose time component is the vacuum.*

*Proof.* (i) $T$ commutes with the same-sign bivectors, hence with $L_k$, so $-TL_k\Phi = -T\Phi L_k = \Phi G L_k = \Phi K_k$ by Proposition 4.8(i) and $K_k = GL_k$. The closed forms use $T(e_j e_l) = -e_k$ for cyclic $(k,j,l)$ and $e_{k+3}\omega = -e_k\omega_j\omega_l$. The values of a solution on the two poles are free, eight components each, four of them of the parity opposite to the pole (r3-15). (ii) $e_{k+3}\omega$ commutes with $e_1, e_2, e_3$ and sends $|\text{vac}\rangle$ to $-e_k|\text{vac}\rangle$, so it acts on $a|\text{vac}\rangle$ as $-a\,e_k$. (iii) $T$ commutes with $\rho_{\text{oct}}(L)$ and $T^2 = -\mathbf{1}$, so the brackets of $\mathfrak{l}_G$ follow from those of $\mathfrak{so}(3)$; $\rho_{\text{oct}}(K)^2 = -L^2$ gives the Casimir, and an intertwiner is a $T$-linear map $\mathbb{C}^3 \to \mathbb{C}^3$ commuting with $\mathfrak{so}(3)$, hence a complex multiple of $\Phi$ (r3-16). (iv) The Casimir is Proposition 13.12 of [1]; $S = 2(\frac{1}{2},\frac{1}{2})$ contains no $(1,0)\oplus(0,1)$; the boost formulas are $K_k = a_j^- a_l^- - a_j^+ a_l^+$ on $f_k$ and $|\text{vac}\rangle$ (r3-16). □

Constructed: the operators $Q_k$ and the representation $\rho_{\text{oct}}$; fixed: the solution counts of (i) and the Casimir values and the intertwiner space of (iii); restated and fixed: (iv). Item (ii) identifies the construction: in textbook variables (Theorem 4.16) the field octants carry Hestenes' Pauli-algebra field $F = \mathbf{E} + I\mathbf{B}_{\text{t}}$ [28], written with $\sigma_k = e_k$ and $I = T$, and the boosts and the duality rotation act as they do in the Pauli algebra, through the commutator with $\sigma_k$ and through multiplication by $I$. The commutator is a left multiplication combined with a right multiplication, and on $S$ the right multiplication is the odd element $e_{k+3}\omega$: this is the reason the boosts are odd. Which symmetries of the field the octant picture carries is now settled.

**Corollary 4.13** (What the octants carry)**.** *By even elements of* $\mathrm{Cl}(3,3)$ *the field octants carry the rotations* $\mathfrak{so}(3)_\Delta$ *and the premetric* $\mathfrak{gl}(3)$ *of Theorem 4.7(iii), through their spin lifts, and the parity, through* $(-1)^N = -\omega$*. They do not carry the boosts or the duality rotations, which act only through the odd operators* $Q_k$ *and* $-T$ *of the representation* $\rho_{\text{oct}}$*, the Pauli algebra of the temporal triple. The spin-lifted Lorentz algebra* $\mathfrak{l}_G$ *makes the field octants parts of two four-vectors, not of a field strength.*

The working note [3] (Section 7) states that there is no nonzero linear Lorentz-equivariant map from the field-strength module to the four-vectors. Item (iv) confirms the same fact for the spinor module, whose modules under $\mathfrak{l}_G$ are four-vectors (for $E_n$ itself the statement is Proposition 3.16); the note's statement that $G$ can represent the Hodge star on the field (its eq. (3) in Section 2 and its verdict table) is confirmed on the octants with a correction: the operator representing it there is the odd element $-T$, the note's $I_T = e_1e_2e_3$ of its Section 5, not an element of $\mathrm{Spin}(3,3)$.

### 4.6 Energies on the octants

**Proposition 4.14** (Energies and the Chevalley–Mukai pairing)**.** *Let* $\psi = \Phi(X) = \psi_{\mathbf{E}} + \psi_{\mathbf{B}}$ *with* $\psi_{\mathbf{E}} = \sum_k E_k f_k$*,* $\psi_{\mathbf{B}} = \sum_m B_m g_m$*.*

*(i)* $|\psi|^2 = |\mathbf{E}|^2 + |\mathbf{B}|^2 = 2E_6$ *and* $\psi^\top(-1)^N\psi = |\mathbf{B}|^2 - |\mathbf{E}|^2 = \mathrm{Re}\, q$*.*

*(ii)* $\psi_{\mathbf{E}}^\top T\psi_{\mathbf{B}} = -\mathbf{E}\cdot\mathbf{B} = -\frac{1}{2}\,\mathrm{Im}\, q$*, while* $\psi^\top T\psi = 0$ *and* $\psi^\top T\Gamma\psi = 2\,\mathbf{E}\cdot\mathbf{B} = \mathrm{Im}\, q$*. Hence*

$$E_0 = \tfrac{1}{2}|q| = \tfrac{1}{2}\Big[\big(\psi^\top(-1)^N\psi\big)^2 + 4\big(\psi_{\mathbf{E}}^\top T\psi_{\mathbf{B}}\big)^2\Big]^{1/2}.$$

*(iii) The* $\mathrm{Spin}(3,3)$*-invariant bilinear forms on* $S$ *are* $\mathrm{span}\{T, T\Gamma\}$*,* $T$ *antisymmetric and* $T\Gamma$ *symmetric. Under* $\Lambda^*(L) \cong S$ *the Chevalley–Mukai pairing* $(\varphi,\chi) = [\tau(\varphi)\wedge\chi]_{\text{top}}$*, with* $\tau$ *the reversal, equals* $-\varphi^\top T\chi$*. Two octants pair to a nonzero value iff their null planes* $N_\sigma, N_\tau$ *are transverse, iff they are antipodal.*

*(iv)* $(-1)^N$ *is invariant under the six compact bivectors and under* $\rho_{\mathrm{oct}}$*, and under none of the nine boosts. The Euclidean norm is invariant under the compact bivectors and the* $\rho_{\mathrm{oct}}(L_k)$*, but not under the boosts* $\rho_{\mathrm{oct}}(K_k) = Q_k$*, which are symmetric:* $|\psi|^2 = 2E_6$ *changes under boosts,* $|\Phi(e^{sK_3}e_1)|^2 = \cosh 2s$*.*

*(v) In SI units, with* $\psi = \Phi(c\mathbf{B}; \mathbf{E})$ $[\mathrm{V\,m^{-1}}]$*,* $u = \frac{\varepsilon_0}{2}|\psi|^2$ *is the energy density and* $\psi^\top(-1)^N\psi = \operatorname{Re} q_c$*,* $q_c = z_c \cdot z_c$*,* $z_c = c\mathbf{B} + i\mathbf{E}$ *(Proposition 2.8 of [2]).*

*Proof.* The $f_k, g_m$ are orthonormal, $(-1)^N = \mp 1$ on them, $Tg_m = -f_m$ and $\Gamma = -(-1)^N$, which gives (i), (ii) and the formula for $E_0$ by $|q|^2 = (\operatorname{Re} q)^2 + (\operatorname{Im} q)^2$. (iii) A form $M$ is invariant iff $\beta^\top M + M\beta = 0$ for the fifteen bivectors, a linear system with a two-dimensional solution space (r3-18, the count of Theorem 13.13 of [1]); $T$ commutes with the antisymmetric (same-sign) and anticommutes with the symmetric (mixed) bivectors (Proposition 2.11 of [2]), and $\Gamma$, which anticommutes with every vector, commutes with all of them. The Mukai identity is checked on all 64 pairs of basis forms, and $\dim(N_\sigma \cap N_\tau)$ is the number of agreeing signs. (iv) $e_k$ and $e_{k+3}\omega$ are symmetric matrices, so $Q_k^\top + Q_k = 2Q_k \neq 0$; (iv), (v): r3-17, r3-18. □

Items (i), (ii), (iv) are fixed, (iii) is inherited in substance and verified here, (v) restates [2]. By (i), the Maxwell Lagrangian $L_M = -\frac{1}{2}\operatorname{Re} q$ of Proposition 2.9 is $-\frac{1}{2}\psi^\top(-1)^N\psi = \frac{1}{2}(|\psi_{\mathbf{E}}|^2 - |\psi_{\mathbf{B}}|^2)$ on the octants, half the squared electric minus the squared magnetic octant amplitudes. The Chevalley form on the spinors of $L \oplus L'$, antisymmetric for $\dim L = 3$ since $(-1)^{3\cdot 2/2} = -1$, and its non-vanishing on pure spinors exactly for transverse null planes are Chevalley's [24]; Gualtieri identifies the form with the Mukai pairing (Section 2.4 of [26], whose Proposition 2.21 restates the transversality criterion). So the $\mathrm{Spin}(3,3)$-invariant mass form $T$ is, on the octants, the Chevalley–Mukai pairing of each octant with its antipode: of the vacuum with the filled pole and of $f_k$ with $g_k$, that is, of $E_k$ with $B_k$, with the sign fixed by (ii): $\psi_{\mathbf{E}}^\top T\psi_{\mathbf{B}} = -\mathbf{E}\cdot\mathbf{B} = -\frac{1}{2}\operatorname{Im} q$. The Lorentz invariant $\operatorname{Re} q$ appears on the octants through the form $(-1)^N$, which is not $\mathrm{Spin}(3,3)$-invariant, while the invariant symmetric form $T\Gamma$ gives $\operatorname{Im} q$; both facts are consistent with Theorem 4.11 and with the representation $\rho_{\mathrm{oct}}$, which preserves $(-1)^N$.

### 4.7 Maxwell's equations on the octants

**Proposition 4.15** (Exterior derivative and codifferential)**.** *On* $S$*-valued fields put* $d := \sum_i a_i^+\partial_i$ *and* $\delta := -\sum_i a_i^-\partial_i$*. Then* $d^2 = \delta^2 = 0$*,*

$$d - \delta = \sum_i e_i\partial_i =: D, \qquad d + \delta = \sum_i e_{i+3}\partial_i =: D', \qquad D^2 = \Delta, \qquad D'^2 = -\Delta,$$

*and under* $f_k \leftrightarrow dx^k$*,* $g_m \leftrightarrow \star dx^m$*,* $|\mathrm{vac}\rangle \leftrightarrow 1$*,* $|\mathrm{full}\rangle \leftrightarrow dx^1 \wedge dx^2 \wedge dx^3$*,* $d$ *and* $\delta$ *are the exterior derivative and the codifferential of the Euclidean* $\mathbb{R}^3$*:* $d(\mathbf{E}\cdot f) = (\nabla\times\mathbf{E})\cdot g$*,* $d(\mathbf{B}\cdot g) = (\nabla\cdot\mathbf{B})|\mathrm{full}\rangle$*,* $\delta(\mathbf{E}\cdot f) = -(\nabla\cdot\mathbf{E})|\mathrm{vac}\rangle$*,* $\delta(\mathbf{B}\cdot g) = (\nabla\times\mathbf{B})\cdot f$*.*

*Proof.* $d^2 = \sum_{ij} a_i^+a_j^+\partial_i\partial_j = 0$ because $a_i^+a_j^+$ is antisymmetric, and likewise for $\delta$; $a_i^+ \pm a_i^-$ are $e_i$ and $e_{i+3}$, and $D^2 = \sum_i e_i^2\partial_i^2$, $D'^2 = \sum_i e_{i+3}^2\partial_i^2$. The four formulas use $a_i^- g_m = \epsilon_{mik} f_k$ and $a_m^+ g_m = |\mathrm{full}\rangle$ (r3-19). □

Inherited, and verified here: $D = d - \delta$ is Kähler's operator on inhomogeneous forms [29], the forms read as spinors [25] and the Dirac–Kähler equation of Becher and Joos [23], here on space.

**Theorem 4.16** (Maxwell's equations as one equation on the octants)**.** *For every smooth field* $X(t, \mathbf{x})$*,*

$$(\partial_t - D)\,\Phi(X) = -(\nabla\cdot\mathbf{E})\,|\mathrm{vac}\rangle + (\partial_t\mathbf{E} + \nabla\times\mathbf{B})\cdot f + (\partial_t\mathbf{B} - \nabla\times\mathbf{E})\cdot g - (\nabla\cdot\mathbf{B})\,|\mathrm{full}\rangle. \quad (13)$$

*Consequently:*

*(i) Maxwell's equations with sources in the frame variables, $\partial_t z = -i\nabla\times z - i\mathbf{j}$ and $\nabla\cdot z = i\rho$ (eq. (16) of [1]), are equivalent to*

$$(\partial_t - D)\,\Phi(X) = -(\rho\,|\mathrm{vac}\rangle + \mathbf{j}\cdot f)\,:$$

*the vacuum component is Gauss's law $\nabla\cdot\mathbf{E} = \rho$, the filled component the absence of magnetic charge, the electric octants carry the Ampère–Maxwell law $\partial_t\mathbf{E} = -\nabla\times\mathbf{B} - \mathbf{j}$ and the magnetic octants Faraday's law $\partial_t\mathbf{B} = \nabla\times\mathbf{E}$.*

*(ii) $(\partial_t + D)(\partial_t - D) = \partial_t^2 - \Delta$ on all eight components, while $(\partial_t + D')(\partial_t - D') = \partial_t^2 + \Delta$. In $D = \sum_i e_i\partial_i$ the derivatives $\partial_i$ are taken along the three physical spatial coordinates $x^i$, the lengths of the classical reading, which sit on $V^-$; the Clifford generators multiplying them are the temporal $e_1, e_2, e_3$, of square $+\mathbf{1}$, the generators that carry $\mathbf{B}$ in the field reading, and not the spatial ones.*

*(iii) In SI units $(\frac{1}{c}\partial_t - D)\Phi(c\mathbf{B};\mathbf{E}) = -(c\rho\,|\mathrm{vac}\rangle + \mathbf{j}\cdot f)/(\varepsilon_0 c)$.*

*(iv) In textbook variables $\Phi_{\mathrm{t}}(X) := \mathbf{E}\cdot f + \mathbf{B}_{\mathrm{t}}\cdot g = -(-1)^N\Phi(X)$ satisfies $T\Phi_{\mathrm{t}} = \Phi_{\mathrm{t}}\circ G$ and*

$$(\partial_t + D)\,\Phi_{\mathrm{t}}(X) = \rho\,|\mathrm{vac}\rangle - \mathbf{j}\cdot f, \qquad \Phi_{\mathrm{t}}(X) = \Big(\sum_k E_k e_k + \sum_k B_{\mathrm{t}k}\,Te_k\Big)|\mathrm{vac}\rangle,$$

*which, under $S\cong\mathrm{Cl}(V^+)$ with $\sigma_k = e_k$ and $I = T$, is the equation $(\partial_t+\nabla)(\mathbf{E}+I\mathbf{B}_{\mathrm{t}}) = \rho-\mathbf{j}$ of the Pauli algebra.*

*(v) With pole components $\psi = \Phi(X) + \phi_0|\mathrm{vac}\rangle + \phi_3|\mathrm{full}\rangle$, $(\partial_t - D)\psi = 0$ reads $\partial_t\phi_0 = \nabla\cdot\mathbf{E}$, $\partial_t\phi_3 = \nabla\cdot\mathbf{B}$, $\partial_t\mathbf{E} = -\nabla\times\mathbf{B} + \nabla\phi_0$, $\partial_t\mathbf{B} = \nabla\times\mathbf{E} + \nabla\phi_3$; every component of a solution obeys the wave equation, so the Gauss constraints hold for all times if $\phi_0, \phi_3$ and the two divergences vanish initially.*

*Proof.* (13) is $\partial_t\Phi(X) - (d-\delta)\Phi(X)$ with the four formulas of Proposition 4.15; it is checked on general symbolic fields $E_k(t,\mathbf{x})$, $B_k(t,\mathbf{x})$ (r3-20), and the real and imaginary parts of the residuals of eq. (16) of [1] are exactly the components of $(\partial_t - D)\Phi(X) + \rho|\mathrm{vac}\rangle + \mathbf{j}\cdot f$. (ii) $(\partial_t + D)(\partial_t - D) = \partial_t^2 - D^2$ (r3-21). (iii) is (i) with $\Phi(\mathbf{E}, c\mathbf{B})$ and $\partial_t\mathbf{E} = -c^2\nabla\times\mathbf{B} - \mathbf{j}/\varepsilon_0$, $\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$ (r3-24, on first-order jets). (iv) $\Phi_{\mathrm{t}}(X) = \mathbf{E}\cdot f - \mathbf{B}\cdot g$; $(-1)^N$ anticommutes with $D$; $Te_k = e_j e_l$ cyclically gives the second form; under $a|\mathrm{vac}\rangle \leftrightarrow a$ left multiplication by $e_k$ is that by $\sigma_k$ (r3-24). (v) is (13) together with $D(\phi_0|\mathrm{vac}\rangle) = \nabla\phi_0\cdot f$ and $D(\phi_3|\mathrm{full}\rangle) = \nabla\phi_3\cdot g$ (r3-21); the wave equation follows from (ii), and $\phi_0$ with $\phi_0 = \partial_t\phi_0 = 0$ at $t = 0$ vanishes identically. □

Fixed in the realisation and inherited in substance. Item (ii) is a statement about the representation and not about spacetime: the octant form of Maxwell's equations uses the Pauli triple $e_1, e_2, e_3$ because it is the triple of square $+\mathbf{1}$ that makes the operator hyperbolic, and this choice is constructed. It does not show that spatial differentiation follows from three temporal directions, and the spatial coordinates $x^i$ remain the lengths of $V^-$; the Dirac operator of the particle reading couples the same derivatives to the spatial generators through $\gamma^0\gamma^a$ instead (Section 5). Item (iv) is the Pauli-algebra form, obtained by multiplying with $\gamma_0$, of the spacetime-algebra equation $\nabla F = J$ of Hestenes [28], which Theorem 13.3 of [1] realises on the interior four-plane; here it is realised on the temporal triple acting on the vacuum. The element $T$ that plays the pseudoscalar $I$ is the $I_T$ of Section 5 of [3], and the properties stated there, $I_T^2 = -\mathbf{1}$ and $[I_T, e_i] = 0$ for the temporal $e_i$, are confirmed (r3-24). The circularly polarised wave of Example 12.11 of [1] solves $(\partial_t - D)\Phi(X) = 0$, and the same wave with $\mathbf{B}$ reversed does not (r3-23). The textbook signs, $\partial_t\mathbf{E} = \nabla\times\mathbf{B}_{\mathrm{t}} - \mathbf{j}$ and $\partial_t\mathbf{B}_{\mathrm{t}} = -\nabla\times\mathbf{E}$, become those of (i) in the frame variable $\mathbf{B} = -\mathbf{B}_{\mathrm{t}}$.

**Proposition 4.17** (Maxwell fixes the scales and leaves a duality angle)**.** *Let* $\Phi_M(X) = (a\mathbf{E} + b\mathbf{B}) \cdot f + (c\mathbf{E} + d\mathbf{B}) \cdot g$ *be the general rotation-equivariant map into the field octants. Then* $(\partial_t - D)\Phi_M(X)$ *vanishes for every source-free Maxwell field iff* $c = -b$ *and* $d = a$*, that is iff* $\Phi_M = r\,\Phi \circ e^{\vartheta G}$ *for some* $r \geq 0$ *and* $\vartheta$*. The map puts* $\mathbf{E}$ *and* $\mathbf{B}$ *on single weights iff* $\vartheta \in \frac{\pi}{2}\mathbb{Z}$*:* $\Phi$ *for* $\vartheta = 0$ *and* $\Phi \circ G$*, with* $\mathbf{E}$ *on weight two, for* $\vartheta = \frac{\pi}{2}$*.*

*Proof.* Substituting $\partial_t \mathbf{E} = -\nabla \times \mathbf{B}$, $\partial_t \mathbf{B} = \nabla \times \mathbf{E}$ and the vanishing divergences, the residual has $f$-component $(d-a)\nabla \times \mathbf{B} + (b+c)\nabla \times \mathbf{E}$, $g$-component $(d-a)\nabla \times \mathbf{E} - (b+c)\nabla \times \mathbf{B}$ and no pole component; it vanishes on all first-order Maxwell jets iff $d = a$, $c = -b$ (r3-22). Conversely $\Phi \circ e^{\vartheta G}$ maps solutions to solutions, since $e^{\vartheta G}$, the duality rotation $z \mapsto e^{i\vartheta} z$, preserves the source-free system. □

Fixed. The two scales of Theorem 4.7(i) are tied by Faraday's and Ampère's laws, $\beta = \alpha$ in Heaviside–Lorentz units and $\beta = c\,\alpha$ in SI, and the energy normalisation of Proposition 4.14(i) fixes $\alpha$ up to sign. What remains is the duality angle. With electric sources, $\Phi$ places the charge density on the vacuum pole and the current on the electric octants, and $\Phi \circ G$ places them on the filled pole and on the weight-two octants; the convention that all charges are electric (Section 13 of [1], [14]) fixes which octants carry the current once the labelling is chosen, not the labelling.

**Remark 4.18** (Octant pooling)**.** Each $e_i$ maps the octant $\sigma$ to the octant with $\sigma_i$ reversed, the image of $\sigma$ under the octant generator $R_i$, and $e_{i+3} = e_i\omega_i$ acts on the same pair of octants (r3-25). Hence the $\sigma$-component of $D\psi$ collects $\partial_i$ of the three neighbouring octants only: the Gauss law on the vacuum collects $\sum_i \partial_i E_i$ from the three adjacent electric octants, and each evolution equation collects a curl from two octants of the other weight. This is the exact content of the proposed "pooling" of Maxwell's equations across octants, and it holds as $D = \sum_i e_i \partial_i$. The proposed mechanism, a per-pair exchange of $e_i$ with $e_{i+3}$ converging on the pole $(-,-,-)$, is not confirmed: the exchange replaces $D$ by $D'$ and the wave operator by the four-dimensional Laplacian (Theorem 4.16(ii)), and nothing converges on one pole, the two poles carrying one Gauss law each.

## 4.8 The answer: $X_{\mathrm{EM}} = (\mathbf{B}; \mathbf{E})$ with each component on an octant

The pair reading has one exact realisation. The field-octant map (Theorem 4.7 and Proposition 4.17), $\Phi(X) = \sum_k (E_k + B_k T) e_k |\mathrm{vac}\rangle$, puts each $E_k$ on the octant with a single minus sign in pair $k$ and each $B_k$ on its antipode; it is unique up to a scale once rotation equivariance, Maxwell's equations and the separation of $\mathbf{E}$ from $\mathbf{B}$ are required, apart from the alternative $\Phi \circ G$ that the earlier labelling uses; in textbook variables the same octants carry, through $\Phi_{\mathrm{t}} = -(-1)^N \Phi$ (Theorem 4.16(iv)), Hestenes' $\mathbf{E} + I\mathbf{B}_{\mathrm{t}}$ on the temporal Pauli algebra, with $I = T$. Clifford multiplication by the field point does not realise the pair: it places the light-cone sums $\mathbf{B} \pm \mathbf{E}$ on the octants (Proposition 4.10), and differs from $\Phi$ by an eighth of a turn of duality. The construction is a representation of the classical field strength; it is not a quantisation of the field.

**The main obstruction.** On the octants the rotations act through the spin representation, but the boosts and the duality rotation of the field act only through the odd operators $Q_k$ and $-T$ of $\rho_{\mathrm{oct}}$ (Theorem 4.11 and Proposition 4.12), whereas a Dirac field transforms under the spin lift of the interior Lorentz algebra. A covariant theory containing both the Dirac field and the field octants must say how $\rho_{\mathrm{oct}}$, the spin lift acting on Dirac spinors and the map $\Phi$ coexist. This is the principal question the octant picture leaves, and the element $T$, which is at once the mass form of the Dirac reading and the duality operator on the octants, is where the two meet (Section 5).

**What the poles carry.** With $\Phi$, the vacuum pole carries Gauss's law and the charge density, the filled pole the absence of magnetic charge (Theorem 4.16); they are the Fock vacuum and

the filled state, the rotation scalars, the eigenlines of $\mathfrak{gl}(3)$ of weights $\Lambda^0$ and $\Lambda^3$, and, under the spin-lifted $\mathfrak{l}_G$, the time components of the two four-vectors $S^{\mp}$ (Proposition 4.12(iv)). The mass form pairs them with each other as it pairs $E_k$ with $B_k$ (Proposition 4.14). Of the earlier labels, "statistics" survives as the constructed correspondence between the Fisher–Rao octant of clocks and the vacuum pole $(+,+,+)$ (Proposition 4.3). "Mass" does not survive as a property of the pole $(-,-,-)$: the mass form treats both poles alike, and the interior Lorentz algebra distinguishes no octant, since the stabiliser in $\mathfrak{l}_{\text{int}}$ of the octant $\sigma$ is the three-dimensional span of $e_1e_4 = \omega_1$ and the null rotations $(e_1 - \sigma_1 e_4)e_5$, $(e_1 - \sigma_1 e_4)e_6$, depending on $\sigma_1$ only, while $g_2, g_3$ commute with $\mathfrak{l}_{\text{int}}$ and $g_1$ normalises it (r3-26). The octant Fock structure is the canonical anticommutation algebra of one constituent; it is not a photon Fock space, and the quantisation of the field stays open, as in [3].

**Open.** Whether a dynamics, rather than a convention, selects $\Phi$ or $\Phi \circ G$; what the two scalar fields $\phi_0, \phi_3$ of the extended system of Theorem 4.16(v) mean, if anything; and how $\rho_{\text{oct}}$, which is not part of the spin representation, enters a covariant dynamics with the Dirac sector of [1]; its covariant couplings linear in the field are determined in Theorem 5.9.

Table 3 lists every claim of the section with its check and its class.

Table 3: The claims of this section, with their checks and classes.

| id | claim | class |
|---|---|---|
| r3-1 | octants = basis lines, sign triples, CAR, $N$ = # minus signs, vacuum $(+,+,+)$ | restated (ES17), fixed |
| r3-2 | octant $\sigma$ is the pure spinor of $N_\sigma = \text{span}\{e_i - \sigma_i e_{i+3}\}$ | fixed; inherited (Cartan–Chevalley) |
| r3-3 | $\Gamma = -\omega_1\omega_2\omega_3$, $(-1)^N = -\Gamma = \gamma_5$; the earlier $I = -\omega_1\omega_2\omega_3$ confirmed | restated, fixed |
| r3-4, 5 | reflections lift to $e_a\omega$, unique; octant group simply transitive; lifted group of order 16, centre $\{\pm\mathbf{1}, \pm e_4e_5e_6\}$ | fixed |
| r3-6 | top eigenline of $B_{\text{diag}\,x}$ is the octant sign $x$; clock octant $\leftrightarrow$ vacuum | fixed; identification constructed |
| r3-7 | $S \cong \Lambda^*(L) \cong \text{Cl}(V^+)\lvert\text{vac}\rangle$, weights $1,3,3,1$; $X$ acts by $2X_L\wedge$ and contraction with $\eta(X_{L'}, \cdot)$ | inherited (Chevalley), verified |
| r3-8 | commutant of $\mathcal{K} = \mathfrak{gl}(3)$, acting as $\Lambda^w(A) - \frac{1}{2}\,\text{tr}\,A$ | inherited (Gualtieri), fixed here |
| r3-9 | diagonal rotations $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$, Fock forms of $L_i, K_i$ | restated, fixed |
| r3-10 | equivariant maps: dimension 4; $\Phi_{\alpha,\beta}$ | fixed; $\Phi$ constructed |
| r3-11 | premetric $\mathfrak{gl}(3)$ on the image; vector action mixes $\mathbf{E}, \mathbf{B}$ | fixed; premetric reading inherited |
| r3-12 | $T$ lifts parity and acts as $-G$ on the octants; $(-1)^N$ acts as parity | fixed |
| r3-13 | $X\lvert\text{vac}\rangle$, $X\lvert\text{full}\rangle$ light-cone amplitudes; (12); $\Phi$ is no Clifford map | fixed; reading interpretive |
| r3-14 | no even intertwiner of boosts or of $G$ | obstructed |
| r3-15, 16 | odd intertwiners $Q_k, -T$; $\rho_{\text{oct}}$; spin lift: four-vectors, no intertwiner | $Q_k$, $\rho_{\text{oct}}$ constructed; counts fixed; restated (iv) |
| r3-17, 18 | energies; invariant forms; $T$ = Chevalley–Mukai pairing of antipodes; $\lvert\psi\rvert^2$ not $\rho_{\text{oct}}$-invariant | fixed; pairing inherited |
| r3-19 | $d, \delta$ on the octants, $d \mp \delta = D, D'$ | inherited (Kähler), verified |
| r3-20, 21, 23, 24 | Maxwell $\Leftrightarrow (\partial_t - D)\Phi(X) = -(\rho\lvert\text{vac}\rangle + \mathbf{j}\cdot f)$; wave operator; SI; Hestenes form | fixed; inherited in substance |
| r3-22 | Maxwell-compatible equivariant maps $= r\Phi \circ e^{\vartheta G}$ | fixed |

| id | claim | class |
|---|---|---|
| r3-25 | pooling = nearest-neighbour structure of $D$ | fixed; mechanism refuted |
| r3-26 | $\mathfrak{l}_{\text{int}}$ distinguishes no octant | fixed; "mass pole" refuted |

## 5 The bridge $T = nJ(n)$ between the two readings

Two sections of this note meet in one element of $\mathrm{Cl}(3,3)$. The clock $n \in V^+$ of Section 3 determines the complex structure $J(n) = Tn$ of the Dirac reading (Proposition 3.4 and Theorem 3.5), while the octant representation of the field in Section 4 uses $T = e_1e_2e_3$ itself, through $-T$, as the duality operator (Propositions 4.8 and 4.12). Since $n$ commutes with $T$ and $n^2 = \mathbf{1}$, $T = nJ(n) = J(n)n$ for every clock. This section asks whether this identity can be built into both readings at once while their distinct Lorentz representations are kept apart, also when the clock varies or the temporal axes are permuted, and it answers the points left open in Section 4: the remark on the spatial derivatives of the octant Maxwell operator after Theorem 4.16, the paragraph "The main obstruction" of Section 4.8, and the question of its paragraph "Open" whether $\rho_{\text{oct}}$ can enter a coupling to the Dirac sector. Three expectations are tested on the way: (E1) that an odd permutation of the temporal axes reverses the mass form and the duality operator together; (E2) that the couplings of the Dirac field to the field, which contain $T$, change sign with it under such a permutation; (E3) that such a coupling can be written through $T$ and $J(n)$. In brief: (1) $T$ is invariant and $J(n)$ covariant under every rotation of the temporal triple and under the diagonal rotations; an odd permutation of the temporal axes reverses $T$ and $J(n)$ together, hence the mass form; paired with the same permutation of the spatial axes, the only form that respects $G$, it also reverses the duality operator and turns the octant map into its textbook-orientation form; reversing the clock is a half-turn, which fixes $T$, and the charge conjugation of [1, 2] is a half-turn composed with $T$ (Section 5.1); (2) $T$ is at once the Dirac mass form, the Dirac parity and the lift of the frame parity, and the duality on the octants; $J(n)$ acts on the octants as a rotation of the transverse field about $n$ and maps the longitudinal field to the two poles that carry the Gauss laws (Section 5.2); (3) the spin lift of $\mathfrak{l}_{\text{int}}(n)$ and the octant representation $\rho_{\text{oct}}$ are inequivalent on the same module, their joint commutant is $\mathbb{R}\mathbf{1}$, no covariant bilinear coupling exists, and the covariant trilinear couplings of a Grassmann Dirac field to the field octants form a two-dimensional space, the Pauli term and its dual, written through $T$ and $J(n)$ (Section 5.3); (4) the two first-order operators use conjugate Pauli triples, related only by elements of mixed parity, and neither derives spatial differentiation from temporal directions (Section 5.4); (5) the field reading does not depend on the clock and the particle reading does, and $T$ is, up to scale, the only element that is the clock times a Dirac complex structure for every clock (Section 5.5). Section 5.6 collects the verdicts.

**Conventions and units.** Those of Sections 3 and 4: $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of eq. (1) of [1], $s_a := e_{a+3}$, $T := e_1e_2e_3$, $\Gamma := \omega = e_1 \cdots e_6$, the binding root $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$, the diagonal rotations $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$ and boosts $K_i = \frac{1}{2}(e_ke_{j+3} - e_je_{k+3})$ of $\mathfrak{l}_G$ for cyclic $(i,j,k)$, acting on $V$ by the commutator, and the octant notation $|\mathrm{vac}\rangle$, $|\mathrm{full}\rangle$, $f_k$, $g_m$, $\Phi(X) = \mathbf{E}\cdot f + \mathbf{B}\cdot g$ of Proposition 4.1, Lemma 4.6, and Theorem 4.7; for $v \in \mathbb{R}^3$ put $f(v) = \sum_k v_k f_k$ and $g(v) = \sum_m v_m g_m$. The field point is $X = \sum_i (B_i e_i + E_i e_{i+3})$ with $\mathbf{B}$ minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$. A *clock* is a unit $n \in V^+$ with an oriented orthonormal frame $(n, u_1, u_2)$ of $V^+$, so $nu_1u_2 = T$ (Proposition 3.4); $\gamma^\mu_n = (n, s_1, s_2, s_3)$, $\mathfrak{l}_{\text{int}}(n)$ is spanned by the bivectors of $E_n = \mathbb{R}n \oplus V^-$, $\Omega_n := ns_1s_2s_3$, and $\phi_n : \mathfrak{l}_G \to \mathfrak{l}_{\text{int}}(n)$ is the dictionary of Proposition 3.15, $L_i \mapsto s_j \wedge s_k$, $K_i \mapsto s_i \wedge n$. A bivector $a \wedge b$ of orthogonal vectors is the Clifford element $\frac{1}{2}ab$, and its spin lift is left multiplication by that element, so the Dirac reading's Lorentz representation of $\mathfrak{l}_G$ on $S$ is

$$\rho_n := \mathrm{spin} \circ \phi_n, \qquad \rho_n(L_i) = \tfrac{1}{2}s_js_k, \qquad \rho_n(K_i) = \tfrac{1}{2}s_in,$$

and the field reading's is $\rho_{\mathrm{oct}}(L_i) = L_i$, $\rho_{\mathrm{oct}}(K_i) = Q_i = -TL_i$ (Proposition 4.12). Further $t_i := \frac{1}{2}e_je_k$ spans the rotations $\mathfrak{so}(3)_+$ of the temporal triple alone, and $A_i := \frac{1}{2}(e_je_k + e_{j+3}e_{k+3})$ rotates the two triples in opposite senses (Lemma 4.6); $L(n) := \sum_i n_iL_i$, $A(n) := \sum_i n_iA_i$. Dimensions: $n$, $J(n)$, $T$, $\Gamma$, all Clifford elements and all intertwiners are dimensionless; $[x^a] = [x^0] = [ct] = \mathrm{m}$ and $[\partial_\mu] = \mathrm{m}^{-1}$; in SI units a Dirac field has $[\psi] = \mathrm{m}^{-3/2}$ and a Lagrangian density has the unit $\mathrm{J\,m}^{-3}$, as in (6); the field enters as $X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$ with $[E_i] = [cB_i] = \mathrm{V\,m}^{-1}$; a magnetic moment $\mu$ has $[\mu] = \mathrm{J\,T}^{-1}$ and an electric dipole moment $d$ has $[d] = \mathrm{C\,m}$. Computations use Heaviside–Lorentz units with $\hbar = c = 1$. Identifiers r4-$n$ refer to `verify/r4_bridge.py`, which checks every finite-dimensional and symbolic claim below in exact arithmetic (r4-1 re-verifies the conventions).

### 5.1 The identity and its covariance

**Proposition 5.1** (The identity)**.** *For every $n \in V^+$: $n^2 = |n|^2\mathbf{1}$, $J(n) := Tn = nT = n_1e_2e_3 + n_2e_3e_1 + n_3e_1e_2$, $J(n)^2 = -|n|^2\mathbf{1}$, and $nJ(n) = J(n)n = |n|^2\,T$. Hence for every clock*

$$T = nJ(n) = J(n)n, \qquad J(n) = Tn = nT, \qquad n^2 = \mathbf{1}, \qquad J(n)^2 = T^2 = -\mathbf{1}.$$

*$J(n)$ commutes with $n, s_1, s_2, s_3$, and $n \mapsto J(n)$ is linear and injective.*

*Proof.* By the blade rule (Lemma 2.1 of [1]) $T$ commutes with each $e_i$ and anticommutes with each $s_a$. So $J(n)^2 = TnTn = T^2n^2$ and $nJ(n) = nnT = |n|^2T$; $T$ and $n$ each anticommute with $s_a$, so $J(n)$ commutes with $s_a$ (r4-2, symbolic in $n$ and on the rational parametrisation $n = (2a, 2b, 1 - a^2 - b^2)/(1 + a^2 + b^2)$ of the clock sphere). □

Restated from Proposition 3.4 and from Remark 4.2 of [1] ($J = \gamma^0T$), re-verified (r4-2). Proposition 5.1 of the working note [3] is confirmed.

**Proposition 5.2** (Rotations)**.** *(i) For $Y \in \mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_\Delta$, $[Y, T] = 0$ and $[Y, J(n)] = J([Y, n])$, with $[Y, n] \in V^+$. Hence for the spin lift $R$ of a rotation of the temporal triple alone, or of a diagonal rotation, $RTR^{-1} = T$ and $RJ(n)R^{-1} = J(Rn)$, where $Rn := RnR^{-1}$.*

*(ii) The rotations $\mathfrak{so}(3)_+$ do not commute with $G$; the diagonal rotations do, and are symmetries of the field reading: $\rho_{\mathrm{oct}}(L_i) = L_i$ and $\Phi$ is equivariant (Theorem 4.7).*

*(iii) A rotor $q \in \mathrm{Spin}(V^+)$ with $qe_1q^{-1} = n$ fixes $T$, the $s_a$ and $e_4e_5e_6$, and it carries $J(e_1)$, $\Omega_{e_1}$, $\mathfrak{l}_{\mathrm{int}}(e_1)$, $\phi_{e_1}$ to $J(n)$, $\Omega_n$, $\mathfrak{l}_{\mathrm{int}}(n)$, $\phi_n$.*

*Proof.* $T$ commutes with every same-sign bivector, hence with $t_i$ and $L_i$, and $[Y, Tn] = T[Y, n]$. The spatial part of $L_i$ commutes with $n$, so $[L_i, n] = -[t_i, n] \in V^+$. A temporal bivector commutes with every $s_a$ and with $e_4e_5e_6$; with $\phi_n(L_i) = s_j \wedge s_k$, $\phi_n(K_i) = s_i \wedge n$ this gives (iii). (ii) is Lemma 4.6. The finite statements are checked with the rational rotors $q = u'(e_1 - n)$, $u' \perp n$, at $n = (2, 3, 6)/7, (1, 2, 2)/3, (-1, 4, 8)/9$ and with an exact Cayley transform of $L_3$ (r4-3, r4-4). □

Fixed. Rotations of the temporal triple move the clock and with it the whole Dirac structure, but they are not symmetries of the field reading; the diagonal rotations move the clock as well and are symmetries of both.

**Proposition 5.3** (Permutations of the temporal axes and the reversal of the clock)**.** *For $\sigma \in S_3$ let $A_\sigma$ be the automorphism of $\mathrm{Cl}(3,3)$ with $e_i \mapsto e_{\sigma(i)}$, $s_a \mapsto s_a$, and $A'_\sigma$ the one with $e_i \mapsto e_{\sigma(i)}$, $s_a \mapsto s_{\sigma(a)}$; write $\sigma n = \sum_i n_ie_{\sigma(i)}$.*

*(i) Both are inner, $A = \mathrm{Ad}_U$ with $U$ unique up to scale and orthogonal up to scale, and*

$$A(T) = \mathrm{sgn}(\sigma)\,T, \qquad A(J(n)) = \mathrm{sgn}(\sigma)\,J(\sigma n), \qquad A(n) = \sigma n.$$

| operation | $T$; $J(n)$ | Dirac reading: rest flow $e^{-m\lambda T}$, senses | field reading: $-T$, $G$, $\Phi$ |
|---|---|---|---|
| rotation $R$ of $V^+$ alone | $T$; $J(Rn)$ | flow invariant; eigenspaces $n=\pm1$ go to $Rn=\pm1$, senses kept | not a symmetry: $[\mathfrak{so}(3)_+, G]\neq 0$ |
| diagonal rotation | $T$; $J(Rn)$ | as above, with $V^-$ rotated alike | symmetry: $\rho_{\mathrm{oct}}(L)=L$, $-T$, $G$, $\Phi$ preserved |
| clock reversal (half-turn) | $T$; $-J(n)$ | flow invariant; the eigenspaces $n=\pm1$ are exchanged, so the particle and antiparticle labels relative to $n$ are exchanged | the octant structure does not involve $n$; the half-turn itself is not a field symmetry |
| odd $\sigma$, temporal alone | $-T$; $-J(\sigma n)$ | $e^{-m\lambda T}\mapsto e^{+m\lambda T}$: sense reversed on each eigenspace; $(m,q)\mapsto(-m,-q)$ | not a symmetry: breaks $G$ |
| odd $\sigma$, paired | $-T$; $-J(\sigma n)$ | as for $\sigma$ alone, spatial axes permuted | $G$ kept, octants permuted, $\Phi\mapsto\Phi_{\mathrm{t}}$, duality $-T\mapsto+T$ |
| $C=RT$, $C\in U_n$ | $T$; $-J(n)$ | same flow; particle $\leftrightarrow$ antiparticle (Proposition 3.8 of [2]) | not a field symmetry ($C$ is a vector) |

Table 4: What the operations of Propositions 5.2 and 5.3 do to $T$, $J(n)$ and the two readings (r4-3 to r4-8).

*(ii)* *$A'_\sigma$ commutes with $G$, permutes the pole operators, $A'(\omega_i)=\omega_{\sigma(i)}$, and with $U|\mathrm{vac}\rangle=|\mathrm{vac}\rangle$ it acts on the octants by $U\Phi(X)=\Phi(\tilde X)$, $\tilde X=(\mathrm{sgn}(\sigma)P_\sigma\mathbf{B};P_\sigma\mathbf{E})$, where $P_\sigma$ permutes the components: $\mathbf{B}$ is transformed as an axial vector. The transported octant map $\Phi':=U\Phi(P_\sigma\oplus P_\sigma)^{-1}$ is $\Phi$ for even $\sigma$ and $\Phi\circ(-\eta)=\Phi_{\mathrm{t}}=-(-1)^N\Phi$, the textbook-orientation map of Theorem 4.16(iv), for odd $\sigma$, and its duality operator is $A'(-T)$: $A'(-T)\Phi'=\Phi' G$.*

*(iii)* *No nontrivial $A_\sigma$ commutes with $G$.*

*(iv)* *Let $R=nu_2$, the rotor of the half-turn about $u_1$. Then $RnR^{-1}=-n$, $RTR^{-1}=T$, $RJ(n)R^{-1}=J(-n)=-J(n)$, and $RT=u_1$: every unit vector $C$ of the surplus plane, the charge conjugation of Theorem 5.1 of [1] at the clock $n$, is the half-turn about $C$, which reverses the clock, composed with $T$.*

*Proof.* (i) $\mathrm{Cl}(3,3)\cong M_8(\mathbb{R})$ is central simple (Skolem–Noether); $e_{\sigma(1)}e_{\sigma(2)}e_{\sigma(3)}=\mathrm{sgn}(\sigma)T$ by reordering anticommuting factors, and $A(J(n))=A(T)A(n)$. (ii) $P_\sigma\oplus P_\sigma$ commutes with the block matrix of $G$. $A'(a_k^+)=a^+_{\sigma(k)}$, so $Uf_k=f_{\sigma(k)}$ and $Ug_m=\frac12\epsilon_{mjk}a^+_{\sigma(j)}a^+_{\sigma(k)}|\mathrm{vac}\rangle=\mathrm{sgn}(\sigma)g_{\sigma(m)}$, since $\epsilon_{\sigma(m)\sigma(j)\sigma(k)}=\mathrm{sgn}(\sigma)\epsilon_{mjk}$. Hence $\Phi'(Y)=\Phi(\mathrm{sgn}(\sigma)\mathbf{B}_Y;\mathbf{E}_Y)$, and $\Phi\circ(-\eta)=-(-1)^N\Phi$ because $(-1)^N$ is $-1$ on the $f$'s and $+1$ on the $g$'s; the duality relation is the transport of $-T\Phi=\Phi G$. (iii) A temporal permutation alone moves $e_i$ and fixes its partner $e_{i+3}$. (iv) $R$ is an even element of $\mathrm{Cl}(V^+)$, so it commutes with $T$; $nu_2nu_1u_2=-u_2u_1u_2=u_1$. All of this is checked for the six permutations, alone and paired, and for three clocks (r4-5, r4-6, r4-7). □

Fixed. Expectation (E1) is confirmed in the form (i): an odd permutation reverses $T$ and $J(n)$ together, and the paired one, which is the only one that respects $G$ and the octants, reverses the mass form $T$ and the duality operator $-T$ together. The reason is (ii): an odd permutation of the axes is an improper rotation, under which $\mathbf{B}$ is axial; on the octants it changes the orientation convention $\mathbf{B}_{\mathrm{t}}=-\mathbf{B}$ of [1] into the textbook one, for which the duality operator is $+T$ (Theorem 4.16(iv)). The working note [3] defers the temporal-clock permutations and rotations (its Section 1) and states that reversing $n$ reverses $J(n)$ (its Section 5); the latter is confirmed and sharpened by (iv): the reversal is a rotation in $\mathrm{Spin}(V^+)$, which leaves $T$, the product of the two reversed objects, fixed.

**Corollary 5.4** (The two senses at every clock)**.** *On the eigenspaces $n=\pm1$ of any clock, $T=\pm J(n)$, so the rest flow $e^{-m\lambda T}$ of Proposition 3.7 of [2] is $e^{\mp m\lambda J(n)}$, the two senses of the*

*charge rotation that are particle and antiparticle there (Proposition 3.8 of [2]). An odd temporal permutation carries the Dirac data $(K^\mu_n, T, J(n))$, with $K^\mu_n = -n\gamma^\mu_n$, to $(K^\mu_{\sigma n}, -T, -J(\sigma n))$: the transported Lagrangian (6) has mass $-m$ and charge $-q$. The axial rotation by $\Omega_{\sigma n}$, which preserves every $K^\mu$ and $J$ and reverses $T$ (Proposition 3.6(iv) of [2]), restores the mass, leaving $q \mapsto -q$, the effect of $C$.*

*Proof.* With $\Pi_\pm = \frac{1}{2}(\mathbf{1} \pm n)$, $T\Pi_\pm = \frac{1}{2}(T \pm J(n)) = \pm J(n)\Pi_\pm$ because $J(n)n = T$. The rest follows from Proposition 5.3(i) and from $\Omega^\top T\Omega = -T$, $\Omega^\top K^\mu \Omega = K^\mu$, $[\Omega, J] = 0$ (r4-8). □

Proposition 3.7 of [2] is restated at every clock (fixed); the effect of the permutations on the Dirac reading is fixed; that a Dirac mass sign is not observable alone is inherited [31]. In the decomposition $C = RT$ of Proposition 5.3(iv), $T$ preserves the flow, the eigenspaces of $n$ and $J(n)$ (it is a parity, Theorem 5.5), so the reversal of the sense by $C$ in Proposition 3.8 of [2] comes entirely from the half-turn that reverses the clock (fixed; the reading of particle and antiparticle as the two clock orientations is interpretive).

## 5.2 The three roles of $T$, and $J(n)$ on the octants

**Theorem 5.5** (The three roles of $T$)**.** *(a)* Mass form. $\gamma^0_n J(n) = T$ *for every clock: the mass kernel of the Dirac reading is the same at every clock, while the kinetic kernels* $K^\mu_n = TJ(n)\gamma^\mu_n = -n\gamma^\mu_n$ *move with it.*

*(b)* Parity. $\mathrm{Ad}_T|_V = \eta$, *the frame parity fixing* $\mathbf{B}$ *and reversing* $\mathbf{E}$, *and* $T$ *is up to scale the only element implementing it. At every clock* $T$ *commutes with* $J(n)$ *and* $TH_n(\mathbf{p})T^{-1} = H_n(-\mathbf{p})$ *for* $H_n(\mathbf{p}) = \sum_a n s_a p_a + mn$: $T = \gamma^0 J$ *is the admissible Dirac parity with* $P^2 = -\mathbf{1}$ *of Theorem 5.3 of [1], at every clock.*

*(c)* Duality. $-T\Phi = \Phi G$, $Tf_k = g_k$, $Tg_k = -f_k$, $T|\mathrm{vac}\rangle = |\mathrm{full}\rangle$: *on the field octants* $-T$ *is the Hodge star.*

*The complex structure* $J(n)$ *is, for the clock* $n$, *the selected imaginary unit and the charge generator of the Dirac reading (Theorem 4.1 of [1] at every clock, Theorem 3.5); in (6) it enters as* $D_\mu = \partial_\mu + qA_\mu J(n)$.

*Proof.* (a) is Proposition 5.1 with $\gamma^0_n = n$, and $TJ(n) = -n$. (b) $T$ commutes with $n$ and anticommutes with $s_a$, so it commutes with $J(n)$ and reverses the momentum term of $H_n$; uniqueness: the solutions of $Ye_a = \eta_{aa} e_a Y$ form a line (Skolem–Noether). (c) is Proposition 4.8(i) (r4-9). □

Restated and fixed: (a) is Remark 4.2 and Proposition 4.5 of [1] and Theorem 3.5(iii) at every clock, (b) is Proposition 2.11 of [2] and Proposition 4.8 together with Theorem 5.3 of [1] at every clock, (c) is Proposition 4.12(i); proved together here (r4-9). The three are compatible: (b) and (a) say that the mass is parity-even, (b) and (c) that the parity lift acts on the field octants as minus the Hodge star, which is Proposition 4.8. Role (a) belongs to the particle reading and uses $T$ as the product $nJ(n)$; (c) belongs to the field reading and uses $T$ alone; (b) belongs to both, since $\mathrm{Ad}_T|_V = \eta$ is the parity of the frame $(\mathbf{B}; \mathbf{E})$ and of the three lengths, and $T$ is the Dirac parity.

**Proposition 5.6** ($J(n)$ relative to the field structure)**.** *With* $L(n)$ *and* $A(n)$ *as in the conventions:*

*(i)* $J(n) = -L(n) + A(n)$, *with* $L(n)A(n) = A(n)L(n) = 0$; *for a clock,* $-L(n)^2$ *and* $-A(n)^2$ *are complementary projectors of rank four,* $L(n)^3 = -L(n)$ *and* $A(n)^3 = -A(n)$.

*(ii)* $L(n) = \rho_n(L(n)) - \frac{1}{2}J(n)$, *in which* $\rho_n(L(n)) = \frac{1}{2}\sum_i n_i s_j s_k$ *is the Dirac spin rotation of* $V^-$ *about the spatial axis* $\sum_i n_i s_i$ *paired with* $n$ *(it acts trivially on* $V^+$*); and* $\rho_{\mathrm{oct}}(L_i) = \rho_n(L_i) - \frac{1}{2}J(e_i)$ *for every* $i$ *and every clock. Hence* $e^{2\pi L(n)} = e^{2\pi\rho_n(L(n))}e^{-\pi J(n)} = (-\mathbf{1})(-\mathbf{1}) = \mathbf{1}$.

*(iii) For every $n$,*

$$J(n)\Phi(X) = -\Phi(n \times \mathbf{B};\, n \times \mathbf{E}) + (\mathbf{E}\cdot n)\,|\mathrm{full}\rangle - (\mathbf{B}\cdot n)\,|\mathrm{vac}\rangle,$$

*and $J(n)|\mathrm{vac}\rangle = g(n)$, $J(n)|\mathrm{full}\rangle = -f(n)$. The kernel of $L(n)$ is spanned by the two poles and the longitudinal octants $f(n)$, $g(n)$; on it $J(n) = A(n)$. On the transverse octants $f(u), g(u)$, $u \perp n$, $J(n) = -L(n)$, the rotation about $n$: $L(n)f(u) = f(n \times u)$, $L(n)g(u) = g(n \times u)$.*

*(iv) $J(n)$ does not commute with $\rho_{\mathrm{oct}}$ and does not preserve the field octants $\Phi(V)$.*

*Proof.* (i) With $t(n) = \sum_i n_i e_j e_k = J(n)$ and $s(n) = \sum_i n_i s_j s_k$: $L(n) = \frac{1}{2}(s(n) - t(n))$, $A(n) = \frac{1}{2}(s(n)+t(n))$, $s(n)$ and $t(n)$ commute and $s(n)^2 = t(n)^2 = -|n|^2\mathbf{1}$, so $L(n)A(n) = \frac{1}{4}(s^2 - t^2) = 0$ and $-L(n)^2 = \frac{1}{2}(\mathbf{1} + st)$, $-A(n)^2 = \frac{1}{2}(\mathbf{1} - st)$ with $(st)^2 = \mathbf{1}$. (ii) $\rho_n(L(n)) = \frac{1}{2}s(n)$, and $L_i = \frac{1}{2}s_j s_k - \frac{1}{2}e_j e_k$ with $e_j e_k = J(e_i)$; $\frac{1}{2}s(n)$ and $\frac{1}{2}J(n)$ commute and square to $-\frac{1}{4}\mathbf{1}$. (iii) $J(n) = \sum_i n_i e_j e_k$ and $e_I|\mathrm{vac}\rangle = a_I^+|\mathrm{vac}\rangle$ (Proposition 4.4) give $J(n)|\mathrm{vac}\rangle = g(n)$; $L(n)$ acts on the $f$'s and $g$'s as the vector rotation (Lemma 4.6) and kills the poles; $A(n)f(u) = (n \cdot u)|\mathrm{full}\rangle$ and $A(n)g(u) = -(n \cdot u)|\mathrm{vac}\rangle$. (iv) follows from (iii), since $\rho_{\mathrm{oct}}$ preserves $\Phi(V)$ (r4-10, r4-11, symbolic in $n$, $\mathbf{E}$, $\mathbf{B}$, $u$). □

Fixed. What this means is the following, and the reading is interpretive. The Dirac charge rotation $e^{\theta J(n)}$, read on the octant module, is not a transformation of the field: on transverse fields it is the rotation by $-\theta$ about the axis paired with the clock, and on the four-dimensional complement it exchanges the longitudinal components with the pole components, $\mathbf{E}\cdot n$ with the slot of $\nabla\cdot\mathbf{B}$ and $\mathbf{B}\cdot n$ with the slot of $\nabla\cdot\mathbf{E}$ (Theorem 4.16). So the complex structure of the particle reading mixes the field with the Gauss constraints, and it is not in the commutant of the field reading's Lorentz representation. Item (ii) shows where integer and half-integer spin part on $S$: the field rotation $L(n)$, which turns $\mathbf{B}$ about $n$ and $\mathbf{E}$ about $\sum_i n_i s_i$, is the Dirac rotation about $\sum_i n_i s_i$ combined with half a turn of the charge, and the two signs $-1$ cancel. The working note [3] asks that $J(n)$ be analysed separately from $G_{\mathrm{EM}} = \star$ (its Section 8, item 5): on the octants $\star$ is $-T$ (Proposition 4.12), $J(n) = nT$ differs from it by the clock, and $J(n)$ commutes with $\mathfrak{l}_{\mathrm{int}}(n)$ while $G$ commutes with $\mathfrak{l}_G$.

### 5.3 Two Lorentz representations on one module, and their couplings

**Proposition 5.7** (Commutants). *(i) The commutant of $\rho_n(\mathfrak{l}_G)$, the spin lift of $\mathfrak{l}_{\mathrm{int}}(n)$, in $\mathrm{End}_{\mathbb{R}}(S)$ is $\mathrm{span}\{\mathbf{1}, u_1, u_2, J(n)\} \otimes \mathrm{span}\{\mathbf{1}, \Omega_n\} \cong M_2(\mathbb{C})$, of dimension 8.*

*(ii) The commutant of $\rho_{\mathrm{oct}}(\mathfrak{l}_G)$ is $\mathrm{span}\{\Pi_F, T\Pi_F\} \oplus \mathrm{End}(\mathrm{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle\})$, of dimension 6, with $\Pi_F$ the projector onto the field octants.*

*(iii) The joint commutant of $\rho_n(\mathfrak{l}_G)$ and $\rho_{\mathrm{oct}}(\mathfrak{l}_G)$ is $\mathbb{R}\mathbf{1}$, for every clock.*

*Proof.* (i) $u_a$ anticommutes with each vector of $E_n$ and so commutes with its bivectors; $\Omega_n$ commutes with the even elements of $\mathrm{Cl}(E_n)$; the eight products are independent, and the commutant has dimension 8 because $S$ complexifies to $2(\frac{1}{2},0)\oplus 2(0,\frac{1}{2})$, whose commutant is $M_2(\mathbb{C})\oplus M_2(\mathbb{C})$ [11]. (ii) $\Phi(V)$ is irreducible of complex type with commutant $\mathrm{span}\{\mathbf{1}, T\}$ (Proposition 4.12(iii)), the poles are two trivial summands, and there is no map between them. (iii) Exact computation at $n = e_1, (2,3,6)/7, (1,2,2)/3$ (r4-12); structurally, $\rho_{\mathrm{oct}}(L_i) - \rho_n(L_i) = -\frac{1}{2}J(e_i)$ spans $\mathfrak{so}(3)_+$, which with $\mathfrak{l}_{\mathrm{int}}(n)$ generates $\mathfrak{so}(3,3)$, whose commutant $\mathrm{span}\{\mathbf{1}, \Gamma\}$ meets that of the odd $Q_i$ in $\mathbb{R}\mathbf{1}$. □

Fixed; (i) contains the commutant $\mathbb{H}$ of the four $\gamma_n^\mu$ of Theorem 3.5(ii). No complex structure, and no operator other than a scalar, commutes with both Lorentz actions.

**Proposition 5.8** (Three inequivalent representations on $S$)**.** *On the same module $S$ the Lie algebra $\mathfrak{l}_G \cong \mathfrak{so}(1,3)$ acts in three ways: through $\rho_n$ (complexification $2(\frac{1}{2},0)\oplus 2(0,\frac{1}{2})$, two Dirac spinors), through $\rho_{\mathrm{oct}}$ ($(1,0)\oplus(0,1)\oplus 2(0,0)$, the field strength and two singlets) and through the spin lift of $\mathfrak{l}_G$ itself ($2(\frac{1}{2},\frac{1}{2})$). The Casimirs $\sum_k(\rho(K_k)^2-\rho(L_k)^2)$ and $(\sum_k \rho(K_k)\rho(L_k))^2$ are $\frac{3}{2}$ and $-\frac{9}{16}$ for $\rho_n$; 4 and $-4$ on the field octants and 0 on the poles for $\rho_{\mathrm{oct}}$; 3 and 0 for the spin lift of $\mathfrak{l}_G$. In $\rho_n$, $\rho_n(L_3)^2 = -\frac{1}{4}\mathbf{1}$ and $e^{2\pi\rho_n(L_3)} = -\mathbf{1}$; in the other two $\rho(L_3)^3 = -\rho(L_3)$ and $e^{2\pi\rho(L_3)} = \mathbf{1}$. The three are pairwise inequivalent, and no invertible $U$ satisfies $U\rho_n(Y)U^{-1} = \rho_{\mathrm{oct}}(Y)$ for all $Y$.*

*Proof.* Equivalent representations have conjugate Casimirs; the values are computed exactly (r4-13). The decompositions are Proposition 13.12 of [1] and Proposition 4.12. □

Restated (the decompositions) and fixed (the Casimirs and the inequivalence). That $\rho_{\mathrm{oct}}$ does not commute with the spin lift of $\mathfrak{l}_{\mathrm{int}}(e_1)$ (r4-13) is the weakest form of this; the proposition says that the two are not even conjugate. They therefore cannot both be the Lorentz action of one field on $S$. A theory containing a Dirac field and the field octants must treat them as two fields, $\psi$ in a copy of $S$ with $\rho_n$ and $\Phi(X)$ in another with $\rho_{\mathrm{oct}}$, and the joint action of $Y \in \mathfrak{l}_{\mathrm{int}}(n)$ is its spin lift $\rho_n(\phi_n^{-1}Y)$ on $\psi$ and $\rho_{\mathrm{oct}}(\phi_n^{-1}Y)$ on the octants; equivalently $Y' \in \mathfrak{l}_G$ acts by $\rho_n(Y')$ and by $\rho_{\mathrm{oct}}(Y')$. Since $\Phi$ intertwines $\mathfrak{l}_G$ on $V$ with $\rho_{\mathrm{oct}}$ (Proposition 4.12(iii)), a coupling to $\Phi(X)$ is a coupling to $X$ with its own action $Y\cdot X = [Y,X]$, and conversely.

For $X \in V$ let $\ell(X) := \sum_i (B_i L_i + E_i K_i) \in \mathfrak{l}_G$, an equivariant isomorphism $V \to \mathfrak{l}_G$ by the brackets and eq. (14) of Theorem 12.3 of [1], and

$$S_n(X) := \rho_n(\ell(X)) = \tfrac{1}{2}\sum_i (B_i\, s_j s_k + E_i\, s_i n),$$

the field strength as a bivector of $E_n$; under the dictionary of Proposition 3.15 it is the Clifford image of the bivector $\sum_{\mu<\nu}(F_X)_{\mu\nu}\,\gamma_\mu\wedge\gamma_\nu = \sum_i (E_i\, s_i \wedge n + B_i\, s_j \wedge s_k)$, $\gamma_\mu := \eta_{\mu\nu}\gamma_n^\nu$, while the same sum with $F_X^{\mu\nu}$ in place of $(F_X)_{\mu\nu}$ is $S_n(\eta X)$ (r4-16).

**Theorem 5.9** (Covariant couplings of the Dirac field to the field octants)**.** *Let $Y \in \mathfrak{l}_G$ act by $\rho_n(Y)$ on $\psi$ and by $\rho_{\mathrm{oct}}(Y)$ on $\Phi(X)$, that is, by $Y\cdot X = [Y,X]$ on $X$.*

*(i) No nonzero bilinear form $\psi^\top M\Phi(X)$ is invariant, nor one on all of $S$ with $\rho_{\mathrm{oct}}$.*

*(ii) The invariant trilinear couplings $\psi^\top M(X)\psi$, $M : V \to \mathrm{End}(S)$ linear with $\rho_n(Y)^\top M(X) + M(X)\rho_n(Y) + M(Y\cdot X) = 0$, are exactly $M(X) = nC\,S_n(X)$ with $C$ in the commutant of Proposition 5.7(i): an eight-dimensional space. For a real Grassmann $\psi$ only antisymmetric kernels contribute, and these form the two-dimensional space spanned by*

$$M_n(X) = T\,J(n)\,S_n(X) = -n\,S_n(X), \qquad M_n^\star(X) = T\,\Gamma\,S_n(X) = M_n(GX),$$

*and $T\Gamma S_n(X) = TJ(n)\Omega_n S_n(X)$. For a commuting $\psi$ the symmetric kernels form the six-dimensional space given by $C \in \mathrm{span}\{u_1,u_2,J(n)\}\otimes\mathrm{span}\{\mathbf{1},\Omega_n\}$. These dimensions hold at every clock.*

*(iii) Explicitly $M_n(X) = \frac{1}{2}\sum_a E_a s_a - \frac{1}{2}n\sum_i B_i s_j s_k$ and $M_n^\star(X) = \frac{1}{2}\sum_a B_a s_a + \frac{1}{2}n\sum_i E_i s_j s_k$. Both commute with $J(n)$ (they carry no charge); under the parity $\psi \mapsto n\psi$ (or $\psi \mapsto T\psi$), $X \mapsto \eta X$, $M_n$ is even, like the mass form $T$, and $M_n^\star$ is odd.*

*Proof.* (i) The rotation $e^{2\pi\rho_n(L_3)}$ is $-\mathbf{1}$ on $\psi$ and $e^{2\pi\rho_{\mathrm{oct}}(L_3)} = \mathbf{1}$ on the octants (Proposition 5.8), so an invariant form satisfies $\beta(-\psi,\chi) = \beta(\psi,\chi)$; the exact solution spaces are zero at $n = e_1, (2,3,6)/7$ (r4-14). (ii) For every $Y$, $\rho_n(Y)^\top = -n\rho_n(Y)n$: the rotations $\frac{1}{2}s_j s_k$ are antisymmetric and commute with $n$, the boosts $\frac{1}{2}s_i n$ are symmetric and anticommute with it. Hence, for $C$ commuting with $\rho_n$,

$$\rho_n(Y)^\top nCS_n(X) + nCS_n(X)\rho_n(Y) = nC\big(S_n(X)\rho_n(Y) - \rho_n(Y)S_n(X)\big) = -nC\,S_n(Y\cdot X),$$

using $[\rho_n(Y), S_n(X)] = \rho_n([Y, \ell(X)]) = \rho_n(\ell(Y \cdot X))$. The map $C \mapsto nCS_n$ is injective, and the space of invariant maps $V \to S^* \otimes S^*$ has dimension 8: complexified, $V = (1,0) \oplus (0,1)$, $\Lambda^2 S = (1,0) \oplus (0,1) \oplus 6(0,0) \oplus 4(\frac{1}{2},\frac{1}{2})$ and $\mathrm{Sym}^2 S = 3(1,0) \oplus 3(0,1) \oplus 2(0,0) \oplus 4(\frac{1}{2},\frac{1}{2})$, so the dimensions are 2 and 6 [11]. The symmetry of $nCS$ follows from the transpose rule and $C^\top$: $(nCS)^\top = -nSnC^\top n$, which is $-nCS$ for $C = \mathbf{1}, \Omega_n$ and $+nCS$ for $C = u_a, J(n)$ and their products with $\Omega_n$. Finally $TJ(n) = -n$, $TJ(n)\Omega_n = T\Gamma$, $T\Gamma = -e_4e_5e_6$, and $M_n(GX) = M_n^\star(X)$ by (iii). The exact row reduction of the $2304 \times 168$ and $2304 \times 216$ linear systems at $n = e_1$ and $n = (2,3,6)/7$ gives 2 and 6, and the displayed kernels span them (r4-15, r4-16). By Proposition 5.10(i) the dimensions are the same at every clock. (iii) $nS_n(X)n$ reverses the electric part; $n\, e_4e_5e_6\, n = -e_4e_5e_6$ (r4-17). □

Fixed: the dimensions, the explicit kernels and their properties. The textbook comparison is inherited. In complex Dirac theory the Lorentz- and gauge-invariant couplings of $\psi$ to the field strength that are bilinear in $\psi$ and contain no derivative are the Pauli term $\bar\psi\sigma^{\mu\nu}\psi F_{\mu\nu}$, the coupling of an anomalous magnetic moment introduced by Pauli [32], and the electric-dipole term $i\bar\psi\sigma^{\mu\nu}\gamma_5\psi F_{\mu\nu}$, odd under parity and time reversal [33, 31]. With $J(n)$ in the place of $i$, $\bar\psi\cdots\psi \leftrightarrow \frac{1}{2}\psi^\top T\cdots\psi$ as in (6), and the real form $\Omega_n$ of $i\gamma_5$ (Proposition 4.6 of [1]), $\sigma^{\mu\nu}F_{\mu\nu}$ becomes a multiple of $J(n)S_n(X)$: $M_n$ is the real form of the Pauli term and $M_n^\star$ that of the electric-dipole term, up to normalisation and the sign convention for $\gamma^a$; their parities in (iii) agree with the textbook ones. The Lagrangian density

$$\mathcal{L}_{\mathrm{P}} = \tfrac{1}{2}\,\psi^\top\Big(\tfrac{\mu}{c}\,M_n(X_{\mathrm{EM}}) + d\,M_n^\star(X_{\mathrm{EM}})\Big)\psi, \qquad X_{\mathrm{EM}} = (c\mathbf{B};\mathbf{E}), \tag{14}$$

has dimension $\mathrm{J\,m^{-3}}$ in SI units for $[\mu] = \mathrm{J\,T^{-1}}$ and $[d] = \mathrm{C\,m}$, and its two couplings have mass dimension $-1$ in natural units: they are not renormalisable [33]. The arena fixes neither constant (open). The kernel $M_n$ is written through $T$ and $J(n)$, as (E3) proposed: $T$ is the adjoint form, $J(n)$ the imaginary unit of $\sigma^{\mu\nu}$, and $S_n(X)$ the field bivector; the dual uses the other $\mathrm{Spin}(3,3)$-invariant form $T\Gamma$ (Proposition 4.14(iii)) and no $J(n)$. With these couplings the Dirac sector reads, at every clock,

$$\mathcal{L}_D + \mathcal{L}_{\mathrm{P}} = \tfrac{1}{2}\,\psi^\top T\Big[J(n)\Big(\gamma_n^\mu D_\mu + \tfrac{\mu}{c}S_n(X_{\mathrm{EM}}) + d\,\Omega_n S_n(X_{\mathrm{EM}})\Big) - m\Big]\psi,$$

in the units of (6): $T$ is the common left factor of the kinetic, mass and Pauli kernels.

**Proposition 5.10** (The couplings when the clock moves or the axes are permuted)**.** *The couplings of Theorem 5.9 transform as follows.*

*(i) Let $q \in \mathrm{Spin}(V^+)$ be a rotor with $qe_1q^{-1} = n$. Then, with $X$ fixed,*

$$M_n(X) = qM_{e_1}(X)q^{-1}, \qquad M_n^\star(X) = qM_{e_1}^\star(X)q^{-1}, \qquad qTq^{-1} = T, \qquad qT\Gamma q^{-1} = T\Gamma :$$

*under $\psi \mapsto q\psi$ the couplings at $e_1$ become those at $n$. The factors $J(n)$, $\Omega_n$, $\gamma_n^\mu$, $S_n$ are conjugated, and $T$, the clock-independent part of every kernel in the display above, is fixed. The couplings themselves depend on the clock: $M_{e_1} \neq M_{e_2}$.*

*(ii) For a diagonal rotation $R$, $M_{Rn}(RX) = RM_n(X)R^{-1}$ and likewise for $M^\star$: clock, spinor and field turn together.*

*(iii) For an odd temporal permutation alone, $A_\sigma(M_n(X)) = M_{\sigma n}(X)$ and $A_\sigma(M_n^\star(X)) = M_{\sigma n}^\star(X)$, while $A_\sigma(T) = -T$. For the paired permutation, $A'_\sigma(M_n(X)) = M_{\sigma n}(\tilde X)$ and $A'_\sigma(M_n^\star(X)) = \mathrm{sgn}(\sigma)M_{\sigma n}^\star(\tilde X)$, with $\tilde X$ of Proposition 5.3(ii).*

*Proof.* (i) $q$ is orthogonal and conjugates $\rho_{e_1}$ to $\rho_n$ (Proposition 5.2(iii)); $S_n$, $J(n)$, $\Omega_n$ follow, and $T$, $\Gamma$ are fixed. (ii) The same with the spin lift of a diagonal rotation, which also rotates $s_a$ and $X$. (iii) $A(TJ(n)) = TJ(\sigma n)$ because the two signs cancel; $A$ fixes $s_a$ and maps $n$ to $\sigma n$, so $A(S_n(X)) = S_{\sigma n}(X)$; $A(\Gamma) = \mathrm{sgn}(\sigma)\Gamma$ for $A_\sigma$ and $\Gamma$ for $A'_\sigma$, and $A'(S_n(X)) = S_{\sigma n}(\tilde X)$ because $s_{\sigma(j)}s_{\sigma(k)}$ is $\mathrm{sgn}(\sigma)$ times the cyclically ordered pair (r4-18, r4-19). □

Fixed. Expectation (E2), that the couplings change sign with $T$ under an odd permutation, is refuted for the Pauli term under both kinds of permutation and for the dual term under the temporal permutation alone; it holds only for the dual term under the paired permutation: $M_n = TJ(n)S_n$ contains the product $TJ(n) = -n$, in which the two reversals cancel. What changes sign is the Pauli term *relative to the mass term*, since $T$ reverses and $M_n$ does not; after the axial rotation that restores the mass (Corollary 5.4), $M_n$ is reversed. Physically the paired odd permutation is an improper rotation, and the electric-dipole coupling is the parity-odd one (Theorem 5.9(iii)).

### 5.4 The two first-order operators and the spatial derivatives

The Dirac operator of the clock $n$ is $\gamma_n^\mu \partial_\mu = n\partial_0 + \sum_a s_a \partial_a$, and $\gamma^0\gamma^\mu\partial_\mu = \partial_0 + \sum_a \alpha_a(n)\partial_a$ with $\alpha_a(n) := ns_a$; the octant Maxwell operator is $\partial_t - D$ with $D = \sum_a e_a\partial_a$ (Theorem 4.16). In both, $\partial_a = \partial/\partial x^a$ differentiates along the three spatial coordinates, the lengths on $V^-$, in $\mathrm{m}^{-1}$.

**Proposition 5.11** (Two Pauli triples)**.** *(i) $\alpha_1(n), \alpha_2(n), \alpha_3(n)$ are symmetric, square to $\mathbf{1}$ and anticommute; so do $e_1, e_2, e_3$. For any such triple $\pi_a$, $(\sum_a k_a\pi_a)^2 = |k|^2\mathbf{1}$, so both $\partial_0 \pm \sum_a \pi_a\partial_a$ factor the wave operator.*

*(ii) $J(n)\alpha_a(n) = Ts_a$ and $e_a = -TJ(e_a)$: the Dirac triple is one clock times the spatial generators, $\alpha_a(n) = -TJ(n)s_a$; the octant triple consists of the three coordinate clocks themselves, $e_a = -TJ(e_a)$, one for each spatial axis.*

*(iii) No fixed element $Z$ gives $\alpha_a(n) = Ze_a$ for all $a$, nor $\alpha_a(n) = e_aZ$; in particular none of $T$, $\Gamma$, $\omega_i$, $J(n)$ does.*

*Proof.* (i) $n$ and $s_a$ are orthogonal matrices, $n$ symmetric, $s_a$ antisymmetric, anticommuting: $(ns_a)^2 = -n^2s_a^2 = \mathbf{1}$ and $ns_ans_b = -s_as_b$. (ii) $J(n)n = T$, and $TJ(e_a) = T^2e_a$. (iii) $\alpha_a(n) = Ze_a$ for all $a$ would give $Z = ns_ae_a$ for $a = 1, 2$, hence $s_1e_1 = s_2e_2$, which is false; $\alpha_a(n) = e_aZ$ would give $e_1ns_1 = e_2ns_2$, whose grade-one part $n_1s_1 = n_2s_2$ forces $n = \pm e_3$, and $e_1e_3s_1 \neq e_2e_3s_2$ (r4-20, symbolic in $n$). □

**Theorem 5.12** (The triples are conjugate only by elements of mixed parity)**.** *For each clock the $U \in M_8(\mathbb{R})$ with $U\alpha_a(n) = e_aU$ $(a = 1, 2, 3)$ form an eight-dimensional space, $\mathcal{C}\,U_0$ for any invertible solution $U_0$, where $\mathcal{C} = \mathrm{span}\{\mathbf{1}, T\} \otimes \mathrm{Cl}^0(V^-) \cong M_2(\mathbb{C})$ is the commutant of the $e_a$; its invertible elements are dense. No nonzero solution is even and none is odd, so no element of the Clifford group, of* $\mathrm{Spin}(3,3)$ *or of* $\mathfrak{so}(3,3)$ *relates the two triples. Every invertible solution carries the Dirac representation to the left-multiplication representation of the temporal Pauli algebra,*

$$U\rho_n(L_i)U^{-1} = -\tfrac{1}{2}e_je_k = -\tfrac{1}{2}J(e_i), \qquad U\rho_n(K_i)U^{-1} = -\tfrac{1}{2}e_i,$$

*whereas $\rho_{\mathrm{oct}}(L_i) = -\frac{1}{2}e_je_k + \frac{1}{2}s_js_k$ and $\rho_{\mathrm{oct}}(K_i) = -\frac{1}{2}e_i - \frac{1}{2}s_i\omega$; by Proposition 5.8 no $U$ relates $\rho_n$ to $\rho_{\mathrm{oct}}$.*

*Proof.* Both triples generate copies of $\mathrm{Cl}(3,0) \cong M_2(\mathbb{C})$, which has a unique simple module, $\mathbb{R}^4$; $S$ is twice it for each, so intertwiners exist and form a torsor of the commutant (inherited: Wedderburn's theorem). If $U$ had definite parity, $U\alpha_a$ would have the parity of $U$ and $e_aU$ the opposite one, so both would vanish, and $U = 0$ since $e_a$ is invertible. The exact solution spaces, their even and odd parts, invertibility of a generic solution and the conjugated generators are computed at $n = e_1$ and $(2, 3, 6)/7$ (r4-21); $\rho_n(L_i) = -\frac{1}{2}\alpha_j\alpha_k$ and $\rho_n(K_i) = -\frac{1}{2}\alpha_i$. □

**Proposition 5.13** (Left and right in the Fock picture)**.** *On $S = \mathrm{Cl}(V^+)|\mathrm{vac}\rangle$ (Proposition 4.4) the temporal generators act by left multiplication, $e_a(x|\mathrm{vac}\rangle) = (e_ax)|\mathrm{vac}\rangle$, and the spatial ones by twisted right multiplication, $s_a(x|\mathrm{vac}\rangle) = (\hat{x}e_a)|\mathrm{vac}\rangle$, $\hat{x} = (-1)^{|x|}x$. Hence $\alpha_a(n)(x|\mathrm{vac}\rangle) = (n\hat{x}e_a)|\mathrm{vac}\rangle$.*

| depends on the clock $n$ | independent of the clock |
|---|---|
| $\gamma_n^0 = n$, $J(n)$, $\Omega_n$; $\mathfrak{l}_{\mathrm{int}}(n)$, $\phi_n$, the boosts $\rho_n(K_i)$; $\alpha_a(n)$; $K_n^\mu$, $H_n(\mathbf{p})$; the Dirac complex structure $\pm J(n)$ and charge; the eigenspaces $n = \pm 1$ that carry the two senses; the couplings $M_n$, $M_n^\star$; the base time $t_n$ | $T$, $T\Gamma = -e_4e_5e_6 = -n\Omega_n$, $\Gamma = J(n)\Omega_n$; the rest flow $e^{-m\lambda T}$; the rotations $\rho_n(L_i) = \frac{1}{2}s_j s_k$; $\rho_{\mathrm{oct}}$, $\Phi$, $G$, $\mathfrak{l}_G$, the octants and pole operators; $D = \sum_a e_a\partial_a$ and the coefficients of the octant Maxwell equation (13) |

Table 5: Clock dependence of the objects of the two readings (r4-2, r4-3, r4-18, r4-23).

*Proof.* On the eight blades $x = e_I$, $I \subset \{1,2,3\}$ (r4-22). □

Restated: the right action is that of Proposition 4.12(ii), where $e_{a+3}\omega$ acts as $x \mapsto -xe_a$, on the Fock model of Chevalley [24] (Proposition 4.4); verified here on all blades. Inherited: in the Pauli and space-time algebras a Dirac spinor transforms by one-sided and the field bivector by two-sided multiplication [28, 13], which is the difference between $\rho_n$ and $\rho_{\mathrm{oct}}$ in Theorem 5.12.

These results answer the remark after Theorem 4.16. The Dirac reading couples $\partial_a$ to the spatial generators, through $\gamma^0\gamma^a = \alpha_a(n) = ns_a$; in the Fock picture this is the clock $n$ on the left and the temporal Pauli generator $e_a$ on the right. The octant field reading couples $\partial_a$ to the temporal generators $e_a$ on the left. In both the index $a$ of the derivative is matched to the index $a$ of a generator by the pairing $a \leftrightarrow a+3$ of the arena, which is background structure, and the derivative is along the length $x^a$ on $V^-$ in both. The two triples are conjugate (fixed), by elements of mixed parity only, which are no symmetry of the arena (fixed), and any Pauli triple would make the operator hyperbolic (fixed). So neither operator shows that spatial differentiation follows from temporal directions: the octant operator uses the three coordinate clocks $e_a$ as spatial Pauli matrices, one per axis, not one clock, and the statement that space is differentiated "through time" is a reading of the choice of representation (interpretive), not a consequence of either equation; as an inference it is refuted, since conjugating by $U$ rewrites Maxwell's equations with the Dirac triple for the octant map $U^{-1}\Phi$.

## 5.5 Varying and permuting the clock

**Proposition 5.14** (The field reading does not see the clock; the particle reading does)**.** *The objects of the right column of Table 5 are defined without $n$; the octant Maxwell equation at clock $n$ is $(\partial_{t_n} - D)\Phi(X) = -(\rho|\mathrm{vac}\rangle + \mathbf{j}\cdot f)$, with the same matrices for every clock, the clock entering only through the time coordinate of the base $E_n$. The objects of the left column depend on $n$: $n \mapsto J(n)$ is injective, $\mathfrak{l}_{\mathrm{int}}(n) = \mathfrak{l}_{\mathrm{int}}(n')$ only for $n' = \pm n$, and $M_{e_1} \neq M_{e_2}$. All of them are carried from one clock to another by the rotors of* $\mathrm{Spin}(V^+)$*, which are not symmetries of the field reading, and, together with the field, by the diagonal rotations, which are.*

*Proof.* By inspection of the definitions in Section 4; Propositions 5.1, 5.2 and 5.10; $\mathfrak{l}_{\mathrm{int}}(n)$ determines $E_n$ and hence $\pm n$ (Proposition 3.2). □

Fixed. This answers the question of Section 8, item 6, of [3], whether different clocks are physically different sectors, gauge-equivalent choices or an order parameter: the diagonal rotations map the coupled data at $(n, \psi, X)$ to those at $(Rn, R\psi, RX)$ (Proposition 5.10(ii)), so for theories built from the structures of this note a change of clock is a change of frame (fixed as a statement of equivalence; its reading as "no sector" is interpretive); whether a dynamics singles out a clock is open, as in Remark 3.14.

**Theorem 5.15** (Uniqueness of $T$)**.** *Let $Y \in \mathrm{Cl}(3,3)$.*

*(i) If $n^{-1}Y$ commutes with $\gamma_n^0, \ldots, \gamma_n^3$ for three linearly independent clocks, then $Y \in \mathbb{R}T$. Three clocks of the Fisher–Rao octant suffice, e.g. $e_1$, $(1,2,2)/3$, $(2,3,6)/7$. For two clocks $n, n'$ the solutions are* $\mathrm{span}\{T,\ (n \times n')\, e_4e_5e_6\}$.

*(ii)* *If $n^{-1}Y$ lies, for two independent clocks, in the joint commutant* $\mathrm{span}\{\mathbf{1}, \Omega_n, J(n), \Omega_n J(n)\}$ *of* $\mathfrak{l}_{\mathrm{int}}(n)$ *and* $J(n)$ *in which Theorem 4.1 of [1] selects the imaginary unit, then* $Y \in \mathrm{span}\{T, e_4e_5e_6\} = \mathrm{span}\{T, T\Gamma\}$*, the two* $\mathrm{Spin}(3,3)$*-invariant forms. Indeed* $nJ(n) = T$ *and* $n\Omega_n = e_4e_5e_6$ *for every clock.*

*(iii)* *Consequently* $T$ *is, up to scale, the only element of* $\mathrm{Cl}(3,3)$ *that is the product of the clock with a Dirac complex structure (a square root of* $-\mathbf{1}$ *in the commutant of the* $\gamma_n^\mu$*) for every clock and that does not depend on the clock: the products are* $\pm nJ(n) = \pm T$*.*

*Proof.* (i) The commutant of the $\gamma_n^\mu$ is $\mathrm{span}\{\mathbf{1}, J(n), u_1\Omega_n, u_2\Omega_n\}$ (Theorem 3.5(ii)), so $Y \in n \cdot (\dots) = \mathrm{span}\{n, T, u_1e_4e_5e_6, u_2e_4e_5e_6\}$, using $nu_a\Omega_n = -u_ae_4e_5e_6$. The four have grades $1, 3, 4, 4$, so for $Y$ fixed the grade-one part lies in $\bigcap \mathbb{R}n = 0$, the grade-four part in $(\bigcap U_n)\, e_4e_5e_6$, which is 0 for three independent clocks and the line of $n \times n'$ for two. (ii) $n \cdot \mathrm{span}\{\mathbf{1}, \Omega_n, J(n), \Omega_n J(n)\} = \mathrm{span}\{n, e_4e_5e_6, T, n\Gamma\}$, of grades $1, 3, 3, 5$; the grades one and five vanish for two independent clocks; $T\Gamma = -e_4e_5e_6$ (r4-1) and $\mathrm{span}\{T, T\Gamma\}$ is the space of invariant forms (Proposition 4.14(iii)). (iii) If $n^{-1}Y$ is a square root of $-\mathbf{1}$ commuting with the $\gamma_n^\mu$ for every clock, then $Y \in \mathbb{R}T$ by (i), so $n^{-1}Y = aJ(n)$, and $(aJ(n))^2 = -a^2\mathbf{1}$ gives $a = \pm 1$; with the selection of Theorem 4.1 of [1] at every clock (Theorem 3.5) the Dirac complex structures are $\pm J(n)$ and the products are $\pm T$ directly. For two clocks the elements $\cos\vartheta\, T + \sin\vartheta\, \hat{w}\, e_4e_5e_6$, $\hat{w}$ the unit vector along $n \times n'$, meet the conditions at both, so (iii), like (i), needs three clocks. Exact solutions in r4-23. □

Fixed; by the two-clock exception in (i) two clocks do not single out $T$, and three are needed. The second element of (ii), $e_4e_5e_6$, is the clock times the other complex structure $\Omega_n$, which Theorem 4.1 of [1] excludes because it anticommutes with the $\gamma_n^\mu$.

**Remark 5.16** (The earlier "covariant mass" $e_4e_5e_6$)**.** It was proposed in an earlier, superseded version of the programme that, of the two trivectors anticommuting with the three Dirac matrices $\alpha_a$, the spatial volume $e_4e_5e_6$ is the scalar mass and $e_1e_2e_3$ its pseudoscalar partner. In the present conventions the first half is confirmed at every clock: the trivectors anticommuting with $\alpha_1(n), \alpha_2(n), \alpha_3(n)$ are $\mathrm{span}\{T, e_4e_5e_6\}$, the clock-independent space of Theorem 5.15(ii) (r4-24, at four clocks; at all clocks by Proposition 5.2(iii), since conjugation by a rotor preserves the grade). The assignment is corrected. As the matrix $\beta$ of $H = \sum_a \alpha_a p_a + m\beta$, $e_4e_5e_6 = n\Omega_n$ is admissible (symmetric, square $\mathbf{1}$, commuting with $J(n)$, $H^2 = (|\mathbf{p}|^2 + m^2)\mathbf{1}$), but relative to the kinetic kernels with $\gamma^0 = n$ it gives the pseudoscalar mass of Proposition 4.6 of [1], whose Grassmann kernel $J(n)e_4e_5e_6 = T\Omega_n$ depends on the clock; the scalar mass has $\beta = n$ and the clock-independent kernel $J(n)n = T$. The element $e_1e_2e_3$ is not a $\beta$ at all (it is antisymmetric with square $-\mathbf{1}$); it is the kernel.

Interpretation. In the particle reading the clock and the complex structure both move, and their product $T$ does not; in the field reading $T$ is the duality on the octants and, as the Chevalley–Mukai pairing, the form pairing $E_k$ with $B_k$ (Proposition 4.14). The mass form is the part of the particle reading that the field reading shares. This is interpretive; what is fixed is Theorems 5.5 and 5.15.

### 5.6 Answer

(1) *Consistency.* $T = nJ(n)$ holds at every clock and is compatible with all operations considered: $T$ is invariant under all rotations of the temporal triple and the diagonal rotations, reversed together with $J(n)$ by odd permutations, and with the duality operator by the paired ones, and fixed by the reversal of the clock (Propositions 5.2 and 5.3; fixed). Its three roles, mass form, parity and duality, are compatible (Theorem 5.5; restated and fixed). It incorporates consistently into both readings.

(2) *The Lorentz representations stay apart.* On $S$ the Dirac representation $\rho_n$ and the octant representation $\rho_{\mathrm{oct}}$ are inequivalent, their joint commutant is $\mathbb{R}\mathbf{1}$, and no operator, $T$ included, relates them (Propositions 5.7 and 5.8 and Theorem 5.12; fixed). A single module carrying both as the Lorentz action of one field is obstructed; the Dirac field and the field octants are two fields.

(3) *The bridge as a coupling.* The answer to "The main obstruction" of Section 4.8: $\rho_{\mathrm{oct}}$, $\rho_n$ and $\Phi$ coexist as the actions on two different fields, and the covariant local couplings without derivatives that are linear in the field strength are trilinear, quadratic in $\psi$: the Pauli kernel $TJ(n)S_n(X)$ and the electric-dipole kernel $T\Gamma S_n(X)$, a two-dimensional space at every clock, bilinear couplings being excluded by the $2\pi$ rotation (Theorem 5.9; fixed, the textbook terms inherited). Couplings of higher order in $X$ exist, for instance $(\psi^\top T\psi)\,\eta(X,X)$, since $T$ is $\rho_n$-invariant and $\eta(X,X)$ is $\mathfrak{l}_G$-invariant (r4-25). Since $\Phi$ is an equivalence onto its image, the octants add no coupling that $X$ itself does not have. $T$ is the clock-independent factor of every Dirac kernel; the couplings are conjugated by the rotation that moves the clock, and an odd permutation paired with the spatial one reverses the electric-dipole term with $T$ but not the Pauli term, and the temporal one alone reverses neither (Proposition 5.10; fixed, (E2) refuted for the Pauli term).

(4) *$J(n)$ on the field.* $J(n) = -L(n) + A(n)$ rotates the transverse octants about the axis with the components of $n$ and maps the longitudinal octants to the Gauss-law poles; this field rotation $L(n)$ is the Dirac rotation about the spatial axis $\sum_i n_i s_i$ combined with half a turn of the charge (Proposition 5.6; fixed; the reading interpretive).

(5) *Spatial derivatives.* The remark after Theorem 4.16 is answered by Propositions 5.11 and 5.13 and Theorem 5.12: both operators differentiate along the lengths of $V^-$, the Dirac reading through the spatial generators and one clock, the octant reading through the three temporal generators; the triples are conjugate only by mixed-parity elements, and nothing derives spatial differentiation from temporal directions (fixed; the inference refuted).

(6) *The clock.* The field reading is clock-independent and the particle reading is not; $T$ is the unique clock-independent product of the clock with a Dirac complex structure, and with the second complex structure $\Omega_n$ the space becomes $\mathrm{span}\{T, T\Gamma\}$ (Proposition 5.14 and Theorem 5.15; fixed). Clocks are related by the diagonal rotations, a symmetry of both readings and of the couplings, so a change of clock is a change of frame (fixed as an equivalence); whether a dynamics selects a clock is open.

(7) *What remains for a covariant Dirac–Maxwell theory in the arena.* The minimal coupling through the potential and $J(n)$, $D_\mu = \partial_\mu + qA_\mu J(n)$ (Proposition 2.13, with the Lagrangian (7); inherited, restated), which uses $A_\mu$ on the base and not the octants; the Pauli and electric-dipole couplings (14), with two constants the arena does not fix (open); the absence of a common Lorentz representation on one $S$ (obstructed at one constituent; the two readings can be housed together at the mixed second rung of Section 13.6 of [1], at the price stated there, restated); and the quantisation of either reading (open, as in [3]).

**On the working note.** Proposition 5.1 of [3] is confirmed; its statement that reversing $n$ reverses $J(n)$ is confirmed and sharpened (Proposition 5.3(iv)); its deferred temporal permutations and rotations are carried out (Propositions 5.2 and 5.3); its item 5 of Section 8 is answered in part by Proposition 5.6 and Section 5.3 ($J(n)$ is not the duality, a common Lorentz representation on $S$ does not exist, and the cross-sector coupling has the units, covariance and parity stated in Theorem 5.9 and (14)), while the projector vacuum and the mixed/spectator embedding mismatch named there are not treated here (open); its item 6 is answered by Proposition 5.14.

Its requirement that a cross-sector coupling specify symplectic and quantum consistency is not addressed here (open).

Table 6: The claims of this section, with their checks and classes.

| id | claim | class |
|---|---|---|
| r4-1 | conventions: Clifford relations, octants, $T$, $\Gamma$, $T\Gamma = -e_4e_5e_6$ | restated |
| r4-2 | $T = nJ(n) = J(n)n$, $J(n)^2 = -\mathbf{1}$ at every clock (Proposition 5.1) | restated ([1], Proposition 3.4); FMB confirmed |
| r4-3, 4 | $T$ invariant, $J(n)$ covariant under $\mathfrak{so}(3)_+$ and $\mathfrak{so}(3)_\Delta$; only $\mathfrak{so}(3)_\Delta$ commutes with $G$ (Proposition 5.2) | fixed |
| r4-5, 6 | odd permutations: $T \mapsto -T$, $J(n) \mapsto -J(\sigma n)$; paired: $\Phi \mapsto \Phi_{\mathrm{t}}$, duality $-T \mapsto T$ (Proposition 5.3) | fixed |
| r4-7 | clock reversal is a half-turn fixing $T$; $C = RT$ (Proposition 5.3(iv)) | fixed; FMB sharpened |
| r4-8 | $T = \pm J(n)$ on $n = \pm 1$; odd permutation $\Rightarrow (m, q) \mapsto (-m, -q)$ (Corollary 5.4) | restated (RE); fixed |
| r4-9 | three roles of $T$; $T$ is the Dirac parity at every clock (Theorem 5.5) | restated, fixed |
| r4-10 | $J(n) = -L(n) + A(n)$, $LA = 0$; $L(n) = \rho_n(L(n)) - \frac{1}{2}J(n)$ (Proposition 5.6) | fixed |
| r4-11 | $J(n)$ rotates transverse octants, maps longitudinal ones to the poles (Proposition 5.6) | fixed; reading interpretive |
| r4-12 | commutants $M_2(\mathbb{C})$, $\mathbb{C} \oplus M_2(\mathbb{R})$, joint $\mathbb{R}\mathbf{1}$ (Proposition 5.7) | fixed |
| r4-13 | three inequivalent Lorentz representations on $S$ (Proposition 5.8) | restated (decompositions), fixed |
| r4-14 | no invariant bilinear coupling $\psi^\top M\Phi(X)$ (Theorem 5.9(i)) | obstructed |
| r4-15, 16 | trilinear couplings: dimension 2 (Grassmann), 6 (commuting); Pauli $TJ(n)S_n$ and dual $T\Gamma S_n$ | fixed; textbook terms inherited |
| r4-17 | Pauli term parity-even, dual parity-odd (Theorem 5.9(iii)) | fixed; inherited in substance |
| r4-18 | couplings conjugated by the clock rotor, $T$ fixed; diagonal covariance (Proposition 5.10) | fixed |
| r4-19 | odd permutation: Pauli term does not change sign, dual does when paired (Proposition 5.10(iii)) | fixed; (E2) refuted for the Pauli term |
| r4-20 | two Pauli triples, $J(n)\alpha_a = Ts_a$, $e_a = -TJ(e_a)$, no one-sided relation (Proposition 5.11) | fixed |
| r4-21 | intertwiners: dimension 8, none of definite parity; Dirac $\mapsto$ left Pauli representation (Theorem 5.12) | fixed; existence inherited |
| r4-22 | temporal = left, spatial = twisted right multiplication (Proposition 5.13) | restated (Proposition 4.12), verified; one-/two-sided inherited |
| r4-23 | uniqueness of $T$; two-clock exception; span$\{T, T\Gamma\}$ (Theorem 5.15) | fixed |
| r4-24 | trivectors anticommuting with $\alpha_a(n)$; $e_4e_5e_6$ as $\beta$ is the pseudoscalar mass (Remark 5.16) | fixed; earlier assignment corrected |
| r4-25 | a covariant quartic coupling $(\psi^\top T\psi)\,\eta(X, X)$: the trilinear couplings are the only ones linear in $X$, not the only ones (Section 5.6) | fixed |
| – | spatial derivatives "from temporal directions" (Section 5.4) | interpretive; inference refuted |
| – | a change of clock is a change of frame (Proposition 5.14) | fixed (equivalence); interpretive |
| – | mass form = the shared part of the two readings | interpretive |

| id | claim | class |
|---|---|---|
| – | values of $\mu$, $d$; a dynamics selecting the clock; quantisation | open |

# 6 Synthesis

The four questions of the introduction have the following answers; each is stated with its class, and the sections hold the proofs.

**(1) Lagrangians.** For the classical reading the free worldline of the arena gives the six-momentum and the shell $E^2 = c^2|\mathbf{p}|^2 + M^2c^4 - c^2|\mathbf{q}|^2$, and the interior shifted Lagrangian gives $\ddot{Y} = -\kappa^2 Y$ with canonical momentum obeying $\dot{\pi} = \kappa G\pi$ (restated, Propositions 2.5 and 2.6). The Dirac Lagrangian $\frac{1}{2}\psi^\top(K^\mu\partial_\mu - mT)\psi$ gives $\gamma^\mu\partial_\mu\psi = -mJ\psi$, its current and its stress tensors (restated and inherited, Proposition 2.8). The energies are governed by the Maxwell Lagrangian $-\frac{1}{2}\operatorname{Re} q$, whose Hamiltonian is the undivided energy $E_6$ and whose first-order form makes duality an off-shell symmetry (inherited, Propositions 2.9 and 2.12). The rest energy $E_0 = \frac{1}{2}|q|$, taken as a Lagrangian, is Lorentz invariant and conformal but not duality invariant in the sense of Gaillard and Zumino, has an indefinite energy, and is singular on radiation; its natural place is the first-order term of the ModMax family. Neither $E_0$ alone nor any Hamiltonian $aE_6 + bE_0$ with $a^2 - b^2 \neq 1$ is Lorentz invariant (fixed, Theorem 2.10 and Proposition 2.11).

**(2) The two $X$'s.** $X_{\mathrm{cl}}$ is a displacement, with converted times on $V^+$ and lengths on $V^-$; $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$ is the pair labelling the six components of the field two-form. They share the vector space and the Clifford action and differ in units, natural Lorentz algebra ($\mathfrak{l}_{\mathrm{int}}$ on a four-plane against $\mathfrak{l}_G$ on the two-form), canonical conjugates and observables. No linear map equivariant for the interior Lorentz algebra relates them in either direction, whatever dictionary is chosen (fixed, Theorem 2.3). The working note's rule of interpretation is confirmed and strengthened.

**(3) The four-momentum.** With a clock $n$ on the temporal triple, $E_n = \mathbb{R}n \oplus V^-$ has signature $(1,3)$; the projection $P_n = (E/c, \mathbf{p})$ of the six-momentum is a four-vector of $\mathrm{SO}^+(1,3)_n$, with $\mathbf{p}$ conjugate to the three lengths and $E/c$ to the time read by $n$, and it is conserved for every dynamics independent of $(ct_n, \mathbf{x})$ (fixed, Propositions 3.3 and 3.7). Of the three mass shells of the working note, the ambient one is the earlier statement that splitting the temporal energy lowers the mass, the first-class reduction is the paper's surplus-time constraint with the clock $n$, and the third is constructed (restated and constructed, Propositions 3.8, 3.10 and 3.11). The four-dimensional mass read by a clock satisfies $m_4(n)^2c^4 = \mu^2c^4 - |n \times \mathbf{E}|^2$, never above the six-dimensional value (fixed, Proposition 3.13). The field has no linear covariant map to a four-vector; its four-momentum is the integral of its stress tensor, a four-vector only for an isolated whole, and the photon's $P = \hbar k$ is not derived in the arena (fixed, inherited and open, Section 3.4). The anti-isometry no-go holds for every clock and obstructs the identification of the interior momentum with the six-momentum, not the duality of the field (fixed, Propositions 3.22 and 3.23).

**(4) The pair on the octants.** The octants are the joint eigenlines of the pole operators, the Fock basis of $S \cong \Lambda^*(L)$. The field-octant map places the electric member of the pair on the three octants of weight one and the magnetic member on the three of weight two, uniquely up to a scale and a quarter turn of duality once rotations and Maxwell's equations are imposed; $\mathbf{E}$ is then a one-form and $\mathbf{B}$ a two-form (fixed and constructed, Theorem 4.7 and Proposition 4.17). Maxwell's four equations are one first-order equation on $S$, with the two Gauss laws on the two poles and the evolution laws on the field octants (fixed, inherited in substance, Theorem 4.16);

the spatial derivatives in it multiply the temporal generators, a property of the representation and not a derivation of space from time. The mass form pairs each electric octant with its magnetic antipode and gives $-\mathbf{E}\cdot\mathbf{B}$ (fixed, Proposition 4.14). Clifford multiplication by the field point mixes the two members of the pair and does not realise it (Proposition 4.10). Boosts and the duality rotation act on the octants only through odd operators (obstructed for even operators, Theorem 4.11 and Proposition 4.12). The construction represents the classical field strength; it quantises nothing.

**The bridge.** The element shared by the two readings is $T = e_1e_2e_3 = nJ(n)$: for every clock it is the Dirac mass form of the particle reading, it lifts the frame parity, and on the octants $-T$ is the duality of the field reading (fixed, Theorem 5.5); it is, up to scale, the only element that is the clock times a Dirac complex structure for every clock (fixed, Theorem 5.15). The field reading does not see the clock and the particle reading does (fixed, Proposition 5.14). The Dirac action, the octant representation of the field and the spin lift of $\mathfrak{l}_G$ are three inequivalent Lorentz representations on one module, whose joint commutant is the scalars; no bilinear coupling of a Dirac field to the field octants is covariant, while the trilinear couplings are exactly the Pauli term and its dual, at every clock (fixed and inherited, Proposition 5.8 and Theorem 5.9). The answer to the question whether $T = nJ(n)$ incorporates consistently into both readings is therefore yes at the level of the module: the two readings keep their Lorentz representations, and they meet in $T$ and in the Pauli coupling. A covariant Dirac–Maxwell dynamics in the arena, with the minimal coupling through the potential and the charge $J(n)$, a dynamics that selects the clock, and the quantisation of either reading remain open.

# 7 Methods

Every finite-dimensional and symbolic claim of Sections 2 to 5 is checked by a Python script in `investigations/two_readings/verify/`, using `numpy`, `sympy` and `scipy` only. Each script prints one line `PASS <id>: <claim>` or `FAIL <id>: <claim>` per claim and exits with status zero only if every check passes; the sections cite the identifiers. The derivation scripts `r1_lagrangians.py`, `r2_fourmomentum.py`, `r3_octants.py` and `r4_bridge.py` were written with the sections. The scripts `s1_check.py`, `s2_check.py`, `s3_check.py` and `s4_check.py` are independent re-checks written by adversarial referees, who did not import the derivation scripts and used different methods where possible (other bases, rank computations over the rationals instead of eigenvalues, symbolic instead of numerical verification); where a referee found a claim false, the corrected statement was checked and the section corrected.

Algebraic identities are checked in exact arithmetic: the generators of the realisation of Appendix A of [1] are signed permutation matrices, and dimensions of commutants, intertwiner spaces and invariant forms are exact ranks of integer or rational linear systems. Lagrangian variations, Euler–Lagrange equations, Noether currents, stress tensors and Maxwell's equations are derived on general symbolic fields; the Grassmann-valued Dirac field is handled by an exact Grassmann jet algebra. Floating point is used only for numerical integrations and random tests, with the tolerance stated at each check. The command

```
python3 investigations/two_readings/verify/run_all.py
```

runs every script and writes a summary to `verify/RESULTS.md`.

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